TET15-P1

A Three-Field Hu–Washizu Tetrahedron

Research note

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Abstract

TET15-P1 is a tetrahedral finite element for the large-deformation plasticity of nearly incompressible materials: materials whose bulk modulus κ, the resistance to a change of volume, is much larger than their shear modulus μ, the resistance to a change of shape, as in rubber and in metals during plastic flow. Three fields — the motion, a scalar volumetric strain and a pressure — are coupled through a Hu–Washizu functional, a functional in which the two auxiliary fields are independent of the motion and tied to it by a constraint term. With both auxiliary fields in one discrete space that is discontinuous between elements, the functional reduces exactly to a functional of the motion alone in which the volumetric strain $\theta(J)$, a function of the Jacobian J of the motion, is replaced element by element by its L^2 projection onto that space (mean dilatation), and the auxiliary fields are eliminated element by element. The deviatoric (shape-changing) response is evaluated at the quadrature points, so the constitutive law and its return map, the algorithm that projects a trial stress onto the yield surface, run unchanged, and no quantity other than θ is projected. In its general form the element calls the material at a modified deformation gradient whose volume is the projected one, and needs nothing of the material beyond its stress and tangent; when the stored energy splits exactly into a volumetric part quadratic in θ and an isochoric (volume-preserving) part, the pressure equals κ times the projected strain at every point and the element needs one projection and one material call per quadrature point. Both forms are implemented; the general one takes one pass more over the quadrature points and, with the J_2 model, 1.4 to 2.1 times the split form's time per explicit step on six devices. The choice of $\theta(J)$ is a parameter of the formulation, and two instances of the split form are developed, $\theta = \log J$ with the Hencky material and $\theta = J - 1$ with the J_2 plasticity model of Simo and Hughes. The element implemented projects $\theta = \log J$, the volumetric part of the logarithmic strain, in both forms, and the J_2 model is used with the volumetric energy $\frac{\kappa}{2}(\log J)^2$ in place of the volumetric energy $\frac{\kappa}{2}(J-1)^2$ of its implementation in Norma (Remark 5). The discrete spaces are the three-dimensional Crouzeix–Raviart pair: the quadratic tetrahedron enriched with four cubic face bubbles and one quartic interior bubble, polynomials that vanish on a prescribed part of the element boundary, carried by a nodal basis on fifteen nodes (the TET15 displacement element), over a pressure that is linear in the reference coordinates of each element and discontinuous between elements; the name of the element states the two spaces, fifteen nodes for the displacement and a linear (P_1) pressure. Volumetric locking, the excess stiffness of a discretization as κ/μ grows, measured here as a loss of the rate of convergence at large κ/μ that does not vanish under refinement, and spurious soft modes, discrete displacements of vanishing energy, are failures of the two hypotheses of Brezzi's theorem for the pair of discrete spaces: the inf-sup condition, that every discrete pressure is resisted by some discrete displacement with a constant independent

of the mesh, and coercivity of the deviatoric form on the discretely isochoric subspace. The inf-sup constant of the pair is measured over a structured mesh sequence at 0.2968, against 0.2216 for the Taylor–Hood pair (continuous quadratic displacement over continuous linear pressure), whose stability is established; of seven element-local pairs two are stable and five decay. On randomly perturbed meshes the constant of the pair lies between 0.287 and 0.295, and on unstructured meshes between 0.319 and 0.332 beyond the coarsest one. Under a plastic tangent that removes the shear stiffness along the flow direction, the coercivity constant cannot separate pairs, and what separates them is the dilatation carried by their softest modes: 0.20 to 0.27 without decay under refinement for elements with a constant pressure, the composite tetrahedron CT10 [16], measured in its Albany implementation with its volume-averaged J, among them, against a decay at local rates of 3.1 to 2.2 in the mesh size h for the Crouzeix–Raviart pair on structured meshes. The smallest energy of a discretely isochoric displacement with a prescribed dilatation, a measure that does not depend on the number of eigenpairs computed, lies above the infimum of the continuum for the Crouzeix–Raviart pair on every mesh except one strongly perturbed mesh, and below it for the constant-pressure pairs. On unstructured meshes the Crouzeix–Raviart pair has element-scale dilatational members within 12% of its lowest eigenvalue. Against a manufactured solution in linear elasticity, the error of the element at $\kappa/\mu = 10^6$ is 1.36 times its error at $\kappa/\mu = 1$ on the finest mesh, and its rate of convergence at $\kappa/\mu = 10^6$ is 0.047 below the rate at $\kappa/\mu = 1$, a difference that decreases under refinement; the quadratic tetrahedron without projection loses 0.94 in rate. On Cook's membrane in three dimensions at Poisson's ratio 0.4999, on three meshes of 327 to 14 475 elements, TET15-P1 and CT10 agree in the tip displacement within 0.6% on every mesh, and their pressure fields are smooth away from the singular clamped edge, where the pressure of the quadratic tetrahedron without projection changes sign inside elements over the whole membrane; in the plastic case at Poisson's ratio 0.29 every element converges to the same tip displacement within 0.5% on the finest mesh. A fourteen-point quadrature rule exact to degree five is the smallest that leaves the element its six rigid-body modes, the row-sum lumped mass of the nodal basis is positive, and the explicit stable step is 3.3 times smaller than that of the linear tetrahedron. The consistent mass matrix requires a rule of degree eight, and both implementations integrate it with a conical product rule of degree nine with 125 points. On the Taylor bar impact in explicit dynamics, with plastic strains above 3, TET15-P1 gives the final radius within 0.07% of the extrapolated converged value on 25 029 elements, where CT10 needs 1.5 million elements for 0.09%; within Sierra/SM the first costs 1/54 of the core-seconds of the second, and per element and time step it takes at most 2.6 times as long. On the indentation of a cylinder by a flat punch at $\kappa/\mu = 10^4$, a plastic zone confined by elastic material, on four levels of refinement, two independent implementations of the element, with different J_2 models, agree in force within 0.06%; its element-to-element pressure jumps are the smallest of the elements compared and decay under refinement, and the quadratic tetrahedron without projection locks and carries a pressure mode that grows under refinement. CT10 shows no element-to-element pressure mode that persists: its jumps decay faster than first order on the finest level in case A. The soft modes of the linearized analysis vary within an element and are not resolved by that measurement. In the perfectly plastic case the assembled tangent of TET15-P1 loses positive definiteness at an unconverged Newton iterate. Its negative eigenvalue comes from the pressure term of the tangent, evaluated at a projected pressure of 316 times the yield stress against 1.8 to 2.3 times at the converged states.

1 Problem statement

Tetrahedra are the only practical choice for meshing complex geometry automatically. Their accuracy is insufficient in four regimes: near-incompressible response at bulk-to-shear modulus ratios $\kappa/\mu \lesssim 10^4$, with κ the bulk modulus and μ the shear modulus; isochoric (volume-preserving) plasticity; large rotation; and weak shocks, pressure waves of small amplitude, in materials of high κ/μ. The target is one spatial formulation for all four, with the time integrator the only difference between the quasi-static and the dynamic regimes.

Two established remedies fail, and each failure is the failure of one hypothesis of the same theorem; this section identifies the hypothesis in each case.

Constant pressure relieves locking and introduces soft modes. CT10, a ten-node tetrahedron subdivided into twelve linear tetrahedra with one element-constant pressure [24, 32], and its five-field extension [16] relieve volumetric locking — the stiffening of a discretization whose displacement space contains too few discrete isochoric fields — with that pressure. Foulk III et al. [16] report that this choice — and it is the choice, not the element — produces a deleterious soft mode "common to both the quadratic and composite tetrahedral element with constant pressure formulations," remedied there by a convex energy penalty. The mechanism is a coercivity failure on the discretely isochoric subspace: an element-constant multiplier does not resist displacement patterns whose volume change varies within the element and averages to zero, so those patterns carry only shear energy. In elasticity every projected-pressure element has such patterns, and a stable pair keeps their energy bounded away from zero; in a confined plastic zone the shear response softens along the flow direction and the pattern can approach zero energy. §9.4 shows that on the quadratic tetrahedron these patterns are not exactly of zero energy, and §6 states the question that count does not settle.

Residual-based stabilization contaminates the deviator. Variational-multiscale stabilization of linear tetrahedra [26] introduces a subgrid pressure of the form

$$p' = c_1\,\tau_{\mathrm{s}}\,(\kappa\,\mathrm{Div}\,\dot{\boldsymbol{\varphi}} - \dot{p})\,, \qquad \tau_{\mathrm{s}} = c_2\,\Delta t, \tag{1}$$

with c_1 and c_2 tuned coefficients, which is added to the stress before the constitutive update is evaluated. Two consequences follow. The spatial solution depends on the time step Δt through τ_{s}, so there is no quasi-static limit independent of it. The coefficient c_2 scales a term that reaches the deviator, and therefore the plastic return map — the algorithm that projects a trial stress onto the yield surface — which is the origin of the oscillations in the equivalent plastic strain, the accumulated norm of the plastic strain increments. No choice of c_2 removes this, because the term enters the deviator by construction.

The common cause. Brezzi's theorem for the mixed problem asks two things of the pair — the discrete space U_h for the displacement together with the discrete space Q_h for the pressure [7, §3.4, §5.2]: the inf-sup condition, that every pressure mode be resisted by some displacement, whose failure is locking on one side and spurious pressure on the other; and coercivity of the deviatoric form on the discretely isochoric subspace $\ker \boldsymbol{G}_h$, that no displacement the pressure does not resist have zero deviatoric energy, whose failure is a soft mode. The first is a property of the pair alone and is settled in §9.5. The second depends on the material tangent as well as

on the pair, and §6 states what this note claims about it. Neither difficulty is a stabilization problem: the remedy for the first is a pair satisfying the inf-sup condition uniformly in the mesh size, and the second must then be checked where the material can violate it. §2 defines both hypotheses.

Organization. §2 defines volumetric locking, the mixed problem, the inf-sup condition and Brezzi's theorem, and gives the construction by which bubble functions satisfy the inf-sup condition. §3 states the continuum formulation and its reduction to mean dilatation. §4 defines the discrete spaces, the element, its quadrature rule and mass, and the element algorithm in the form an implementation needs. §5 gives the plasticity model. §6 states the claim about soft modes under a plastic tangent, §7 the explicit time integration, and §8 an optional extension. §9 reports the measurements on the linearized operator, §10 the nonlinear results on Cook's membrane, §11 the Taylor bar impact in explicit dynamics, and §13 and §14 summarize the evidence, the open measurements and the limitations.

2 Background: locking, the mixed problem and the inf-sup condition

This section defines the terms the rest of the note uses. The setting is small-strain isotropic elasticity, because every stability property below is measured on the linearized operator at the undeformed state (§4.1); the notation $\boldsymbol{\epsilon}(\boldsymbol{v}) := \operatorname{sym}\operatorname{Grad}\boldsymbol{v}$ is the small strain of a displacement $\boldsymbol{v}$, $\operatorname{dev}\boldsymbol{\epsilon} := \boldsymbol{\epsilon} - \frac{1}{3}(\operatorname{tr}\boldsymbol{\epsilon})\boldsymbol{I}$ its deviatoric part, and $\operatorname{Div}\boldsymbol{v} = \operatorname{tr}\boldsymbol{\epsilon}(\boldsymbol{v})$ its volume change.

2.1 Why low-order tetrahedra lock

The stored energy of an isotropic linear elastic material,

$$W(\boldsymbol{\epsilon}) = \mu|\operatorname{dev}\boldsymbol{\epsilon}|^2 + \tfrac{\kappa}{2}(\operatorname{tr}\boldsymbol{\epsilon})^2, \tag{2}$$

assigns the energy $\kappa/2$ per unit squared volume change and μ per unit squared shape change. As κ/μ grows, minimizing the energy over a discrete displacement space drives $\operatorname{Div}\boldsymbol{u}_h$ toward zero at every quadrature point, a point of the integration rule: in the limit the discrete solution is confined to the discrete displacements whose volume change vanishes at every quadrature point. Whether that subspace is rich enough to represent the deformation is a matter of counting. The divergence of a linear tetrahedron is constant, so each element imposes one constraint; a mesh with six tetrahedra per vertex, the count of the structured meshes of §9.1, has about six constraints per vertex against three displacement unknowns per vertex, two constraints per unknown. Few displacements satisfy all of them, so the discrete solid is far stiffer than the material, by an amount that grows with κ/μ and does not decrease with the mesh size. That is volumetric locking. The quadratic tetrahedron has a linear divergence, four constraints per element against about $2 \cdot 3 = 6$ unknowns per element on the same meshes (§9.3), and locks as well.

The remedy is to impose the constraint in an average sense, through an independent pressure field that acts as its Lagrange multiplier, the force that enforces it. This is a mixed method: a method with two independent discrete fields, displacement and pressure, whose behavior depends on the pair of discrete spaces chosen for them.

2.2 The mixed problem and the two hypotheses of Brezzi's theorem

In the incompressible limit the mixed problem reads: find the displacement $\boldsymbol{u}_h \in U_h$ and the pressure $p_h \in Q_h$ with

$$\begin{aligned} a(\boldsymbol{u}_h, \boldsymbol{v}) + b(\boldsymbol{v}, p_h) &= f(\boldsymbol{v}) \quad \text{for all } \boldsymbol{v} \in U_h, \\ b(\boldsymbol{u}_h, q) - \tfrac{1}{\kappa}\,(p_h, q) &= 0 \qquad \text{for all } q \in Q_h, \end{aligned} \tag{3}$$

with

$$a(\boldsymbol{u}, \boldsymbol{v}) := 2\mu \int_\Omega \operatorname{dev} \boldsymbol{\epsilon}(\boldsymbol{u}) : \operatorname{dev} \boldsymbol{\epsilon}(\boldsymbol{v})\, \mathrm{d}V, \qquad b(\boldsymbol{v}, q) := \int_\Omega q \operatorname{Div} \boldsymbol{v}\, \mathrm{d}V, \tag{4}$$

with f the work of the external loads and $(\cdot,\cdot)$ the L^2 inner product; a is the deviatoric energy form, b the work of the pressure on the volume change, and the second equation is the constitutive relation $p = \kappa \operatorname{Div} \boldsymbol{u}$ tested against the pressure space, which becomes the constraint $b(\boldsymbol{u}_h, q) = 0$ as $\kappa \to \infty$. Brezzi's theorem [8], [7, Theorem 5.2.1 and §5.2], gives two conditions under which (3) has a unique solution that converges to the continuum one with constants independent of the mesh size h.

The inf-sup condition. There is a $\beta > 0$, independent of h, with

$$\beta_h := \inf_{q \in Q_h} \sup_{\boldsymbol{v} \in U_h} \frac{\displaystyle\int_\Omega q \operatorname{Div} \boldsymbol{v}\, \mathrm{d}V}{\|q\|_{L^2}\, \|\operatorname{Grad} \boldsymbol{v}\|_{L^2}} \;\geq\; \beta. \tag{5}$$

In words: every discrete pressure does work on the volume change of some discrete displacement, in a ratio bounded below independently of the mesh. The condition is also named after Ladyzhenskaya, Babuška and Brezzi (LBB). Its failure has two forms. If $\beta_h = 0$ there is a pressure mode that no displacement resists: the pressure is not determined, modes of alternating sign between neighboring elements appear, and the system becomes singular as $\kappa/\mu \to \infty$; the pair P_1/P_0 (linear displacement, constant pressure) has 45 such modes on the coarsest mesh of §9.5 and 558 on the finest. If $\beta_h > 0$ on every mesh but $\beta_h \to 0$ under refinement, the pressure is determined but the error grows with $1/\beta_h$: the stresses oscillate and the displacement is too stiff, and locking returns as the mesh is refined.

Coercivity on the discretely isochoric subspace. There is an $\alpha > 0$, independent of h, with

$$a(\boldsymbol{v}, \boldsymbol{v}) \;\geq\; \alpha\, \|\operatorname{Grad} \boldsymbol{v}\|_{L^2}^2 \quad \text{for all } \boldsymbol{v} \in \ker \boldsymbol{G}_h := \{\boldsymbol{v} \in U_h : b(\boldsymbol{v}, q) = 0 \text{ for all } q \in Q_h\}. \tag{6}$$

In words: every nonzero discrete displacement whose volume change the pressure space does not see has a deviatoric energy bounded below. A displacement in $\ker \boldsymbol{G}_h$ with zero deviatoric energy has no stiffness at all; it is a spurious soft mode.

The error estimate. Under both hypotheses [7, Theorem 5.2.2],

$$\|\operatorname{Grad}(\boldsymbol{u} - \boldsymbol{u}_h)\| + \|p - p_h\| \;\leq\; C\Big(1 + \frac{1}{\alpha}\Big)\Big(1 + \frac{1}{\beta_h}\Big)\Big[\inf_{\boldsymbol{v} \in U_h} \|\operatorname{Grad}(\boldsymbol{u} - \boldsymbol{v})\| + \inf_{q \in Q_h} \|p - q\|\Big] : \tag{7}$$

the error is at most a factor times the best approximation the two spaces allow, and the factor is what the two constants control.

Measuring β_h. With $\{\boldsymbol{\phi}_i\}$ a basis of U_h, the functions of nodes on the fixed boundary excluded, and $\{\chi_m\}$ a basis of Q_h, let

$$K_{ij} = \int_\Omega \operatorname{Grad}\boldsymbol{\phi}_i : \operatorname{Grad}\boldsymbol{\phi}_j \,\mathrm{d}V, \qquad G_{mi} = \int_\Omega \chi_m \operatorname{Div}\boldsymbol{\phi}_i \,\mathrm{d}V, \qquad M_{mn} = \int_\Omega \chi_m\chi_n \,\mathrm{d}V, \tag{8}$$

the Gram matrix of the norm $\|\operatorname{Grad}\boldsymbol{v}\|$, the discrete constraint operator and the pressure mass matrix. For a fixed pressure vector q the largest value over $\boldsymbol{v}$ of $(q^{\mathsf{T}}\boldsymbol{G}\boldsymbol{v})^2/\boldsymbol{v}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{v}$ is $q^{\mathsf{T}}\boldsymbol{G}\boldsymbol{K}^{-1}\boldsymbol{G}^{\mathsf{T}}q$, so

$$\beta_h^2 = \lambda_{\min}^{\neq 0}(\boldsymbol{G}\boldsymbol{K}^{-1}\boldsymbol{G}^{\mathsf{T}},\ \boldsymbol{M}), \tag{9}$$

the smallest nonzero generalized eigenvalue [11]. With the whole boundary fixed the constant pressure does no work on any admissible displacement, so one eigenvalue is always zero and is excluded; any further zero is an unresisted pressure mode. §9.1 gives the computation.

2.3 How the enriched pair satisfies the inf-sup condition

A sufficient condition for (5) is the existence of a Fortin operator [15], [7, §5.4]: a linear map Π_h from the continuum displacement space into U_h with

$$b(\boldsymbol{v} - \Pi_h\boldsymbol{v}, q) = 0 \text{ for all } q \in Q_h, \qquad \|\operatorname{Grad}\,\Pi_h\boldsymbol{v}\| \le C_\Pi \,\|\operatorname{Grad}\boldsymbol{v}\|\,, \tag{10}$$

which gives $\beta_h \ge \beta/C_\Pi$ with β the continuum inf-sup constant. The operator must reproduce, for every q in Q_h, the work of q on the volume change of $\boldsymbol{v}$, with a bounded gradient. For a pressure that is linear on each element T, a tetrahedron of the mesh, and discontinuous between elements, integration by parts on one element,

$$\int_T q \operatorname{Div}\boldsymbol{w}\,\mathrm{d}V = -\int_T \operatorname{Grad}q\cdot\boldsymbol{w}\,\mathrm{d}V + \int_{\partial T} q\,\boldsymbol{w}\cdot\boldsymbol{n}\,\mathrm{d}S, \tag{11}$$

shows what has to be matched: the flux $\boldsymbol{w}\cdot\boldsymbol{n}$ weighted by q on each face, and the interior integral of $\boldsymbol{w}$ against the constant $\operatorname{Grad}q$. Two kinds of bubble supply the two parts. A face bubble b_F, a cubic polynomial that vanishes on the three other faces of the element, times the face normal $\boldsymbol{n}_F$, changes the boundary term on face F alone and controls the mean of q on that face, and with it the jump of the pressure between the two elements sharing F; four face bubbles per element control the element mean of q and its values on the faces. The interior bubble b_T, a quartic polynomial that vanishes on the whole element boundary, times $\operatorname{Grad}q$, changes the interior term alone: $\int_T q\operatorname{Div}(b_T\operatorname{Grad}q)\,\mathrm{d}V = -\int_T b_T|\operatorname{Grad}q|^2\,\mathrm{d}V < 0$ for $\operatorname{Grad}q \neq 0$, so it controls the gradient of the pressure inside each element. Face bubbles alone leave the linear part of the pressure uncontrolled, and the interior bubble alone leaves the constant part uncontrolled: it vanishes on the boundary, so its volume change has zero mean on the element and does no work on a constant pressure (Remark 7). The displacement space with both enrichments over the linear discontinuous pressure is the three-dimensional Crouzeix–Raviart pair [13], [7, Example 8.7.2], whose stability in three dimensions, for the whole family of Crouzeix–Raviart elements, is also proved by Sauter and Torres [25], and the construction above is the proof of its stability on any shape-regular family of meshes. §9.5 measures its constant and confirms that neither enrichment alone suffices.

3 Continuum formulation

This section develops the formulation for a material given only through its stored energy, its stress and its tangent, and then specializes it twice: to a material whose stored energy splits exactly into a volumetric and an isochoric part, and, within those, to a material whose volumetric energy is quadratic in the volumetric strain. Each specialization adds a property of the material and removes work from the element; the general form contains both.

3.1 Kinematics and the volumetric strain

A body $\Omega \subset \mathbb{R}^3$ undergoes a motion $\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X}, t)$ with deformation gradient $\boldsymbol{F} := \operatorname{Grad} \boldsymbol{\varphi}$ and Jacobian $J := \det \boldsymbol{F} > 0$. The multiplicative factorization

$$\boldsymbol{F} = J^{1/3} \bar{\boldsymbol{F}}, \qquad \bar{\boldsymbol{F}} := J^{-1/3} \boldsymbol{F}, \qquad \det \bar{\boldsymbol{F}} = 1, \tag{12}$$

separates the two kinematic contributions this formulation treats differently: the mixed treatment acts on J alone, and the plasticity on $\bar{\boldsymbol{F}}$ alone.

The mixed field is a scalar function of J alone. Let

$$\theta = \theta(J), \qquad \theta'(J) > 0, \qquad \theta(1) = 0, \qquad \theta'(1) = 1, \tag{13}$$

be a strictly increasing volumetric strain measure, normalized to vanish with unit slope in the undeformed state. The normalization has two consequences. It makes the bulk modulus κ of a quadratic volumetric energy in θ (§3.7) the physical bulk modulus for every admissible θ, so that two instances of the formulation are calibrated to the same material rather than to two different ones. And since $\partial J / \partial \boldsymbol{F} = J \boldsymbol{F}^{-\mathsf{T}}$,

$$\frac{\partial \theta}{\partial \boldsymbol{F}} = \theta'(J)\, J\, \boldsymbol{F}^{-\mathsf{T}}, \tag{14}$$

so (13) also forces $\partial \theta / \partial \boldsymbol{F} = \boldsymbol{I}$ at $\boldsymbol{F} = \boldsymbol{I}$ for every admissible θ (§4.1).

Two choices are discussed below: the logarithmic measure $\theta = \log J$, the volumetric part of the logarithmic strain, and $\theta = J - 1$. Both satisfy (13), and by Remark 1 they nevertheless give different elements. The element implemented projects $\theta = \log J$.

Remark 1 (The volumetric variable is a modeling choice)**.** Projecting $\log J$ and projecting $J - 1$ do not give the same element. Measured on one tetrahedron under a displacement quadratic in $\boldsymbol{X}$, with the element mean as the projection and a Hencky material so that both variants use the same exact W^{vol}, the element energies differ by 9.4×10^{-4} at 0.5% nominal strain, 1.0×10^{-2} at 5%, 5.6×10^{-2} at 20% and 2.7×10^{-1} at 50% (`prototype/materials.jl`). The difference is $O(\text{strain})$ and vanishes with it, so both are consistent discretizations of the same continuum problem; they are different elements, and they differ by 5.6% to 27% at 20% to 50% strain, the finite-deformation regime of §1. Which variable is projected is therefore a modeling decision.

3.2 Three-field functional for a general material

The material is given as every finite-element code obtains it from a material model: the stored energy $W(\boldsymbol{F})$, the first Piola–Kirchhoff stress $\boldsymbol{P}(\boldsymbol{F}) = \partial W / \partial \boldsymbol{F}$ and the tangent $\mathbb{A}(\boldsymbol{F}) =$

$\partial^2 W/\partial \boldsymbol{F}\partial \boldsymbol{F}$, with the internal variables the model carries and updates. Only the volumetric response is treated mixedly. Introduce an independent volumetric strain $\bar{\theta}$ and a deformation gradient whose isochoric part is the pointwise one and whose volume is that of $\bar{\theta}$,

$$\tilde{J} := \theta^{-1}(\bar{\theta}), \qquad \tilde{\boldsymbol{F}} := (\tilde{J}/J)^{1/3}\,\boldsymbol{F}, \qquad \det \tilde{\boldsymbol{F}} = \tilde{J}, \qquad \tilde{J}^{-1/3}\tilde{\boldsymbol{F}} = \bar{\boldsymbol{F}}, \tag{15}$$

with θ^{-1} the inverse of the strictly increasing $\theta(J)$, and constrain $\bar{\theta}$ to $\theta(J)$ through a multiplier $\bar{p}$:

$$\Pi[\boldsymbol{\varphi}, \bar{\theta}, \bar{p}] := \int_\Omega W(\tilde{\boldsymbol{F}}(\boldsymbol{\varphi}, \bar{\theta}))\,\mathrm{d}V + \int_\Omega \bar{p}\left(\theta(\boldsymbol{\varphi}) - \bar{\theta}\right)\,\mathrm{d}V - \int_\Omega \rho_0\,\boldsymbol{b}\cdot\boldsymbol{\varphi}\,\mathrm{d}V - \int_{\partial_T\Omega} \boldsymbol{T}\cdot\boldsymbol{\varphi}\,\mathrm{d}S, \tag{16}$$

with ρ_0 the reference density, $\boldsymbol{b}$ the body force per unit mass, and $\boldsymbol{T}$ the traction prescribed on the part $\partial_T\Omega$ of the boundary. The material is evaluated at the pointwise shape change $\bar{\boldsymbol{F}}(\boldsymbol{\varphi})$ and at the volume change of $\bar{\theta}$; no other quantity carries an auxiliary field, and Remark 3 gives the reason. At stationarity $\bar{p}$ is the pressure: with $\tilde{\boldsymbol{P}} := \boldsymbol{P}(\tilde{\boldsymbol{F}})$ and the mean Cauchy stress at $\tilde{\boldsymbol{F}}$,

$$\tilde{p} := \frac{1}{3\tilde{J}}\,\tilde{\boldsymbol{P}} : \tilde{\boldsymbol{F}} = \tfrac{1}{3}\operatorname{tr}\tilde{\boldsymbol{\sigma}}, \tag{17}$$

positive in expansion, the derivative of the energy density with respect to $\bar{\theta}$ is $\partial W(\tilde{\boldsymbol{F}})/\partial\bar{\theta} = \tilde{\boldsymbol{P}} : \partial\tilde{\boldsymbol{F}}/\partial\bar{\theta} = \tilde{p}/\theta'(\tilde{J})$, which stationarity in $\bar{\theta}$ equates to $\bar{p}$.

3.3 Reduction to a functional of the motion

Put $\bar{\theta}$ and $\bar{p}$ in the same discrete space Q_h, as §4.1 requires. Stationarity in $\bar{p}$ gives $\bar{\theta} = \mathcal{P}_h\,\theta$, the L^2 projection of θ onto Q_h, and stationarity in $\bar{\theta}$ gives

$$\bar{p} = \mathcal{P}_h\left(\frac{\tilde{p}}{\theta'(\tilde{J})}\right), \tag{18}$$

the projection of a quantity the material returns at the already computed $\bar{\theta}$. The constraint term of (16) then vanishes identically,

$$\int_\Omega \bar{p}\left(\theta - \bar{\theta}\right)\,\mathrm{d}V = \int_\Omega \bar{p}\,(\theta - \mathcal{P}_h\theta)\,\mathrm{d}V = 0, \tag{19}$$

because $\bar{p} \in Q_h$ while $(\theta - \mathcal{P}_h\theta) \perp Q_h$ by the definition of the projection. What is left is a functional in the motion alone,

$$\Pi[\boldsymbol{\varphi}] = \int_\Omega W(\tilde{\boldsymbol{F}}(\boldsymbol{\varphi}))\,\mathrm{d}V - \Pi_{\mathrm{ext}}, \qquad \tilde{J} = \theta^{-1}(\mathcal{P}_h\theta(J)), \tag{20}$$

with Π_{ext} the work of the external forces. For $\theta = J - 1$, $\tilde{J} = 1 + \mathcal{P}_h(J-1) = \mathcal{P}_h J$, and for $\theta = \log J$, $\tilde{J} = \exp(\mathcal{P}_h \log J)$. In the notation of the element,

$$\tilde{J} := \theta^{-1}(\mathcal{P}_h\theta(J)), \qquad s := (\tilde{J}/J)^{1/3}, \qquad \tilde{\boldsymbol{F}} := s\,\boldsymbol{F}. \tag{21}$$

Equation (20) is the mean-dilatation formulation of Simo et al. [30], in which the volumetric strain is replaced by its projection onto an element-wise space, with the projection $\mathcal{P}_h$ in place of the element mean; with the constant projection and $\theta = J - 1$ it is that construction, in

which the $\bar{F}$ element of de Souza Neto et al. [14] replaces the element mean of J by its value at the centroid, and it is the mechanism of the $\bar{B}$ family (projected strain–displacement operator) and the $\bar{F}$ family (projected deformation gradient) of elements; Boffi et al. [7, §8.12] set out the relation between reduced integration, projected pressure, and the mixed method it is equivalent to. The reduction is exact by (19), and no property of the material or of θ enters it: the exactness is a property of the discretization, that the two auxiliary fields share one space. The quantity the formulation leaves open is Q_h, which §4.1 selects and §9 measures.

3.4 Requirements on the material

The general form requires of the material its energy, stress and tangent and nothing else, and the choice of $\theta(J)$ is free within (13). The return map runs at $\tilde{\boldsymbol{F}}_q$, and the internal variables are updated there, so strain and stress are evaluated at the same point and the same deformation, as §5 requires. For a material whose isochoric response depends on J, that response is evaluated at $\tilde{J}$ instead of J; the difference is of the order of $\theta - \mathcal{P}_h\theta$ and vanishes under refinement. The inf-sup constant of the pair depends only on the spaces and is unchanged. Coercivity of the linearized operator requires the material's volumetric tangent, $\partial\tilde{p}/\partial\tilde{J}$ at fixed shape, to be positive over the states reached. The logarithmic measure itself is required by nothing in the present section; it is required by §5.

3.5 Residual, passes and tangent

The residual. With $\tilde{\boldsymbol{P}}$ the stress the material returns at $\tilde{\boldsymbol{F}}$ of (21), its internal variables updated there, and $\tilde{p}$ of (17), the variation of (21) is $\delta\tilde{\boldsymbol{F}} = s\,\delta\boldsymbol{F} + \delta s\,\boldsymbol{F}$ with $\delta s = \frac{s}{3}\,(\delta\tilde{J}/\tilde{J} - \delta J/J)$, $\delta J = J\,\boldsymbol{F}^{-\mathsf{T}} : \delta\boldsymbol{F}$ and $\delta\tilde{J} = \delta\bar{\theta}/\theta'(\tilde{J})$, $\delta\bar{\theta} = \mathcal{P}_h(\theta'(J)\,\delta J)$. Substituting into $\delta\Pi = \int \tilde{\boldsymbol{P}} : \delta\tilde{\boldsymbol{F}}\,\mathrm{d}V$, using $\tilde{\boldsymbol{P}} : \tilde{\boldsymbol{F}} = 3\tilde{J}\tilde{p}$ and moving the projection onto the other factor by self-adjointness gives the stress that multiplies $\operatorname{Grad}\boldsymbol{\psi}$ at a quadrature point,

$$\boldsymbol{P}_q = s_q\,\tilde{\boldsymbol{P}}_q - \tilde{p}_q\,\tilde{J}_q\,\boldsymbol{F}_q^{-\mathsf{T}} + \bar{p}_q\,\theta'(J_q)\,J_q\,\boldsymbol{F}_q^{-\mathsf{T}}, \qquad \bar{p} := \mathcal{P}_h\!\left(\frac{\tilde{p}}{\theta'(\tilde{J})}\right), \tag{22}$$

and $\boldsymbol{R}_a = \sum_q w_q j_q\,\boldsymbol{P}_q \operatorname{Grad} N_a(\boldsymbol{\xi}_q)$ as in (44). The first term is the material's stress scaled to the pointwise volume, the second removes its mean part, and the third puts back the projected mean stress with the pointwise volumetric kinematics. For a material with the split, $\tilde{\boldsymbol{P}} = \boldsymbol{P}^{\mathrm{iso}}(\bar{\boldsymbol{F}})\,s^{-1} + W^{\mathrm{vol}\prime}(\bar{\theta})\,\theta'(\tilde{J})\,\tilde{J}\,\tilde{\boldsymbol{F}}^{-\mathsf{T}}$, the isochoric stress does no work on $\tilde{\boldsymbol{F}}$, and (22) reduces to (44) with $\bar{p} = \mathcal{P}_h W^{\mathrm{vol}\prime}(\bar{\theta})$; with a quadratic W^{vol} the projection is the identity and $\bar{p} = \kappa\bar{\theta}$. For $\theta = J - 1$ and the constant projection, $\bar{p} = \tilde{p}$ on the element and (22) is $\boldsymbol{P}_q = s_q\tilde{\boldsymbol{P}}_q + \tilde{p}\,(J_q - \tilde{J})\,\boldsymbol{F}_q^{-\mathsf{T}}$.

The passes. The element computes this residual in three passes over its quadrature points (§4.4): the projection of θ, the material at $\tilde{\boldsymbol{F}}_q$ with the projection of $\tilde{p}_q/\theta'(\tilde{J}_q)$, and the stress (22).

The tangent. The tangent is the second derivative of (20). Along two increments with gradients $\delta_1\boldsymbol{F}$ and $\delta_2\boldsymbol{F}$ at the quadrature points,

$$D^2\Pi[\delta_1\boldsymbol{\varphi}, \delta_2\boldsymbol{\varphi}] = \sum_q w_q j_q \Big[\delta_1\tilde{\boldsymbol{F}}_q : \mathbb{A}_q : \delta_2\tilde{\boldsymbol{F}}_q + \tilde{\boldsymbol{P}}_q : (\delta_1 s_q\,\delta_2\boldsymbol{F}_q + \delta_2 s_q\,\delta_1\boldsymbol{F}_q + \delta_1\delta_2 s_q\,\boldsymbol{F}_q)\Big], \tag{23}$$

with $\mathbb{A}_q$ the material tangent at $\tilde{\boldsymbol{F}}_q$, the variations $\delta\tilde{\boldsymbol{F}}$ and δs as above, and

$$\begin{aligned}
\delta_1\delta_2 s &= \frac{\delta_1 s}{3}\Big(\frac{\delta_2\tilde{J}}{\tilde{J}} - \frac{\delta_2 J}{J}\Big) + \frac{s}{3}\Big(\frac{\delta_1\delta_2\tilde{J}}{\tilde{J}} - \frac{\delta_1\tilde{J}\,\delta_2\tilde{J}}{\tilde{J}^2} - \frac{\delta_1\delta_2 J}{J} + \frac{\delta_1 J\,\delta_2 J}{J^2}\Big),\\
\delta_1\delta_2 J &= J\,(\boldsymbol{F}^{-\mathsf{T}} : \delta_1\boldsymbol{F})(\boldsymbol{F}^{-\mathsf{T}} : \delta_2\boldsymbol{F}) - J\ \mathrm{tr}\,(\boldsymbol{F}^{-1}\delta_1\boldsymbol{F}\,\boldsymbol{F}^{-1}\delta_2\boldsymbol{F}),\\
\delta_1\delta_2\tilde{J} &= \frac{\mathcal{P}_h\big(\theta''(J)\,\delta_1 J\,\delta_2 J + \theta'(J)\,\delta_1\delta_2 J\big)}{\theta'(\tilde{J})} - \frac{\theta''(\tilde{J})}{\theta'(\tilde{J})}\,\delta_1\tilde{J}\,\delta_2\tilde{J}.
\end{aligned} \tag{24}$$

The projection in $\delta\tilde{J}$ couples the quadrature points of the element, as $\delta\bar{\theta}$ does in (45): in matrix form the tangent is the sum of the quadrature-point blocks and of terms of the form $\boldsymbol{D}\boldsymbol{M}^{-1}(\cdot)\boldsymbol{M}^{-1}\boldsymbol{D}^{\mathsf{T}}$ and $\boldsymbol{D}\boldsymbol{M}^{-1}(\cdot)$ with the matrix $\boldsymbol{D}$ of §4.4. The tangent is symmetric, and an implementation is verified by the difference checks of §4.5: the residual against the energy (20) and the tangent against the residual, which detect any omitted term.

What the general element changes. The requirements of §3.4 disappear: the material supplies W, $\boldsymbol{P}$ and $\mathbb{A}$ and nothing else, and the choice of $\theta(J)$ is free within (13). The return map runs at $\tilde{\boldsymbol{F}}_q$, and the internal variables are updated there, so strain and stress are evaluated at the same point and the same deformation, as §5 requires. For a material whose isochoric response depends on J, a material without the split, that response is evaluated at $\tilde{J}$ instead of J; the difference is of the order of $\theta - \mathcal{P}_h\theta$ and vanishes under refinement. The pressure entering the residual is the projection of the material's own mean stress at $\tilde{\boldsymbol{F}}$, with the pointwise relation between pressure and projected strain recovered only when W^{vol} is quadratic in θ. The inf-sup constant of the pair is unchanged. Coercivity of the linearized operator requires the material's volumetric tangent, $\partial\tilde{p}/\partial\tilde{J}$ at fixed shape, to be positive over the states reached. The element needs the element-level assembly of §4.4 and one pass more, and its cost per element is one material call per quadrature point plus the two 4×4 solves.

3.6 Materials with a volumetric–isochoric split

The first specialization is to a stored energy that splits exactly:

$$W(\boldsymbol{F}) = W^{\mathrm{vol}}(J) + W^{\mathrm{iso}}(\bar{\boldsymbol{F}}), \qquad \bar{\boldsymbol{F}} := J^{-1/3}\boldsymbol{F}, \qquad \det\bar{\boldsymbol{F}} = 1. \tag{25}$$

Whether a material satisfies (25) is decidable without knowing W^{vol}: if the split exists then $W(\boldsymbol{F}) - W(\bar{\boldsymbol{F}}) = W^{\mathrm{vol}}(J) - W^{\mathrm{vol}}(1)$ depends on J alone, so holding J fixed while varying the isochoric part must leave the difference unmoved. Column 2 of Table 1 is that test, run over seven material models: linear elasticity, Saint-Venant–Kirchhoff, neo-Hookean, reciprocal neo-Hookean, Seth–Hill, Hencky and the J_2 plasticity model of Simo and Hughes [29], as implemented in the finite-element code Norma. Five of the seven satisfy it.

For such a material $W(\tilde{\boldsymbol{F}}) = W^{\mathrm{vol}}(\theta(\tilde{J})) + W^{\mathrm{iso}}(\bar{\boldsymbol{F}}) = W^{\mathrm{vol}}(\mathcal{P}_h\theta) + W^{\mathrm{iso}}(\bar{\boldsymbol{F}})$, with W^{vol} written as a function of θ, and (20) becomes

$$\Pi[\boldsymbol{\varphi}] = \int_\Omega W^{\mathrm{iso}}\Big(\bar{\boldsymbol{F}}(\boldsymbol{\varphi})\Big)\,\mathrm{d}V + \int_\Omega W^{\mathrm{vol}}(\mathcal{P}_h\theta)\,\mathrm{d}V - \Pi_{\mathrm{ext}}. \tag{26}$$

The isochoric energy is evaluated at the pointwise $\bar{\boldsymbol{F}}$ and is never projected; plasticity acts entirely inside W^{iso}, through an isochoric internal variable evolved by a return map. The pressure (18) becomes

$$\bar{p} = \mathcal{P}_h\,W^{\mathrm{vol}\prime}(\bar{\theta}), \tag{27}$$

Table 1: What seven material models satisfy. Column 2 is the spread of $W(\boldsymbol{F}) - W(\bar{\boldsymbol{F}})$ across twenty random isochoric parts at each of four fixed $J \in [0.85, 1.2]$, relative to its own magnitude; a spread below 10^{-12} means the split (25) holds to rounding error. Columns 3 and 4 are the relative residual of a least-squares fit of the extracted $W^{\mathrm{vol}}(J) = W(J^{1/3}\boldsymbol{I}) - W(\boldsymbol{I})$ over $J \in [0.8, 1.25]$ to a quadratic in each candidate volumetric variable; in both exact cases the fitted coefficient is $\kappa/2$ to five digits. Two materials admit a quadratic W^{vol}, and they disagree on the variable. Script: `prototype/materials.jl`, which needs Norma.

	split (25)	W^{vol} quadratic in	
material	spread at fixed J	$J - 1$	$\log J$
Linear elastic	2.0×10^{0}	—	—
Saint-Venant Kirchhoff	3.0×10^{1}	—	—
Neo-Hookean	4.2×10^{-13}	5.9×10^{-2}	1.3×10^{-1}
Reciprocal neo-Hookean	4.8×10^{-13}	1.8×10^{-1}	6.5×10^{-3}
Seth–Hill ($m = n = 1$)	4.3×10^{-13}	1.8×10^{-1}	6.5×10^{-3}
Hencky	1.4×10^{-13}	1.8×10^{-1}	$\mathbf{1.7 \times 10^{-15}}$
J_2, Simo–Hughes	5.9×10^{-13}	$\mathbf{5.3 \times 10^{-15}}$	1.8×10^{-1}

the projection of a known function of $\bar{\theta}$, since the isochoric stress does no work on $\tilde{\boldsymbol{F}}$; the derivative of (26) reproduces it, because $\mathcal{P}_h$ is self-adjoint in L^2: $\int W^{\mathrm{vol}\prime}(\bar{\theta})\,\mathcal{P}_h(\delta\theta)\,\mathrm{d}V = \int \mathcal{P}_h W^{\mathrm{vol}\prime}(\bar{\theta})\,\delta\theta\,\mathrm{d}V$. The stress at a quadrature point is (44) with this $\bar{p}$, the material is called at the pointwise deformation for its isochoric part, and the third pass of §3.5 is unnecessary, since $\bar{p}$ follows from $\bar{\theta}$ alone. Four things follow.

1. With the constant projection nothing changes: $\bar{\theta}$ is constant on the element, so $W^{\mathrm{vol}\prime}(\bar{\theta})$ is constant and the projection in (27) is the identity. This is the mean-dilatation element of Simo et al. [30], which was formulated for a general W^{vol}.

2. With the linear projection the pressure is projected twice, and $\bar{p}(\boldsymbol{\xi}_q) \neq W^{\mathrm{vol}\prime}(\bar{\theta}(\boldsymbol{\xi}_q))$: the pressure at a quadrature point is not the constitutive response of the projected strain at that point. Using $W^{\mathrm{vol}\prime}(\bar{\theta}(\boldsymbol{\xi}_q))$ there instead is not the derivative of any functional, gives an unsymmetric tangent and fails the consistency checks of §4.5; it is not admissible.

3. The tangent replaces the constant matrix $\kappa \boldsymbol{D}\boldsymbol{M}^{-1}\boldsymbol{D}^{\mathsf{T}}$ of §4.4 by $\boldsymbol{D}\boldsymbol{M}^{-1}\tilde{\boldsymbol{U}}\boldsymbol{M}^{-1}\boldsymbol{D}^{\mathsf{T}}$ with $\tilde{U}_{mn} = \sum_q w_q j_q\, W^{\mathrm{vol}\prime\prime}(\bar{\theta}_q)\,\chi_m(\boldsymbol{\xi}_q)\chi_n(\boldsymbol{\xi}_q)$, a weighted mass matrix assembled at every iteration, and the volumetric stress $\bar{p}_q\,\partial \boldsymbol{g}/\partial \boldsymbol{F}$ of (45) uses the projected $\bar{p}_q$; the tangent stays symmetric. In the linearized problem of §2.2 the compressibility term $-(1/\kappa)(p, q)$ becomes a term weighted by $1/W^{\mathrm{vol}\prime\prime}(\bar{\theta})$, which varies inside the element and with the deformation; the inf-sup constant β_h is unchanged, since it depends only on the spaces and on the constraint operator, and the constants of Brezzi's theorem then involve the smallest and largest values of $W^{\mathrm{vol}\prime\prime}$ over the body, which must be positive: W^{vol} must be convex in θ over the range of $\bar{\theta}$ reached.

4. The second projection can be avoided by a change of variable. Any $W^{\mathrm{vol}}(J)$ with a single minimum at $J = 1$ and $W^{\mathrm{vol}\prime\prime}(1) = \kappa > 0$ is quadratic in

$$\theta(J) := \operatorname{sign}(J - 1)\,\sqrt{2W^{\mathrm{vol}}(J)/\kappa}, \tag{28}$$

which satisfies (13) with $\theta = J - 1 + O((J-1)^2)$ and is convex-admissible wherever W^{vol} is. The element of §4.4 then applies unchanged with this θ. For the volumetric law $W^{\mathrm{vol}} = \frac{\kappa}{4}(J^2 - 1 - 2\log J)$ of Albany-LCM, for example, $\theta^2 = (J^2 - 1 - 2\log J)/2$. What distinguishes this choice from a non-quadratic W^{vol} in the variable $J-1$ is the quantity averaged over the element, $\mathcal{P}_h\theta$ against $\theta(\mathcal{P}_h(J-1))$; by Remark 1 the two are different elements that converge to the same solution.

3.7 Quadratic volumetric energy

The second specialization is to a split material whose volumetric part is quadratic in θ:

$$W(\boldsymbol{F}) = W^{\mathrm{vol}}(\theta) + W^{\mathrm{iso}}(\bar{\boldsymbol{F}}), \qquad W^{\mathrm{vol}}(\theta) = \tfrac{\kappa}{2}\theta^2. \tag{29}$$

The volumetric constitutive relation is then linear,

$$p = \partial W^{\mathrm{vol}}/\partial\theta = \kappa\theta, \tag{30}$$

$\kappa\bar{\theta}$ already lies in Q_h, the projection in (27) is the identity, and

$$\bar{p} = \kappa\bar{\theta}, \qquad \Pi[\boldsymbol{\varphi}] = \int_\Omega W^{\mathrm{iso}}\Big(\bar{\boldsymbol{F}}(\boldsymbol{\varphi})\Big)\,\mathrm{d}V + \int_\Omega \tfrac{\kappa}{2}\,(\mathcal{P}_h\theta)^2\,\mathrm{d}V - \Pi_{\mathrm{ext}}\,: \tag{31}$$

the discrete pressure equals $\kappa\bar{\theta}$ at every point, and it is this linearity — not the choice of θ — that gives the pointwise relation of §4.2 and one projection fewer. With $\bar{\theta}$ and $\bar{p}$ in Q_h the functional (31) is exactly the three-field functional with $W^{\mathrm{vol}}(\bar{\theta})$ evaluated at the independent strain, whose stationarity conditions are those of §3.9.

The case developed in detail. Table 1 admits two materials with a quadratic W^{vol}, and they do not agree on the variable: Hencky is quadratic in $\log J$ to 1.7×10^{-15}, and the Simo–Hughes J_2 model as implemented in Norma [22] is quadratic in $J-1$ to 5.3×10^{-15}. Each is exact in its own variable and wrong by 18% in the other. The implemented J_2 model replaces its volumetric energy by $\frac{\kappa}{2}(\log J)^2$ (Remark 5), so that both materials are quadratic in the projected $\log J$. The three remaining split materials — neo-Hookean, its reciprocal, and Seth–Hill — use §3.6 with one projection more, and any other material uses §3.5.

3.8 The two admissible instances

Table 1 leaves two materials. They agree on everything §3.1–§3.5 requires; Table 2 lists where they differ.

The Hencky instance is the one (32) defines: with

$$\boldsymbol{\epsilon} := \tfrac{1}{2}\log\boldsymbol{C} \in \mathbb{S}, \qquad \boldsymbol{C} := \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}, \tag{32}$$

evaluated by inverse scaling and squaring with Padé approximants — repeated square roots of $\boldsymbol{C}$ followed by a rational approximant of the logarithm [3] — the volumetric–deviatoric decomposition

$$\boldsymbol{\epsilon} = \tfrac{1}{3}\theta\boldsymbol{I} + \boldsymbol{e}, \qquad \theta = \operatorname{tr}\boldsymbol{\epsilon} = \log J, \qquad \boldsymbol{e} = \operatorname{dev}\boldsymbol{\epsilon}, \tag{33}$$

is orthogonal in the Frobenius inner product, and the split of (29) is inherited from the strain measure rather than imposed on it. The Simo–Hughes instance imposes it instead, through the $J^{-2/3}$ factor in $\bar{\boldsymbol{b}}^{\mathrm{e}}$; $\det\bar{\boldsymbol{b}}^{\mathrm{e}} = 1$ then makes W^{iso} independent of J by construction. Both are exact, and Table 1 confirms both to machine precision. Three consequences follow.

Table 2: The two instances of (29). Everything above the rule is fixed by the generic development; everything below it is where the instances differ. $\boldsymbol{b}^{\mathrm{e}} := \boldsymbol{F}^{\mathrm{e}}(\boldsymbol{F}^{\mathrm{e}})^{\mathsf{T}}$ is the elastic left Cauchy–Green tensor, with $\boldsymbol{F}^{\mathrm{e}}$ the elastic part of the multiplicative decomposition $\boldsymbol{F} = \boldsymbol{F}^{\mathrm{e}}\boldsymbol{F}^{\mathrm{p}}$, and $\bar{\boldsymbol{b}}^{\mathrm{e}} := J^{-2/3}\boldsymbol{b}^{\mathrm{e}}$ its isochoric part.

	Hencky	Simo–Hughes (original)	Simo–Hughes (implemented)
volumetric variable θ	$\log J$	$J-1$	$\log J$
$\theta'(J)$	$1/J$	1	$1/J$
$\partial\theta/\partial\boldsymbol{F}$	$\boldsymbol{F}^{-\mathsf{T}}$	$J\,\boldsymbol{F}^{-\mathsf{T}}$	$\boldsymbol{F}^{-\mathsf{T}}$
$W^{\mathrm{vol}}(\theta)$	$\frac{\kappa}{2}(\log J)^2$	$\frac{\kappa}{2}(J-1)^2$	$\frac{\kappa}{2}(\log J)^2$
isochoric argument	$\boldsymbol{e} = \operatorname{dev}\boldsymbol{\epsilon}$	$\bar{\boldsymbol{b}}^{\mathrm{e}}$	
W^{iso}	$\mu\,\lVert\boldsymbol{e}\rVert^2$	$\frac{\mu}{2}(\operatorname{tr}\bar{\boldsymbol{b}}^{\mathrm{e}} - 3)$	
return map	radial return in $\boldsymbol{\epsilon}$	radial return on $\bar{\boldsymbol{b}}^{\mathrm{e}}$	
small-strain models usable	yes	no	
implemented in Carina	no	no	yes, consistent tangent

The generic development covers both. Nothing in §3.1–§3.5, and nothing in the discretization of §4.1–§4.2, refers to either column of Table 2. The reduction (31), the pointwise relation $\bar{p} = \kappa\bar{\theta}$, and the element-local elimination hold for any θ satisfying (13) and any W^{iso} satisfying (29).

The pairing at $\boldsymbol{F} = \boldsymbol{I}$ is instance-independent. By (14), $\partial\theta/\partial\boldsymbol{F} = \boldsymbol{I}$ at $\boldsymbol{F} = \boldsymbol{I}$ in both columns; §4.1 draws the consequence for the stability results.

The instances differ in plasticity only. The last two rows of Table 2 are the entire difference, and they follow from (33), as §5 develops. Remark 5 states what the implemented instance, Simo–Hughes, entails.

3.9 Variations

With $\boldsymbol{\varphi} \in \mathcal{U}$, the space of motions of finite deformation energy that take the prescribed values on $\partial_{\boldsymbol{\varphi}}\Omega$, the part of the boundary on which the motion is prescribed; $\bar{\theta}, \bar{p} \in \mathcal{Q}$, the space of square-integrable scalar fields; test functions $\boldsymbol{\psi} \in \mathcal{U}$ vanishing on $\partial_{\boldsymbol{\varphi}}\Omega$; and $\vartheta, \pi \in \mathcal{Q}$:

$$
\begin{aligned}
D_{\boldsymbol{\varphi}}\Pi(\boldsymbol{\psi}) &= \int_\Omega \bar{p}\,\frac{\partial\theta}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}]\,\mathrm{d}V + \int_\Omega \boldsymbol{s}(\boldsymbol{e}(\boldsymbol{\varphi})) : \frac{\partial\boldsymbol{e}}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}]\,\mathrm{d}V - \int_\Omega \rho_0\,\boldsymbol{b}\cdot\boldsymbol{\psi}\,\mathrm{d}V - \int_{\partial_T\Omega} \boldsymbol{T}\cdot\boldsymbol{\psi}\,\mathrm{d}S = 0,\\
D_{\bar{\theta}}\Pi(\vartheta) &= \int_\Omega \Big(\kappa\bar{\theta} - \bar{p}\Big)\,\vartheta\,\mathrm{d}V = 0,\\
D_{\bar{p}}\Pi(\pi) &= \int_\Omega \Big(\theta(\boldsymbol{\varphi}) - \bar{\theta}\Big)\,\pi\,\mathrm{d}V = 0.
\end{aligned}
\tag{34}
$$

Here $\boldsymbol{s}$ is the deviatoric stress conjugate to $\boldsymbol{e}$, given by the material law (§5). The kinematic derivatives are

$$
\frac{\partial\theta}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}] = \theta'(J)\,J\,\boldsymbol{F}^{-\mathsf{T}} : \operatorname{Grad}\boldsymbol{\psi}, \qquad \frac{\partial\boldsymbol{e}}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbb{L}^{\mathrm{dev}}(\boldsymbol{C}) : \operatorname{sym}\Big(\boldsymbol{F}^{\mathsf{T}}\operatorname{Grad}\boldsymbol{\psi}\Big), \tag{35}
$$

with $\mathbb{L}(\boldsymbol{C}) := \partial \log \boldsymbol{C}/\partial \boldsymbol{C}$ the Fréchet derivative of the matrix logarithm [3] and $\mathbb{L}^{\mathrm{dev}} := \mathbb{L} - \frac{1}{3}\boldsymbol{I}\otimes\boldsymbol{C}^{-1}$. The volumetric derivative follows from Jacobi's formula and (14). The deviatoric expression is written for the Hencky instance of Table 2; the Simo–Hughes instance replaces it with the variation of $\bar{\boldsymbol{b}}^{\mathrm{e}}$, and nothing outside this equation changes.

3.10 Stress reconstruction

Collecting $(34)_1$ into the form $\int \boldsymbol{P} : \operatorname{Grad}\boldsymbol{\psi}\,\mathrm{d}V$, with $\boldsymbol{P}$ the first Piola–Kirchhoff stress, the stress conjugate to $\boldsymbol{F}$, gives in general

$$\boldsymbol{P} = \theta'(J)\, J\, \bar{p}\, \boldsymbol{F}^{-\mathsf{T}} + \boldsymbol{F}\,(\mathbb{L}(\boldsymbol{C}) : \boldsymbol{s})\,. \tag{36}$$

The deviatoric term is $\mathbb{L}$ and not $\mathbb{L}^{\mathrm{dev}}$ applied to $\boldsymbol{s}$. Moving $\mathbb{L}^{\mathrm{dev}}$ off the strain variation in (35) and onto $\boldsymbol{s}$ takes its transpose, which carries the $\frac{1}{3}$ term onto $\operatorname{tr}\boldsymbol{s} = 0$, and $\mathbb{L}$ is self-adjoint because $\log\boldsymbol{C}$ is the gradient of $\operatorname{tr}(\boldsymbol{C}\log\boldsymbol{C} - \boldsymbol{C})$.

When the deviatoric response is elastic, $\boldsymbol{s} = 2\mu\,\boldsymbol{e}$ is a function of the current $\boldsymbol{C}$ alone through an isotropic law, so $\boldsymbol{s}$ is coaxial with $\boldsymbol{C}$ at every quadrature point; the off-diagonal blocks of $\mathbb{L}$ in the principal basis then act on nothing and (36) collapses to

$$\boldsymbol{P} = (\varpi\boldsymbol{I} + \boldsymbol{s}_x)\,\boldsymbol{F}^{-\mathsf{T}} = \boldsymbol{\tau}\,\boldsymbol{F}^{-\mathsf{T}}, \qquad \boldsymbol{\tau} := \varpi\boldsymbol{I} + \boldsymbol{s}_x = J\,\boldsymbol{\sigma}, \qquad \varpi := \theta'(J)\,J\,\bar{p}, \tag{37}$$

in which $\boldsymbol{\tau} = J\,\boldsymbol{\sigma}$ is the Kirchhoff stress, $\boldsymbol{\sigma}$ the Cauchy stress, and $\boldsymbol{s}_x := \boldsymbol{R}\,\boldsymbol{s}\,\boldsymbol{R}^{\mathsf{T}}$ the spatial deviator, with $\boldsymbol{R}$ the rotation of the polar decomposition $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$. The material $\boldsymbol{s}$ of (33) and the spatial $\boldsymbol{s}_x$ are different tensors; in material form the same statement reads $\boldsymbol{P} = \boldsymbol{F}^{-\mathsf{T}}\boldsymbol{s}$, which agrees with (37) because $\boldsymbol{s}$ and $\boldsymbol{U}$ commute when coaxial.

The Kirchhoff pressure ϖ carries the whole of the instance dependence. For the Hencky instance $\theta'(J)J = 1$ and $\varpi = \bar{p}$; for the Simo–Hughes instance in its original form $\theta'(J)J = J$ and $\varpi = J\bar{p}$, giving $\boldsymbol{\tau} = J\bar{p}\,\boldsymbol{I} + \boldsymbol{s}$ with $\bar{p} = \kappa(J-1)$, the Kirchhoff stress of Simo and Hughes [29, BOX 9.1]; with the logarithmic volumetric energy, as implemented in both codes, $\varpi = \bar{p}$ as for the Hencky instance. Agreement of (36) with both implementations confirms the normalization (13).

Remark 2 (Coaxiality is an elastic property)**.** Equation (37) rests on $\boldsymbol{s}$ being coaxial with $\boldsymbol{C}$, which holds only when $\boldsymbol{s}$ is a function of the current $\boldsymbol{C}$ alone. It fails in two settings, and the second is the one this formulation exists for. With an independent discrete stress field, $\boldsymbol{s}$ is tied to $\boldsymbol{C}$ only weakly and is not coaxial with it at an arbitrary iterate. With plasticity, the return map (47) evaluates $\boldsymbol{s} = 2\mu(\boldsymbol{e} - \boldsymbol{e}^{\mathrm{p}}_{n+1})$ against a stored plastic strain $\boldsymbol{e}^{\mathrm{p}}_n$, which is coaxial with the current $\boldsymbol{C}$ only under proportional loading; whenever the principal axes rotate — the large-rotation regime of §1 — $\boldsymbol{s}$ is not coaxial with $\boldsymbol{C}$, the off-diagonal blocks of $\mathbb{L}$ do not vanish, and (37) is wrong. In both settings (37) holds at best at a converged proportional state, and assembling the residual from it gives a residual that is not the derivative of the energy, with an error that is undetected under proportional loading. The residual is assembled from (36); Miehe et al. [20] carry the full projection tensor for this reason.

Remark 3. No deviatoric mixed pair appears because none is needed. Volumetric locking is a constraint phenomenon: at $\kappa/\mu \gg 1$ the near-isochoric constraint over-determines the displacement space, and relieving it requires a multiplier. The deviatoric response is not constrained in the same sense and admits no analogous locking mode. Adding a deviatoric

pair adds degrees of freedom, an inf-sup condition that has to be established separately, and — if strain and stress are placed in different spaces — a constitutive law evaluated across two discretizations, which is the error that corrupts plastic strain fields when the strain is interpolated in one space and the stress evaluated in another. §8 gives the construction that adds strain as an independent field should a later requirement demand it.

3.11 Implementation in Carina

Both forms are implemented in Carina in one physics type with a form parameter, chosen at setup: the split form when the material exposes the split, the general form otherwise, and the general form on request for any material (input key `volumetric form`). The volumetric variable is a property of the element, not of the material, and is $\theta = \log J$ in both forms; the split form applies to a material whose own volumetric strain is $\log J$, since its pressure $\kappa\bar{\theta}$ is the stationarity condition of the energy $\frac{\kappa}{2}\theta^2$ written in the material's θ, and the general form to any other (Remark 5). The general residual takes the three passes above with the fourteen stresses $\tilde{\boldsymbol{P}}_q$ held on the stack. Calling the material again in the third pass instead of storing them gives the same residual to rounding; on the Taylor bar mesh at $h = 0.75$ mm with the J_2 model it takes 1.30 times as long on a CPU with 12 threads, 1.35 on an AMD RX 7600 and 1.48 on an NVIDIA L4, and 0.87 times on a V100, 0.75 on an A100 and 0.74 on an H100 (`benchmark/tet15-p1/taylor/implicit_timing.jl`). Carina stores them. The element matrix and the diagonal and nodal-block kernels of the preconditioners assemble (23) in closed form. The stress $\boldsymbol{P}_q$ of (22) depends on the deformation gradient at its own point and on the element only through $\bar{\theta}$ and $\bar{p}$, so that its variation is $\delta\boldsymbol{P}_q = \mathbb{C}_q : \delta\boldsymbol{F}_q + \mathbb{E}_q\,\delta\bar{\theta} + \mathbb{H}_q\,\delta\bar{p}$, with $\mathbb{C}_q$ a 9×9 matrix and $\mathbb{E}_q$, $\mathbb{H}_q$ $9 \times n_p$ matrices, n_p the number of projection functions; both $\delta\bar{\theta}$ and $\delta\bar{p}$ are sums over the points of the element. The element matrix is then the sum of the quadrature-point matrices $w_q j_q\, \mathbb{G}_q \mathbb{C}_q \mathbb{G}_q^{\mathsf{T}}$, $\mathbb{G}_q$ the discrete gradient, and of two corrections of rank at most n_p, and it uses the material tangent $\mathbb{A}_q$ at $\tilde{\boldsymbol{F}}_q$ once per quadrature point. It equals the derivative of the element residual by forward-mode differentiation to rounding, which the kernels used before: that derivative took one pass with three directions per node, fifteen passes per element, each evaluating $\mathbb{A}_q$ at every point. The action on a vector remains one forward-mode pass with one direction, in which the material's stress and tangent at $\tilde{\boldsymbol{F}}_q$ supply the value and the derivative of $\tilde{\boldsymbol{P}}_q$, so that a material that writes its internal variables into real-valued storage is never called with derivative-carrying numbers. On the Taylor bar mesh at $h = 0.75$ mm with the J_2 model in plastic flow, the closed form reduced the time of the element matrix from 1078 to 133 ms on a CPU with 12 threads, against 147 ms for the split form, and the time of the diagonal from 1032 to 89 ms against 75 ms; on an AMD RX 7600 the diagonal went from 483 to 54 ms, against 22 ms for the split form, and the action takes 30 ms against 21 ms. Measured on the Taylor bar mesh at $h = 0.75$ mm (§11, 339 993 unknowns) with the J_2 model, the general residual takes 1.36 times the split form's time on a CPU with 12 threads (16.2 against 11.9 ms) and 1.60 times on an AMD RX 7600 (13.6 against 8.5 ms); per explicit step of a run the ratio is 1.36 on a CPU with 16 threads, 1.42 on the RX 7600, 1.71 on an NVIDIA L4, 2.18 on an A100, 2.27 on a V100 and 2.43 on an H100 (Table 21). The split form remains the default for that model.

The diagonal and nodal-block kernels of the preconditioners first held, per element, the fourteen stresses of the second pass and four $45 \times n_p$ coupling matrices, and evaluated the material twice per quadrature point (the stress in the second pass, the stress and the tangent

in the third); on the GPUs they took 3.2 to 5.7 times the split form's diagonal. Two of the four matrices are the same matrix, $\mathbb{D} = \sum_q w_q j_q \theta'(J_q) J_q (\mathbb{G}_q \boldsymbol{F}_q^{-\mathsf{T}}) \chi_q^{\mathsf{T}}$, which depends on the geometry and J only; with $\mathbb{D}M^{-1}$ formed before the material loop, the contribution of each quadrature point is added to the diagonal at that point, and the second pass accumulates $\bar{\boldsymbol{b}}$ without storing its stresses. The material returns its stress and tangent from one call (one return map for J_2), and the point matrix is formed without a $9 \times 9 \times 9$ product. The general diagonal then took 38.9 instead of 53.1 ms on the RX 7600, and on NVIDIA cards 8.9 instead of 25.1 ms (H100), 20.0 instead of 43.2 ms (A100), 34.6 instead of 112.4 ms (V100, the same card for both) and 58.2 instead of 54.5 ms (L4); on a CPU it is unchanged. The register counts of the compiled kernels explain the spread: the NVIDIA compiler receives no register limit, but the kernels call the element loop and the material as separate device functions, and the compiler chose 128 registers per thread for the restructured kernel on every card, against 255 (L4) and 32 (H100) for the former one, whose local frame was 50 kB per thread on the H100. Inlining every call into the kernel (CUDA.jl's `always_inline`) gives the same kernel on the V100, A100, L4 and H100 (128 registers, a local frame of 13.7 kB instead of 22 kB) and shortens the general diagonal by a further 26 to 44%, the general action by 14 to 57%, the split action by 41 to 71% and the split diagonal by 28 to 48% (except on the L4, +4.5%); against the former kernel the general diagonal is then 3.9 (A100), 5.3 (V100), 5.0 (H100) and 1.3 (L4) times faster. Marking only the material functions inline is not safe: on the A100 the compiler then gave the general diagonal 32 registers and it became 44% slower. Since the finite-element library launches its kernels with a backend derived from its arrays, the launcher of Carina makes that backend the inlining one for CUDA arrays (`bin/cuda_always_inline.jl`), so that runs on NVIDIA cards are compiled with `always_inline`; AMDGPU.jl compiles with it by default, which is why inlining the material functions changed nothing on the RX 7600.

4 Discretization

This section selects the pair of discrete spaces, defines the element that carries the displacement, gives the element-local elimination of the auxiliary fields, and states the element algorithm in the form an implementation needs. P_k denotes continuous piecewise polynomials of degree k on the mesh, P_k^{disc} the discontinuous ones, and a bubble a polynomial on one element that vanishes on a prescribed part of its boundary; U_h/Q_h denotes the pair with displacement space U_h and pressure space Q_h.

4.1 The pair

Let $U_h \subset \mathcal{U}$ carry the motion and $Q_h \subset \mathcal{Q}$ carry both $\bar{\theta}$ and $\bar{p}$. Two requirements fix the choice.

Same space for both scalars. $\bar{\theta}$ and $\bar{p}$ are placed in one space Q_h. This is what makes the constraint term vanish identically and the functional reduce to (20) for any material; it is the only thing that reduction needs. Given it, $\bar{p}$ is the projection (18) of the material's mean stress, and for a split material with a quadratic W^{vol} (§3.7) $\bar{p} = \kappa\bar{\theta}$ holds pointwise rather than in a projected sense. Splitting the two scalars across different spaces makes the constitutive relation hold only after a projection, determines one field only through its projection onto the other's space, and can leave the discrete system singular.

Inf-sup stability, in three dimensions. By (35) the pressure–displacement pairing is $\int \varpi\, \boldsymbol{F}^{-\mathsf{T}} : \mathrm{Grad}\, \boldsymbol{\psi}$ with $\varpi = \theta'(J)\, J\, \bar{p}$, and not $\int \bar{p}\, \boldsymbol{F}^{-\mathsf{T}} : \mathrm{Grad}\, \boldsymbol{\psi}$; the two coincide at $\boldsymbol{F} = \boldsymbol{I}$ and nowhere else. The normalization (13) sets $\theta'(1) = 1$, so $\varpi = \bar{p}$ there and the pairing reduces to the divergence pairing $\int \bar{p}\, \mathrm{Div}\, \boldsymbol{\psi}\, \mathrm{d}V$ — in both instances of Table 2, which is what makes the stability results of §9.5 a property of the pair rather than of the material. That pairing is the one of incompressible linear elasticity, in which the pressure is the multiplier of the constraint $\mathrm{Div}\, \boldsymbol{\varphi} = 0$, and its discrete inf-sup condition,

$$\inf_{q \in Q_h} \sup_{\boldsymbol{v} \in U_h} \frac{\int_\Omega q\ \mathrm{Div}\, \boldsymbol{v}\, \mathrm{d}V}{|\boldsymbol{v}|_{H^1}\, \|q\|_{L^2}} \ \geq\ \beta_h, \tag{38}$$

is the condition on the pair. The admissible pairs are therefore those for which β_h in (38) is bounded below uniformly in h, and stability must also hold on the target range of κ/μ [7].

The pair has been used before in large-deformation hyperelasticity: Chamberland et al. [10], comparing discretizations of a nearly incompressible Mooney–Rivlin material against manufactured solutions on tetrahedral and hexahedral meshes, recommend it and its hexahedral counterpart for nearly incompressible large-deformation problems for accuracy, run time and memory, and observe on some manufactured solutions a larger pressure error than that of Taylor–Hood, which they attribute to a smaller inf-sup constant of the richer pressure space (the constants measured in §9.5 are 0.2968 for this pair and 0.2216 for Taylor–Hood on the Freudenthal meshes). Their formulation is the two-field displacement–pressure form, whose pressure equation $\int_{\Omega_0} (J-1)\, q\, \mathrm{d}V - \int_{\Omega_0} \kappa^{-1} p\, q\, \mathrm{d}V = 0$ for all q in the discontinuous space Q_h gives $p = \kappa\, \mathcal{P}_h(J-1)$ element by element; in elasticity their element is therefore the element of this note with $\theta = J - 1$ and a volumetric energy quadratic in θ. They also report that on meshes of curved elements a pressure linear in the coordinates of the physical element is more accurate than one linear in the coordinates of the parent element. The element of this note takes the pressure linear in the parent coordinates $\boldsymbol{\xi}$ (§4.4); on an element with straight edges these are affine functions of the reference coordinates $\boldsymbol{X}$ and the two choices coincide, and they differ only on elements with curved edges, such as those on the lateral surface of the Taylor bar (§11). The effect of that choice on the element of this note is not measured. No use of the pair in finite-deformation plasticity is known to the author.

Stability of a pair depends on the spatial dimension. The conforming Crouzeix–Raviart pair [13] is, in two dimensions, P_2 enriched with a cubic interior bubble over P_1^{disc}; its three-dimensional counterpart below carries a quartic interior bubble and cubic face bubbles, and the quadrature degree the element needs follows from the quartic. Unenriched P_2/P_1^{disc} is stable in two dimensions only on special meshes. In three dimensions Boffi et al. [7, §8.7] state that a constant discontinuous pressure is not controlled by P_2: "to control piecewise constant pressure, we need some degrees of freedom on the faces," and the quadratic space has none. The stable three-dimensional choices with a discontinuous pressure therefore require bubble enrichment of the displacement space, and the enrichment has two parts with two functions: face bubbles control the constant pressure, and an interior bubble lifts it to a linear one [7, Example 8.7.2]. Table 3 lists the candidates, and §9.5 measures every one of them.

The pair this formulation requires is P_2 enriched with face and interior bubbles over P_1^{disc}. The two enrichments are different kinds of object. The interior bubble vanishes on the whole element boundary: it is never shared, never reached by a boundary condition, and condenses away, adding no global unknown. A face bubble vanishes on three faces and not on its own, so the two elements meeting there share it; it is a global unknown and does not condense. The

Displacement	Pressure	3D stable	Pressure elimination
P_1	P_0	no	local
P_2	P_0	**no**	local
$P_2 \oplus b_{\text{face}}$	P_0	yes	local
P_2	P_1^{disc}	**no**	local
$P_2 \oplus b_{\text{int}}$	P_1^{disc}	no	local
$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$	P_1^{disc}	yes	local
P_2	P_1 continuous	yes	global

Table 3: Displacement–pressure pairs on tetrahedra, with three-dimensional stability as Boffi et al. [7, §8.7, Example 8.7.2] give it. Local elimination requires a discontinuous pressure; three-dimensional stability with a discontinuous pressure requires face bubbles, and a linear pressure requires an interior bubble as well.

enrichment therefore adds global degrees of freedom, which §9.3 counts. The pressure elimination of §4.2 nevertheless stays element by element, because it rests on Q_h being discontinuous and not on the displacement space.

The linear pressure is not forced by the stability results. $P_2 \oplus b_{\text{face}}/P_0$ is also stable (§9.5), at a higher constant and with the same number of global unknowns, since the interior bubble condenses. The count of §9.2 applies to the unenriched displacement space, and §9.4 shows that its modes do not persist after assembly. What separates the two pressure spaces is the claim of §6: a constant pressure leaves in $\ker \boldsymbol{G}_h$ the element-scale dilatational modes that a plastic tangent can free, and a linear one does not. §9.7 measures both pairs, and that measurement is the reason for the linear pressure.

Both that pair and the Taylor–Hood pair — continuous P_2 over continuous P_1, whose stability in three dimensions is established [7] and which serves as the positive control of the measurements — are measured in §9.5, and every other element-local pair decays. The difference between the two stable pairs is not stability but where the coupling is — a global pressure that cannot be eliminated element by element against global face bubbles that cannot be condensed — and §14 quantifies it.

4.2 Local elimination

Because Q_h is discontinuous, the mass matrix $[M]_{mn} = \int_\Omega \chi_m \chi_n \,\mathrm{d}V$, with $\{\chi_m\}$ a basis of Q_h, is block diagonal by element and constant in the reference configuration. One factorization at setup serves every Newton iteration of every step, and

$$M\{\bar{\theta}\} = \int_\Omega \chi_m \, \theta(\boldsymbol{\varphi}) \,\mathrm{d}V, \qquad M\{\bar{p}\} = \int_\Omega \chi_m \, \frac{\tilde{p}}{\theta'(\tilde{J})} \,\mathrm{d}V \tag{39}$$

are element-local solves with the same factorization, the second the projection (18) of the material's mean stress. For a split material the second integrand is $W^{\text{vol}\prime}(\bar{\theta})$, and for a quadratic W^{vol} the second solve is unnecessary, since $\bar{p} = \kappa\bar{\theta}$ pointwise. Substituting $\bar{p}(\boldsymbol{\varphi})$ into the residual leaves a displacement-sized nonlinear problem whose tangent retains the sparsity of the displacement mesh — the auxiliary fields never couple two elements. This is the property that keeps CT10's tangent as sparse as a displacement element's, and it is preserved here.

Remark 4. The locality is a consequence of the discontinuous pressure, not of the Hu–Washizu structure. A conforming auxiliary field — one requiring inter-element continuity — would make $\bar{p}(\boldsymbol{\varphi})$ a global function of $\boldsymbol{\varphi}$, and every tangent–vector product would then require a global solve with its mass matrix, which CT10 does not require. That is the main reason the pressure is kept discontinuous.

4.3 The element

The displacement space of the pair is carried by a fifteen-node tetrahedron, `TETRA15` in the naming of the Exodus mesh format, with three displacement unknowns at each node. This section defines it in full.

Nodes. With $\lambda_1, \dots, \lambda_4$ the barycentric coordinates of the tetrahedron, the fifteen nodes are the four vertices (nodes 1–4); the six edge midpoints (nodes 5–10), on the edges 1–2, 2–3, 3–1, 1–4, 2–4, 3–4 in that order; the four face centroids (nodes 11–14), on the faces with vertices 1–3–2, 2–3–4, 1–4–3 and 1–2–4 in that order, which is the side-node convention of the Exodus library (node 14 on side 1, node 12 on side 2, node 13 on side 3, node 11 on side 4); and the centroid (node 15). The first ten are the nodes of the quadratic tetrahedron TET10 in the same order, so a `TETRA10` mesh is converted by adding one node at the centroid of every face and of every element; for a mesh with straight edges the added nodes are the averages of the face and element vertices. Each face is a seven-node triangle: three vertices, three edge midpoints and the centroid.

The space. The displacement space on one element is

$$\left[P_2 \oplus \operatorname{span}\{b_{F_1}, \dots, b_{F_4}\} \oplus \operatorname{span}\{b_T\}\right]^3, \qquad b_F = 27\,\lambda_a\lambda_b\lambda_c, \qquad b_T = 256\,\lambda_1\lambda_2\lambda_3\lambda_4, \tag{40}$$

with P_2 the ten quadratic Lagrange functions of the tetrahedron, b_F the cubic bubble of face F with vertices a, b, c, which vanishes on the three other faces and equals one at the centroid of F, and b_T the quartic interior bubble, which vanishes on the whole boundary and equals one at the centroid. The space has dimension fifteen per component and contains the complete quadratic polynomials, so the element is conforming and its interpolation error is that of the quadratic tetrahedron.

The nodal basis. The functions of (40) in the order listed — the ten Lagrange functions of TET10, then $b_{F_1}, \dots, b_{F_4}$ in the face order of the nodes, then b_T — form the hierarchical basis $\{\hat{N}_j\}$, so called because the functions of the lower-degree element remain in it and the bubbles are appended. It is not nodal: the Lagrange functions are nonzero at the face centroids and at the centroid. The nodal basis $\{N_a\}$, with $N_a(\boldsymbol{x}_b) = \delta_{ab}$ at the fifteen nodes, is obtained from it by the constant matrix

$$N_a = \sum_j A_{aj}\hat{N}_j, \qquad \boldsymbol{A} = (\boldsymbol{V}^{-1})^{\mathsf{T}}, \qquad V_{bj} = \hat{N}_j(\boldsymbol{x}_b), \tag{41}$$

with $\boldsymbol{V}$ the values of the hierarchical functions at the nodes, a 15×15 matrix with rational entries that is inverted exactly once. The gradients transform by the same matrix. Every unknown is then the displacement of a node, so a code that numbers unknowns by node treats

the element like any Lagrange element: the four face-centroid unknowns are shared between the two elements of a face and the centroid unknown belongs to one element. §9.9 measures the two bases; the nodal one is used.

Quadrature. The element is integrated with the fourteen-point rule of Walkington [33], exact for polynomials of degree five, with positive weights. Its points are three orbits of the symmetry group of the tetrahedron: two orbits of four points with barycentric coordinates $(1-3a, a, a, a)$ and its permutations, and one orbit of six points with coordinates $(b, b, \frac{1}{2}-b, \frac{1}{2}-b)$ and its permutations,

$$\begin{aligned} a_1 &= 0.0927352503108912, \quad w_1 = 0.01224884051939366, \\ a_2 &= 0.3108859192633006, \quad w_2 = 0.01878132095300264, \\ b &= 0.4544962958743503, \quad w_3 = 0.007091003462846911, \end{aligned} \tag{42}$$

with the weights normalized to the reference volume 1/6. The gradient of b_T is cubic and the stiffness integrand of its block is of degree six, which the rule integrates with an error of 35% on that block (§9.9); the rules of degree two and three (four and five points) leave the element twenty-one zero-energy modes beyond the six rigid-body modes and cannot be used. The five-point rule of degree three of Keast [19] has a negative weight at the centroid and gives an indefinite element stiffness, with 15 zero and 6 negative eigenvalues. The same fourteen-point rule integrates the projection (39), the residual and the row-sum lumped mass. The row sums are integrals of quartic functions and are exact under the rule.

Consistent mass. The consistent mass is not exact under the fourteen-point rule, and it is singular under it. Under a rule with fourteen points the scalar consistent mass of the element is a sum of fourteen matrices of rank one, so its rank is at most fourteen for fifteen functions. The function of the space that vanishes at the fourteen points is in its null space. Assembled on Freudenthal meshes of the unit cube with $N = 1$, 2 and 3 and no boundary condition, the scalar mass under the fourteen-point rule has one zero eigenvalue, and the exactly integrated mass has none. The integrand of the mass is a polynomial of degree eight, so the consistent mass requires a rule of degree eight. The conical product rule with five Gauss–Jacobi points in each direction, of degree nine with 125 positive weights, integrates it exactly. The mass is computed once per element in the reference configuration, so the cost of the rule is incurred once. Table 4 compares the rules on the reference element in the nodal basis.

Both implementations, Carina TET15-P1 and Sierra/SM TET15-P1, form the consistent mass of the element with the 125-point rule and the lumped mass with the row sums of the fourteen-point rule. In both, the reference element mass agrees with the mass computed in rational arithmetic, or with a rule of degree fifteen, to within 2.1×10^{-15} in each entry, against a largest entry of 0.026. Its condition number is 93.6, and it has no zero eigenvalue. On an affine element the mass equals the reference mass scaled by $\rho \det \nabla \boldsymbol{x}$, its sum equals ρV, and its row sums equal the lumped mass, each to within 6×10^{-16} relative to ρV in Sierra/SM and 10^{-14} in Carina. In Carina the mass assembled on a mesh of the unit cube with 891 unknowns and no boundary condition has no zero eigenvalue. Its matrix-free action agrees with the assembled matrix, and between the CPU and the GPU, to within 3×10^{-15} relative. The action takes 5.3$\boldsymbol{\mu}$s per element on one CPU core, against 3.5$\boldsymbol{\mu}$s with the fourteen-point rule. The plain fifteen-node tetrahedron, without the projection, still uses the fourteen-point rule

Table 4: Rules for the consistent mass of the enriched element, nodal basis, reference element: zero eigenvalues of the scalar element mass, its condition number, the largest error of an entry against exact integration relative to the largest entry, and the number of zero eigenvalues of the assembled scalar mass on the Freudenthal meshes with $N = 2$ and 3 and no boundary condition. The row sums are exact under every rule, to 3.3×10^{-15}. Script: `prototype/mass_rule.jl` and `prototype/mass14.jl`.

rule	degree	points	zero eigenvalues	condition number	entry error
Walkington	5	14	1 (assembled 1, 1)	∞	1.7×10^{-1}
conical	5	27	0 (0, 0)	123.0	7.3×10^{-2}
conical	7	64	0 (0, 0)	94.4	1.0×10^{-3}
conical	9	125	0 (0, 0)	93.6	2.6×10^{-15}

for its consistent mass in Carina. No result of this note uses the consistent mass: the explicit computations use the row-sum lumped mass, and the quasi-static ones have no mass. With the implicit integrator the effective matrix $\boldsymbol{K} + c\,\boldsymbol{M}$ of a step is positive definite when $\boldsymbol{K}$ is, under either rule. An operation that inverts the consistent mass alone, such as the computation of an initial acceleration, requires the rule of degree eight or higher.

Mass. The consistent mass matrix in the nodal basis has condition number 93.6 on the reference element with exact integration. Its row sums, $m_a = \rho \int_e N_a \,\mathrm{d}V$, are positive at every node because the nodal basis is a partition of unity with positive integrals: as fractions of the element mass they are 0.020 at each vertex, 0.038 at each edge midpoint, 0.096 at each face centroid and 0.305 at the centroid. The row-sum lumped mass is therefore admissible for explicit time integration (§7), which it is not for TET10, whose vertex row sums are negative.

4.4 Element algorithm

This section states what an implementation computes on one element, with the nodal displacements $\boldsymbol{u}_a$ of its fifteen nodes given. The pressure space Q_h on the element is spanned by $\chi_m(\boldsymbol{\xi})$, $m = 1, \dots, 4$, with $\chi = (1, \xi_1, \xi_2, \xi_3)$ in the reference coordinates $\boldsymbol{\xi}$ of the element; the constant projection, the element TET15-P0, which §10 also measures, has $\chi = (1)$. The mass matrix of the projection is taken with the reference-configuration volume element, so it does not change with the deformation. The quadrature points $\boldsymbol{\xi}_q$ have weights w_q and reference Jacobian determinants j_q. The residual takes three passes over them for a general material, and two for a split material.

First pass: the projected volumetric strain. At each quadrature point, $\boldsymbol{F}_q = \boldsymbol{I} + \sum_a \boldsymbol{u}_a \otimes \operatorname{Grad} N_a(\boldsymbol{\xi}_q)$, $J_q = \det \boldsymbol{F}_q$ and $\theta_q = \theta(J_q)$. Then

$$M_{mn} = \sum_q w_q j_q \, \chi_m(\boldsymbol{\xi}_q) \, \chi_n(\boldsymbol{\xi}_q), \qquad b_m = \sum_q w_q j_q \, \chi_m(\boldsymbol{\xi}_q) \, \theta_q, \qquad \bar{\theta} = \boldsymbol{M}^{-1} \boldsymbol{b}, \tag{43}$$

a 4×4 solve, and the projected strain at each point is $\bar{\theta}_q = \sum_m \chi_m(\boldsymbol{\xi}_q) \, \bar{\theta}_m$, from which $\tilde{J}_q = \theta^{-1}(\bar{\theta}_q)$ and $s_q = (\tilde{J}_q / J_q)^{1/3}$. This pass is the same for every material.

Second pass, general material: the material at $\tilde{\boldsymbol{F}}_q$. At each point the element forms $\tilde{\boldsymbol{F}}_q = s_q \boldsymbol{F}_q$, calls the material once with its internal variables, stores $\tilde{\boldsymbol{P}}_q$ and $\tilde{p}_q$ of (17), and accumulates $\bar{b}_m = \sum_q w_q j_q \, \chi_m(\boldsymbol{\xi}_q) \, \tilde{p}_q / \theta'(\tilde{J}_q)$, from which $\bar{p} = \boldsymbol{M}^{-1}\bar{\boldsymbol{b}}$ with the mass matrix of (43).

Third pass, general material: the residual. From the stored $\tilde{\boldsymbol{P}}_q$ the element forms the stress (22) at each point and the residual $\boldsymbol{R}_a = \sum_q w_q j_q \, \boldsymbol{P}_q \operatorname{Grad} N_a(\boldsymbol{\xi}_q)$. The storage is one stress tensor per quadrature point; the material is called once per point per residual. The tangent is (23), assembled in closed form (§3.11).

Second pass, split material: the stress and the residual. For a material with the split (25) the pressure follows from $\bar{\theta}$ alone, $\bar{p} = \mathcal{P}_h W^{\mathrm{vol}\prime}(\bar{\theta})$, and for a quadratic W^{vol} it is $\bar{p}_q = \kappa \bar{\theta}_q$ at each point, with no second projection. At each quadrature point the material supplies the isochoric first Piola–Kirchhoff stress $\boldsymbol{P}_q^{\mathrm{iso}}$ from $\boldsymbol{F}_q$ and its internal variables, with its own return map, and the element adds the volumetric part with the projected pressure:

$$\boldsymbol{P}_q = \boldsymbol{P}_q^{\mathrm{iso}} + \bar{p}_q \, \boldsymbol{g}_q, \qquad \boldsymbol{g}_q := \frac{\partial \theta}{\partial \boldsymbol{F}}\Big|_q = \theta'(J_q) \, J_q \, \boldsymbol{F}_q^{-\mathsf{T}}, \qquad \boldsymbol{R}_a = \sum_q w_q j_q \, \boldsymbol{P}_q \operatorname{Grad} N_a(\boldsymbol{\xi}_q). \tag{44}$$

The residual is the pointwise one with the projected pressure in place of the pointwise one; for a quadratic W^{vol} it is the derivative of the element energy $\sum_q w_q j_q [W^{\mathrm{iso}}(\bar{\boldsymbol{F}}_q) + \frac{\kappa}{2}\bar{\theta}_q^2]$ with respect to the nodal displacements, which §4.5 verifies by differences. With the same material the general form gives the same residual to rounding (§4.5) at one pass more.

The tangent of the split form. For a quadratic W^{vol} the derivative of (44) along a nodal displacement increment $\boldsymbol{v}$ with gradient $\operatorname{Grad} \boldsymbol{v}_q$ at the quadrature points has three parts: the isochoric material tangent of the model, the geometric term of the volumetric stress, and the coupling through the projection,

$$\begin{aligned} \delta \boldsymbol{P}_q &= \frac{\partial \boldsymbol{P}^{\mathrm{iso}}}{\partial \boldsymbol{F}}\Big|_q : \operatorname{Grad} \boldsymbol{v}_q + \bar{p}_q \, \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{F}}\Big|_q : \operatorname{Grad} \boldsymbol{v}_q + \kappa \, \delta\bar{\theta}_q \, \boldsymbol{g}_q, \\ \delta\bar{\theta} &= \boldsymbol{M}^{-1} \sum_q w_q j_q \, \chi(\boldsymbol{\xi}_q) \, (\boldsymbol{g}_q : \operatorname{Grad} \boldsymbol{v}_q), \qquad \delta\bar{\theta}_q = \sum_m \chi_m(\boldsymbol{\xi}_q) \, \delta\bar{\theta}_m, \end{aligned} \tag{45}$$

with, for $\phi(J) := \theta'(J)J$ and ϕ' its derivative,

$$\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{F}} : \delta\boldsymbol{F} = \phi'(J) \, J \, (\boldsymbol{F}^{-\mathsf{T}} : \delta\boldsymbol{F}) \, \boldsymbol{F}^{-\mathsf{T}} - \phi(J) \, \boldsymbol{F}^{-\mathsf{T}} \delta\boldsymbol{F}^{\mathsf{T}} \boldsymbol{F}^{-\mathsf{T}}. \tag{46}$$

The third term couples the quadrature points of the element: in matrix form it adds $\kappa \, \boldsymbol{D}\boldsymbol{M}^{-1}\boldsymbol{D}^{\mathsf{T}}$ to the sum of the quadrature-point matrices, with $\boldsymbol{D} = \sum_q w_q j_q \, (\operatorname{Grad} N \cdot \boldsymbol{g}_q) \, \chi(\boldsymbol{\xi}_q)^{\mathsf{T}}$ the matrix that maps nodal displacements to the right-hand side of (43). For a non-quadratic W^{vol} the factor κ becomes the weighted mass matrix of §3.6. The tangent is symmetric, since it is the second derivative of the element energy. The matrix-free tangent–vector product is (45) applied to $\boldsymbol{v}$ with the two passes; the assembled tangent, its diagonal and its nodal blocks are formed from the same three terms.

What the material supplies. For the general form, the material supplies at a quadrature point, with its internal variables, the stress $\tilde{\boldsymbol{P}}$ at $\tilde{\boldsymbol{F}}$ together with the update of the internal variables it entails, and the tangent $\mathbb{A}$ at $\tilde{\boldsymbol{F}}$; the energy is needed only to verify the residual. For the split form it supplies the volumetric strain measure $\theta(J)$ with $\theta'(J)$ and $\theta''(J)$; W^{vol} or, when quadratic, the bulk modulus κ; the isochoric energy $W^{\mathrm{iso}}(\bar{\boldsymbol{F}})$; the isochoric first Piola–Kirchhoff stress with the update of its internal variables; and the isochoric tangent $\partial \boldsymbol{P}^{\mathrm{iso}}/\partial \boldsymbol{F}$. For the Simo–Hughes J_2 model as implemented, $\theta = \log J$, $\theta' = 1/J$, $\theta'' = -1/J^2$, $\kappa = \lambda + 2\mu/3$ from the Lamé constants, and the isochoric stress and tangent are those of BOX 9.1 of Simo and Hughes [29] with the volumetric term omitted; the return map on $\bar{\boldsymbol{b}}^{\mathrm{e}}$ and the internal variables $\boldsymbol{F}^{\mathrm{p}}$ and $\bar{\epsilon}^{\mathrm{p}}$ are unchanged.

Assembly. Because the first pass needs every quadrature point of the element before the second can begin, the element is assembled as a unit, not quadrature point by quadrature point. Internal variables are stored per quadrature point as for any element. Tractions on a face are integrated over the seven-node triangle with the triangle rule of the same degree as the element rule. The element runs under the quasi-static, the implicit and the explicit integrators of the code without change.

4.5 Consistency checks

An implementation of §4.4 is verified by the following checks, which the implementation in Carina passes (`test/projected-element.jl`, forty-six checks) on a cube of TET15 elements with the J_2 model in its elastic range at a converged nonlinear state, so that the geometric and coupling terms of the tangent are nonzero, for both the linear and the constant projection: the residual equals the derivative of the assembled energy by central differences, and the assembled tangent the derivative of the residual, to the truncation error of the differences; the matrix-free action, the diagonal and the nodal-block kernels reproduce the assembled tangent to 10^{-12}; an affine displacement field is reproduced exactly (patch test, 10^{-10}), because a constant J is its own projection and the projected element must then agree with the pointwise one; and Newton converges quadratically (relative residual 1, 8.6×10^{-3}, 8×10^{-7}, 2×10^{-14}).

The general form of §3.5 passes the same checks (`test/projected-general.jl`, 84 checks) with the neo-Hookean material, which has no accessible split in the material library, with $\theta = \log J$ and both projections, on the CPU and on a GPU, where the Newton history and the displacements agree with the CPU's within 10^{-8}. Its reduction to the split form is checked on the same cube with the J_2 model forced through the general form: the Newton history, the displacements, the internal variables, the energy, the residual and the assembled tangent agree with the split form's to between 10^{-15} and 10^{-13} relative, in the elastic range and in a plastic step with an equivalent plastic strain of 0.021. The formulas themselves were checked on one distorted element before the kernels were written (`prototype/general.jl`): the hand-derived residual against the gradient of the energy by automatic differentiation to 4×10^{-16}, the closed-form tangent against its Hessian to 1.2×10^{-14}, and the three reductions of §3.5 to rounding. The same Julia prototype found that the tangent of the J_2 model in the material library, the consistent tangent of BOX 9.2 of Simo and Hughes [29], divided its coefficient β_2 by μ instead of the effective modulus $\bar{\mu} = \mu\,\mathrm{tr}(\bar{\boldsymbol{b}}^{e}_{\mathrm{tr}})/3$; the two agree at small strain and the term vanishes without hardening, which is why the library's own checks had not seen it, and the symmetric part of the tangent was off by 5×10^{-2} at an equivalent plastic strain of 0.6 reached

in one increment. With the correction the symmetric part agrees with the central-difference Jacobian of the stress to 10^{-10}; what remains, 4×10^{-3} to 5×10^{-3}, is the antisymmetric part that BOX 9.2 drops by construction, so Newton converges linearly, not quadratically, on steps that yield with this model.

5 Plasticity

This section develops the logarithmic instance of Table 2; the Simo–Hughes column differs here and nowhere else (§3.8), and Remark 5 states what the implemented instance, Simo–Hughes, entails.

The deviatoric response is evaluated at quadrature points of the displacement element, on $\boldsymbol{e}(\boldsymbol{\varphi})$ computed there. The plastic strain is carried as a logarithmic strain $\boldsymbol{e}^{\mathrm{p}}$ and subtracted additively — the construction of Miehe et al. [20], in which a material geometric pre-processing $\boldsymbol{C} \mapsto \boldsymbol{\epsilon}$ and a post-processing through $\mathbb{L}(\boldsymbol{C})$ precede and follow an unmodified small-strain algorithm. This is a different constitutive model from the multiplicative $\boldsymbol{F} = \boldsymbol{F}^{\mathrm{e}}\boldsymbol{F}^{\mathrm{p}}$ plasticity of Simo [28], which also preserves the small-strain return map but applies it to the elastic logarithmic strain of a trial elastic state with an exponential update; the two coincide under proportional loading with fixed principal axes and differ when the axes rotate. Because the constitutive law is linear in logarithmic strain, the J_2 return map is the small-strain radial return — the return map that scales the trial deviatoric stress toward the yield surface along its own direction — unchanged:

$$
\begin{aligned}
&\boldsymbol{e}^{\mathrm{tr}} := \boldsymbol{e}(\boldsymbol{\varphi}) - \boldsymbol{e}^{\mathrm{p}}_n, \qquad \boldsymbol{s}^{\mathrm{tr}} := 2\mu\, \boldsymbol{e}^{\mathrm{tr}},\\
&\Delta\gamma: \quad \left\|\boldsymbol{s}^{\mathrm{tr}}\right\| - 2\mu\,\Delta\gamma - \sqrt{2/3}\left[\sigma_y + \mathcal{H}'\left(\bar{\epsilon}^{\mathrm{p}}_n + \sqrt{2/3}\,\Delta\gamma\right)\right] = 0 \quad \text{if } f^{\mathrm{tr}} > 0, \quad \text{else } \Delta\gamma := 0,\\
&\boldsymbol{e}^{\mathrm{p}}_{n+1} := \boldsymbol{e}^{\mathrm{p}}_n + \Delta\gamma\,\boldsymbol{n}, \qquad \boldsymbol{n} := \boldsymbol{s}^{\mathrm{tr}} / \left\|\boldsymbol{s}^{\mathrm{tr}}\right\|, \qquad \bar{\epsilon}^{\mathrm{p}}_{n+1} := \bar{\epsilon}^{\mathrm{p}}_n + \sqrt{2/3}\,\Delta\gamma,\\
&\boldsymbol{s} := 2\mu\left(\boldsymbol{e}(\boldsymbol{\varphi}) - \boldsymbol{e}^{\mathrm{p}}_{n+1}\right).
\end{aligned}
\tag{47}
$$

Here σ_y is the initial yield stress, $\bar{\epsilon}^{\mathrm{p}}$ the equivalent plastic strain, $\mathcal{H}'$ the hardening function (the increase of the yield stress with $\bar{\epsilon}^{\mathrm{p}}$) and $\mathcal{H}''$ its derivative, the hardening modulus, and $f^{\mathrm{tr}} := \|\boldsymbol{s}^{\mathrm{tr}}\| - \sqrt{2/3}\,[\sigma_y + \mathcal{H}'(\bar{\epsilon}^{\mathrm{p}}_n)]$ the trial yield function. The consistency condition is a scalar equation in $\Delta\gamma$. For linear hardening it closes in one step, $\Delta\gamma = f^{\mathrm{tr}}/(2\mu + \frac{2}{3}\mathcal{H}'')$, which is the form both codes implement; for nonlinear hardening it is a scalar Newton iteration [29, BOX 3.2]. Internal variables are stored at quadrature points and are never projected, so no interpolation error enters the history. Frame-indifferent interpolation, when tensorial internal variables must be transferred, follows Mota et al. [21].

Three properties follow. They are the motivation for the formulation rather than incidental consequences of it.

The constitutive law is untouched. All finite-deformation content is absorbed into two material-independent geometric transforms: the forward map $\boldsymbol{C} \mapsto \boldsymbol{\epsilon}$ and the tangent map $\mathbb{L}(\boldsymbol{C})$ [20]. Between them is an unmodified small-strain algorithm. Any model already written for small strain — J_2, kinematic hardening, viscoplasticity, damage — is usable without reformulation. This is the practical content of preferring a Hu–Washizu structure over one built

around a particular stress–strain pairing. This property belongs to the logarithmic measure and not to the Hu–Washizu structure, so it is the one requirement beyond those of §3.4: an admissible material in the sense of (25) need not have it, and the Simo–Hughes model both codes implement does not (Remark 5).

Strain and stress never cross discretizations. $\boldsymbol{e}$ is evaluated and $\boldsymbol{s}$ returned at the same quadrature point of the same element. There is no projection between the two, so the interpolation error between two discrete spaces that corrupts the equivalent plastic strain in stabilized formulations has no mechanism here.

No stabilization enters the deviator. There is no subgrid term, no penalty and no tuned coefficient anywhere in the formulation, so nothing of the form (1) can reach the return map.

Remark 5 (The instance implemented)**.** The instance implemented in Carina, and the one every nonlinear result of §10 uses, is the Simo–Hughes J_2 model [29, BOX 9.1]: the multiplicative decomposition $\boldsymbol{F} = \boldsymbol{F}^{\mathrm{e}}\boldsymbol{F}^{\mathrm{p}}$ with the volumetric–isochoric split $W = \frac{\kappa}{2}(\log J)^2 + \frac{\mu}{2}(\operatorname{tr}\bar{\boldsymbol{b}}^{\mathrm{e}} - 3)$, and the radial return taken on the isochoric elastic left Cauchy–Green tensor $\bar{\boldsymbol{b}}^{\mathrm{e}}$ rather than on the logarithmic strain. It satisfies (25) exactly — $\det\bar{\boldsymbol{b}}^{\mathrm{e}} = 1$ makes the isochoric term independent of J by construction — and its W^{vol} is quadratic in $\log J$, so the element of §4 uses it in the variable $\theta = \log J$ with no change other than the omission of the volumetric term from the stress the model returns. The implementation of the model in Norma [22], the Julia finite element code from which Carina descends and whose material library it shares, has $W^{\mathrm{vol}} = \frac{\kappa}{2}(J-1)^2$, and Simo and Hughes [29, BOX 9.1] use $W^{\mathrm{vol}} = \frac{\kappa}{2}\left[\frac{1}{2}(J^2-1) - \log J\right]$; these volumetric laws and the logarithmic one agree to second order in $J-1$, and the logarithmic one is used so that the element projects the volumetric part of the logarithmic strain whatever the material, which is also what Sierra/SM TET15-P1 projects. Until 2026-10-16 the implemented instance projected $J-1$. The difference is measurable where the pointwise J varies within an element: on the indentation benchmark at $h = a/2$ projecting $J-1$ changed the punch force by 0.21% (case A) and 0.41% (case B) against $\log J$, and with $\log J$ the two implementations agree to 0.06%; on the Taylor bar at $h = 0.38$ mm it changed the final radius by 0.06%, and on Cook's membrane the tip displacement by at most 0.03%. The two instances share the mixed treatment of the volume and differ in the plasticity model: additive plasticity in logarithmic strain in (47), multiplicative plasticity with the return map on $\bar{\boldsymbol{b}}^{\mathrm{e}}$ in the implemented instance; the two coincide under proportional loading with fixed principal axes and differ when the axes rotate. The property that a small-strain model is usable without reformulation belongs to the logarithmic instance alone; the Simo–Hughes return map is specific to its own kinematics. The smoothness of the equivalent plastic strain is a question of whether strain and stress cross discretizations, which §4.4 settles for either instance, and it is to be measured with the implemented one (§13).

6 The claim

This section states the claim of the note: which hypothesis of Brezzi's theorem a plastic tangent can violate, and what distinguishes pairs under it. The second hypothesis is coercivity of the deviatoric form on $\ker \boldsymbol{G}_h$ (§1).

In elasticity the second hypothesis holds for every conforming pair: by Korn's inequality, which bounds $\int |\mathrm{Grad}\, \boldsymbol{v}|^2 \,\mathrm{d}V$ by a multiple of $\int |\mathrm{dev}\, \boldsymbol{\epsilon}(\boldsymbol{v})|^2 \,\mathrm{d}V$ for $\boldsymbol{v} \in H^1_0$, the deviatoric form is coercive on all of U_h, so it is coercive on $\ker \boldsymbol{G}_h$, the set of discrete displacements whose divergence has zero projection onto Q_h (§9.1), and no elastic soft mode is a failure of the theorem. A count of exact zero-energy modes on one element is therefore not an argument against a constant pressure in elasticity; §9.4 shows that those modes do not persist after assembly.

Under plastic loading the continuum elastoplastic tangent of J_2 plasticity at infinitesimal strain is

$$\mathbb{C}^{\mathrm{ep}} = 2\mu \left(\mathbb{I}^{\mathrm{dev}} - \eta\, \boldsymbol{n} \otimes \boldsymbol{n} \right), \qquad \mathbb{I}^{\mathrm{dev}} := \mathbb{I} - \tfrac{1}{3} \boldsymbol{I} \otimes \boldsymbol{I}, \qquad \boldsymbol{n} := \boldsymbol{s}^{\mathrm{tr}} / \left\| \boldsymbol{s}^{\mathrm{tr}} \right\|, \qquad \eta = \frac{2\mu}{2\mu + \frac{2}{3}\mathcal{H}''}, \tag{48}$$

with $\eta = 1$ for perfect plasticity: the shear stiffness along the flow direction is removed. The algorithmic tangent of the radial return (47) differs from (48): it also scales the deviatoric part by $1 - 2\mu\Delta\gamma / \left\| \boldsymbol{s}^{\mathrm{tr}} \right\|$, and at finite strain it has further terms. The measurements of §9.7 use (48) at $\boldsymbol{F} = \boldsymbol{I}$, with one flow direction and one value of η in the whole domain.

Read as a constant, the second hypothesis cannot separate pairs. For every $\boldsymbol{v} \in H^1_0(\Omega)$, integration by parts gives $\int |\mathrm{dev}\, \boldsymbol{\epsilon}|^2 \,\mathrm{d}V = \frac{1}{2} \int |\mathrm{Grad}\, \boldsymbol{v}|^2 \,\mathrm{d}V + \frac{1}{6} \int (\mathrm{Div}\, \boldsymbol{v})^2 \,\mathrm{d}V$, and $(\boldsymbol{n} : \boldsymbol{\epsilon})^2 \le |\mathrm{dev}\, \boldsymbol{\epsilon}|^2$ since $\boldsymbol{n}$ is a unit deviator, so

$$\frac{1}{2\mu} \int \boldsymbol{\epsilon} : \mathbb{C}^{\mathrm{ep}} : \boldsymbol{\epsilon} \,\mathrm{d}V \;\ge\; (1 - \eta) \int |\mathrm{dev}\, \boldsymbol{\epsilon}|^2 \,\mathrm{d}V \;\ge\; \frac{1 - \eta}{2} \int |\mathrm{Grad}\, \boldsymbol{v}|^2 \,\mathrm{d}V \tag{49}$$

for every conforming pair and every member of its kernel, isochoric or not. For $\eta < 1$ the coercivity constant on $\ker \boldsymbol{G}_h$ is therefore bounded below uniformly in h and in the pair. For $\eta = 1$ the continuum operator loses strong ellipticity, positivity of the energy of every displacement jump across a plane: a shear band, a displacement jump $\boldsymbol{e}_1$ across a plane with normal $\boldsymbol{e}_2$, has $\boldsymbol{\epsilon} \propto \mathrm{sym}(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2)$, to which (48) with that $\boldsymbol{n}$ assigns zero energy, so the constant tends to zero under refinement for every pair as the discrete space resolves narrower bands (§9.7). Neither limit says anything about the pair.

What distinguishes the pairs is what the soft directions are. A member of $\ker \boldsymbol{G}_h$ that is nearly isochoric is a shear band the continuum problem also has, and its low energy is physical. A member with pointwise nonzero dilatation of zero element mean has the volumetric energy $\kappa \int (\mathrm{Div}\, \boldsymbol{v})^2 \,\mathrm{d}V$ in the continuum and none in the element, so once the tangent removes its shear energy the element assigns it zero energy. That is the soft mode reported for constant-pressure elements, and whether $\ker \boldsymbol{G}_h$ contains such members is a property of the pair.

The inf-sup condition does not exclude such members. By scaling, a member of $\ker \boldsymbol{G}_h$ supported on a few elements can carry a dilatation of order one at every h, for any pair. An earlier version of this note cited Boffi et al. [7, Prop. 5.1.3] for the statement that the dilatation of every member of $\ker \boldsymbol{G}_h$ vanishes with h for a stable pair. That proposition bounds the distance from the continuous kernel to the discrete one and not the reverse, and Taylor–Hood, which is stable, has members of $\ker \boldsymbol{G}_h$ that are dilatation almost entirely (§9.7); the statement is withdrawn. A bound specific to the pair applies to members that are smooth on the scale of the mesh. For a member $\boldsymbol{v}$ of $\ker \boldsymbol{G}_h$ of the Crouzeix–Raviart pair, $\mathrm{Div}\, \boldsymbol{v}$ is orthogonal to the linear functions on every element K, so $\|\mathrm{Div}\, \boldsymbol{v}\|_{L^2(K)} \le C h_K^2 |\mathrm{Div}\, \boldsymbol{v}|_{H^2(K)}$. For a constant pressure the corresponding bound is $C h_K |\mathrm{Div}\, \boldsymbol{v}|_{H^1(K)}$. The bound gives no information on members that vary on the scale of the element.

A family of members of $\ker \boldsymbol{G}_h$ whose dilatation $r_v = \int (\operatorname{Div} \boldsymbol{v})^2 \,\mathrm{d}V / \int |\operatorname{Grad} \boldsymbol{v}|^2 \,\mathrm{d}V$ does not decrease under refinement is called here a family of spurious dilatational modes. Its eigenvalue need not tend to zero; it matters when it is among the softest modes. The claim is therefore the following, and it is a claim about measurements. Under a plastic tangent, the softest modes of the pair with a linear discontinuous pressure carry a dilatation that decreases under refinement, and those of a constant pressure carry a dilatation that does not; the soft modes reported for constant-pressure elements are of the latter kind. Both operators are linear at a fixed plastic state, so the claim is tested without a nonlinear solve, and §9.7 tests it, on structured, perturbed and unstructured meshes. If the dilatation of the soft modes does not separate the pairs, the formulation has no advantage over the constant-pressure elements it is meant to replace.

7 Dynamics

The spatial formulation is unchanged; the mass matrix enters only the displacement block. Two implementation choices are specified.

Lumping. Explicit integration divides by a diagonal mass, so every lumped mass must be positive. Row-sum lumping assigns to each node the row sum of the consistent mass matrix, $m_a = \rho \int N_a \,\mathrm{d}V$, which for a basis whose functions sum to one adds up to the element mass. On the quadratic tetrahedron TET10 the vertex functions have negative integrals, so the row-sum mass is negative at the vertices and cannot be used. In the nodal basis of the enriched element (§4.3) every row sum is positive: 0.020 of the element mass at each vertex, 0.038 at each edge midpoint, 0.096 at each face centroid and 0.305 at the centroid, on the reference element and on the distorted and Kuhn elements of §9.9. The row-sum lumped mass is used; it conserves the momentum of a rigid translation exactly, because the functions sum to one. The alternative is diagonal-scaling lumping [17] (HRZ: Hinton, Rock and Zienkiewicz), which takes the diagonal of the consistent mass, positive for any basis, and scales it to preserve the element mass. In the nodal basis HRZ also conserves momentum. In the hierarchical basis it does not: a rigid translation has zero coefficient on every bubble unknown, so the mass HRZ assigns to the bubbles, 51% of the element mass on the reference tetrahedron, carries no momentum, unless the scaling runs over the ten Lagrange functions alone with each bubble keeping its consistent diagonal (§9.9).

Critical time step. From the element eigenvalue bound at $\kappa/\mu = 2$ (§9.9), the Courant number $c_p \,\Delta t/h$, with $c_p = \sqrt{(\kappa + 4\mu/3)/\rho}$ the dilatational wave speed, ρ the density, Δt the stable step and h the element edge length, is given in Table 16 for the four-node linear tetrahedron (TET4), for TET10 under HRZ, and for the enriched element in both bases. The quadratic element reduces the stable step by a factor of 2.4 against the linear one on the same mesh, and the enrichment a further 1.4 with the nodal basis; the ordering is the same on the distorted element and on a Kuhn tetrahedron of the Freudenthal mesh (§9.1). A P_2 mesh resolves a given feature with fewer elements; on the Taylor bar (§11) the smaller element count more than compensates for the smaller step. These Courant numbers hold on the undeformed element: on a deformed mesh the smallest distance between two nodes does not bound the critical step, which is then computed from the largest eigenvalue of $M_L^{-1} K$ (§11).

8 Optional five-field variant

This section gives the extension to an independent deviatoric strain field, for non-local regularization or for internal variables requiring their own discrete representation: it introduces $\bar{\boldsymbol{e}}$ and $\bar{\boldsymbol{s}}$ in the same discontinuous space, of the same degree as Q_h. Sharing the space keeps the return map at shared quadrature points, keeps every elimination element-local, leaves the discrete system non-singular, and — for a linear deviatoric law, and only then — makes the constitutive relation hold pointwise rather than after a projection; with plasticity $\bar{\boldsymbol{s}} = \mathcal{P}_h\, \boldsymbol{s}(\bar{\boldsymbol{e}})$ is projected. Placing $\bar{\boldsymbol{e}}$ and $\bar{\boldsymbol{s}}$ in different spaces loses each of these properties. Remark 2 also applies: with an independent $\bar{\boldsymbol{s}}$, the residual must be assembled from (36) and not from (37).

9 Numerical results

Every measurement below is on a linear operator, and all but those of §9.6 are at $\boldsymbol{F} = \boldsymbol{I}$: the condensed operator (52) for the count and the locking sweep, the H^1 seminorm for the stability sweep, and the deviatoric form under a plastic tangent for the soft-mode measurements. None needs a solver, a material model, or a mesh reader. The scripts are in `prototype/`, each reports its own result, and `checks.jl` verifies the quadrature, the bubbles, and the conformity of the enriched spaces against values known in advance.

9.1 Method

Meshes are generated rather than read, so the refinement sequence is exact. The unit cube is divided into N^3 cubes and each cube into six tetrahedra by the Freudenthal (Kuhn) subdivision — the six permutations (a, b, c) giving the Kuhn tetrahedra $\{\mathbf{0}, \boldsymbol{e}_a, \boldsymbol{e}_a + \boldsymbol{e}_b, (1,1,1)\}$. That decomposition is conforming across shared cube faces, which the five-tetrahedron split is not. Quadratic meshes add edge midpoints in the node order the reference element expects.

For the stability test the displacement is fixed on the entire boundary and three operators are assembled,

$$K_{ij} = \int \operatorname{Grad} N_i : \operatorname{Grad} N_j \,\mathrm{d}V, \qquad G_{mi} = \int \chi_m \operatorname{Div} N_i \,\mathrm{d}V, \qquad M_{mn} = \int \chi_m \chi_n \,\mathrm{d}V, \tag{50}$$

with N_i the displacement basis functions, χ_m the pressure basis functions, and K the H^1 seminorm, the norm on the displacement in (38); $\ker \boldsymbol{G}$ is the discretely isochoric subspace, the set of displacements whose divergence has zero projection onto Q_h. The discrete inf-sup constant is $\beta_h = \sqrt{\lambda_{\min}}$ over the nonzero spectrum of

$$\boldsymbol{G}\boldsymbol{K}^{-1}\boldsymbol{G}^{\mathsf{T}}\, q = \lambda \boldsymbol{M} q. \tag{51}$$

Boundary data fixed everywhere always leaves one zero mode, the hydrostatic one, because $\int \operatorname{Div} \boldsymbol{v} = 0$ for $\boldsymbol{v} \in H^1_0$; any zero beyond that is a spurious pressure mode.

9.2 Spurious modes on one element

For linear response the count of zero-energy modes is a rank count on a single element. Condensing the three fields gives

$$\boldsymbol{K}_e = \boldsymbol{K}^{\mathrm{dev}} + \boldsymbol{K}^{\mathrm{vol}}, \qquad \boldsymbol{K}^{\mathrm{vol}} := \kappa\, \boldsymbol{G}^{\mathsf{T}} \boldsymbol{M}^{-1} \boldsymbol{G}, \qquad M_{mn} = \int_e \chi_m \chi_n \,\mathrm{d}V, \qquad G_{mi} = \int_e \chi_m \left(\operatorname{tr} \boldsymbol{B}\right)_i \,\mathrm{d}V, \tag{52}$$

with $\boldsymbol{B}$ the strain–displacement operator of the element and $m = \dim Q_h$ on one element. $\boldsymbol{K}^{\mathrm{dev}}$ annihilates every mode with zero deviatoric strain at quadrature: the six rigid-body modes, plus the volumetric directions the displacement space can produce. The second term has rank equal to the number of those directions the pressure space resists, at most m. Anything left over carries neither deviatoric nor volumetric energy and is exactly zero-energy:

$$n_{\mathrm{spurious}} = \dim\ker \boldsymbol{K}^{\mathrm{dev}} - 6 - \operatorname{rank} \boldsymbol{K}^{\mathrm{vol}}. \tag{53}$$

For the quadratic tetrahedron $\dim\ker \boldsymbol{K}^{\mathrm{dev}} = 10$: six rigid modes and four volumetric directions, the latter because $\operatorname{tr} \boldsymbol{\epsilon}(\boldsymbol{\varphi})$ is a P_1 field. Table 5 reports (53) against the measured spectrum.

Pressure space	m	$\dim\ker \boldsymbol{K}^{\mathrm{dev}}$	$\operatorname{rank} \boldsymbol{K}^{\mathrm{vol}}$	predicted	observed
P_0 constant	1	10	1	**3**	**3**
P_1 discontinuous	4	10	4	0	0
P_2 discontinuous	10	10	4	0	0

Table 5: Spurious zero-energy modes of the three-field quadratic tetrahedron at $\kappa/\mu = 10^4$, on the affine element. Script: `prototype/softmode.jl`.

Four consequences follow.

The constant-pressure element carries three modes that are exactly zero-energy on one element, not merely soft. Whether they persist after assembly is the question of §9.4.

A P_1-discontinuous pressure removes them without an added energy term and leaves the six rigid-body modes. That is an element-level fact: §9.4 and §9.5 show that the same pair locks and is unstable once assembled.

Enriching the pressure further changes nothing: the rank saturates at four, because four is all the displacement space can produce. Enriching the pressure beyond P_1^{disc} adds unknowns and leaves the rank unchanged.

The count is a property of the affine element. On a curved element, with the midside nodes displaced so that $\det \boldsymbol{J}$ ranges over $[0.55, 1.19]$, the isoparametric displacement space is not polynomial in $\boldsymbol{x}$, only the dilatation of its four volumetric directions remains free of deviatoric strain, $\dim\ker \boldsymbol{K}^{\mathrm{dev}} = 7$, and the three constant-pressure modes are soft rather than exactly zero: the softest nonzero mode of the constant-pressure element lies one to two orders of magnitude below the softest mode of the linear-pressure element on the same curved element. A four-point rule samples a curved element at four points and reports the affine counts there as well; the count uses a 27-point rule. The assembled behavior of §9.4 rests on soft rather than exact modes.

Remark 6. This identifies the mechanism. Inf-sup stability is a global property of a mesh sequence and is invisible on one element; the rank argument above says which modes exist, not whether the constant in the stability estimate persists under refinement; §9.5 measures it.

9.3 A necessary condition from degree-of-freedom counts

Whether the constant in the stability estimate persists under refinement is a property of a mesh sequence, and a necessary condition for it follows from counting.

Write n_u for the number of free displacement degrees of freedom and n_p for the pressure degrees of freedom. The Schur complement of the mixed system is $\boldsymbol{S} = \boldsymbol{G}\boldsymbol{K}^{-1}\boldsymbol{G}^{\mathsf{T}}$, an $n_p \times n_p$ matrix of rank at most n_u. If $n_p > n_u$ then $\boldsymbol{S}$ is singular by counting, the pressure space carries modes no displacement can resist, and no inf-sup constant exists — regardless of the values those constants would otherwise take.

Table 6 counts them on a Freudenthal-subdivided cube.

Pair	N	n_u	n_p	n_p/n_u
P_2/P_0	12	36,501	10,368	0.284
P_2/P_1^{disc}	12	36,501	41,472	**1.136**
$P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$	12	67,605	41,472	0.613
$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	12	127,221	41,472	0.326

Table 6: Degree-of-freedom counts on the unit cube, displacement fixed on the boundary. Script: `prototype/infsup.jl`.

For P_2/P_1^{disc} the ratio exceeds one, and its limit is computable. The quadratic nodes of this mesh family fill the $(2N+1)^3$ grid, so with the whole boundary fixed $n_u = 3(2N-1)^3 \to 24N^3$, while $n_p = 4 \cdot 6N^3 = 24N^3$ regardless of boundary data. The ratio tends to one: the pair is critically determined in the limit, which excludes a uniform inf-sup constant.

The finite-N ratio depends on the boundary condition: with the whole boundary fixed, $n_u = 3(2N-1)^3$ and the ratio exceeds one at every N; with one face fixed, $n_u = 6N(2N+1)^2$ and it is below one at every N. Both tend to one.

Proposition 1. *On a tetrahedral mesh family with at least six tetrahedra per vertex asymptotically — the Freudenthal family among them — and the boundary fixed, P_2/P_1^{disc} is not inf-sup stable, and refinement does not restore stability.*

The restriction to the mesh family is necessary. On a general tetrahedral mesh with n_V vertices, n_E edges and n_T tetrahedra, Euler's relation for a simply connected three-dimensional mesh gives $n_E \approx n_V + n_T$, so $n_u \approx 3(2n_V + n_T)$, $n_p = 4n_T$, and $n_p/n_u \to 4/(3(2n_V/n_T + 1))$, which exceeds one only when $n_T/n_V > 6$. Below that the count refutes nothing, and on other mesh families the pair can be stable: Zhang [35] shows P_2/P_1^{disc} stable on Powell–Sabin tetrahedral meshes once their known spurious pressure modes are filtered. The general three-dimensional result is Boffi et al. [7, Remark 8.6.2]: with a P_k displacement, a discontinuous pressure is stable on general meshes only at degree $k-3$, so for $k=2$ nothing is admissible without enrichment.

Adding one interior bubble per element supplies $3n_{\text{el}}$ further displacement degrees of freedom and moves the ratio to 0.613; adding the four face bubbles as well — the enrichment §4.1 calls for — supplies $3n_{\text{f}}$ more, one vector unknown per interior face, and moves it to 0.326. Both satisfy $n_p < n_u$. The enrichment also fixes the number of global unknowns. The interior bubble condenses; the face bubbles are shared and do not. The global displacement count of the full enrichment is therefore 36,501 + 59,616 = 96,117, which is 2.6 times the 36,501 of P_2/P_0 — also the nodal count of CT10 on the same `TETRA10` mesh — and 2.5 times the 38,698 global unknowns of Taylor–Hood, whose 2,197 vertex pressures cannot be eliminated. The 127,221 in Table 6 is the dimension of the space, not the number of global unknowns.

The count has precedents. n_p/n_u is the constraint count of Nagtegaal et al. [23], whose subject — locking of simplicial elements in the fully plastic range — is this note's, and its reciprocal $\varrho = n_u/n_p$ is the constraint ratio of Hughes [18, §4.3.7], with $\varrho = 3$ the optimum in three dimensions. At $N = 12$: $\varrho = 3.52$ for P_2/P_0, 0.88 for P_2/P_1^{disc}, 1.63 with the interior bubble, and 3.07 for the full enrichment. The stable pair has the optimal value, and the isochoric fraction $1 - n_p/n_u$ of §9.4 is $1 - 1/\varrho$, with 2/3 the optimum.

Whether that limiting degeneracy is the decisive objection on the meshes tested is a separate question, which §9.4 answers.

9.4 Assembled behavior

The element count of §9.2 is a local statement. An element-local mode persists after assembly only if its trace is compatible with every neighbor across every shared face, so the global count can be smaller than $3n_{\text{el}}$, and it is what the argument against a constant pressure rests on.

Table 7 assembles $\boldsymbol{K}^{\text{dev}} + \kappa \boldsymbol{G}^{\mathsf{T}} \boldsymbol{M}^{-1} \boldsymbol{G}$ over meshes under two boundary conditions: the whole boundary fixed, and the base $z = 0$ alone fixed. The second is closer to the conditions of a body confined on part of its boundary and free elsewhere, under which soft modes are reported.

The locking measure is again a rank. A displacement mode has volumetric energy exactly when it lies outside $\ker \boldsymbol{G}$, so as $\kappa \to \infty$ the reachable deformations collapse onto that kernel and $\dim \ker \boldsymbol{G} = n_u - \operatorname{rank} \boldsymbol{G}$ is the discretely isochoric subspace. A large fraction means the element still deforms; a collapse toward zero is volumetric locking.

The fraction is compared with a bound. Since $\operatorname{rank} \boldsymbol{G} \le n_p$, it cannot fall below $1 - n_p/n_u$. A pair that attains that bound has $\operatorname{rank} \boldsymbol{G} = n_p$: every pressure mode is resisted by a distinct displacement direction. A fraction above the bound means that $n_p - \operatorname{rank} \boldsymbol{G}$ pressure modes are unresisted; a pair that cannot reach it, because $n_p > n_u$, is over-determined. The difference $n_p - \operatorname{rank} \boldsymbol{G}$ counts the pressure modes no displacement resists, which with the whole boundary fixed should be exactly one, the hydrostatic mode; Table 8 carries it as a column.

Three findings follow.

The constant-pressure soft modes do not persist after assembly. No spurious mode is present under either boundary condition at any mesh. Nor are they present as low-energy modes: the smallest eigenvalue relative to the deviatoric scale is 2.39×10^{-3} for P_0 against 2.74×10^{-3} for P_1^{disc} on the same mesh with the base alone fixed, and both are the long-wavelength modes of that block. The three modes per element of §9.2 exist on each element and are element-local: their traces do not match across faces.

What this test does not detect is the mode Foulk III et al. [16] report, which is remedied there by a penalty and therefore has energy, small and not zero. Those are the displacement modes in $\ker \boldsymbol{G}_{P_0}$ whose divergence is nonzero pointwise, which carry shear energy alone in the element; whether they are spurious is the coercivity question of §6, which §9.7 measures, and not a count of zeros. CT10 of Foulk III et al. [16] is built on a subdivided-linear displacement space rather than P_2, which is a different kernel and is not modeled here; the soft mode of that element may be specific to it. Nothing above contradicts their observation; it does not reproduce it on a quadratic tetrahedron.

boundary	Pair	N	n_u	n_p	rank $\boldsymbol{G}$	fraction	bound
whole	P_2/P_0	3	375	162	158	0.579	0.568
		4	1029	384	380	0.631	0.627
	P_2/P_1^{disc}	3	375	648	363	**0.032**	—
		4	1029	1536	957	**0.070**	—
	$\oplus\, b_{\mathrm{int}}$	3	861	648	644	0.252	0.247
		4	2181	1536	1532	0.298	0.296
	$\oplus\, b_{\mathrm{face}}$	3	1185	648	646	0.455	0.453
		4	3045	1536	1534	0.496	0.496
	$\oplus\, b_{\mathrm{int}}, b_{\mathrm{face}}$	3	1671	648	647	**0.613**	0.612
		4	4197	1536	1535	**0.634**	0.634
	$P_2 \oplus b_{\mathrm{face}}/P_0$	3	1185	162	161	0.864	0.863
		4	3045	384	383	0.874	0.874
base	P_2/P_0	3	882	162	162	0.816	0.816
		4	1944	384	384	0.802	0.802
	P_2/P_1^{disc}	3	882	648	522	0.408	0.265
		4	1944	1536	1216	0.374	0.210
	$\oplus\, b_{\mathrm{int}}$	3	1368	648	648	0.526	0.526
		4	3096	1536	1536	0.504	0.504
	$\oplus\, b_{\mathrm{face}}$	3	1962	648	648	0.670	0.670
		4	4440	1536	1536	0.654	0.654
	$\oplus\, b_{\mathrm{int}}, b_{\mathrm{face}}$	3	2448	648	648	0.735	0.735
		4	5592	1536	1536	0.725	0.725
	$P_2 \oplus b_{\mathrm{face}}/P_0$	3	1962	162	162	0.917	0.917
		4	4440	384	384	0.914	0.914

Table 7: Assembled count at $\kappa/\mu = 10^4$. The bound is $1 - n_p/n_u$ and is undefined where $n_p > n_u$. Assembled zero-energy modes were zero in every row. Script: `prototype/locking.jl`.

The unenriched pair locks. With the boundary fixed, P_2/P_1^{disc} retains 3–7% of its deformation modes as κ grows, against 58–63% for P_0; with the base alone fixed, 37–41% against 80%. Under either boundary condition it is the pair that over-constrains, and locking is the more serious failure because the element returns a stiff response and no spurious mode indicates the error.

The interior bubble raises the rank of $\boldsymbol{G}$ over a linear pressure. As Remark 7 states, the interior bubble adds nothing to the rank over P_0. Against the linear pressure modes its columns are not zero: it raises rank $\boldsymbol{G}$ from 957 to 1532 at $N = 4$, and the interior-only pair has an isochoric fraction of 0.252–0.298. The full enrichment reaches 0.61–0.63, equal to P_0's to two digits and without P_0's element-level degeneracy; it attains the bound $1 - n_p/n_u$ to three digits and the interior-only pair to two, so every pressure mode is resisted by a distinct displacement direction; and for the full pair $n_p - \operatorname{rank} \boldsymbol{G} = 1$ at every mesh, the hydrostatic mode alone, where P_2/P_0 and the interior-only pair carry four: the enrichment raises rank $\boldsymbol{G}$ to $n_p - 1$.

9.5 Stability, measured against a validated control

The count of §9.3 is a necessary condition; Table 8 shows that it is not sufficient.

Two properties of the computation allow meshes to $N = 8$. With $\boldsymbol{K} = \boldsymbol{P}^{\mathsf{T}}\boldsymbol{L}\boldsymbol{L}^{\mathsf{T}}\boldsymbol{P}$ from a sparse Cholesky, $q^{\mathsf{T}}\boldsymbol{G}\boldsymbol{K}^{-1}\boldsymbol{G}^{\mathsf{T}}q = \|\boldsymbol{L}^{-1}(\boldsymbol{G}^{\mathsf{T}}q)_{[p]}\|^2$, so β_h is the smallest nonzero singular value of $\boldsymbol{W} = \boldsymbol{L}^{-1}\boldsymbol{G}^{\mathsf{T}}\boldsymbol{M}^{-1/2}$ — never forming $\boldsymbol{G}\boldsymbol{K}^{-1}\boldsymbol{G}^{\mathsf{T}}$, and so never squaring the condition number of the quantity measured. And the null dimension is fixed independently as $n_p - \operatorname{rank}\boldsymbol{G}$, so the cut is taken by index rather than by selecting a threshold. The gap is unambiguous: $O(10^{-15})$ below it against $O(10^{-2})$ above.

The sweep carries a positive control, the Taylor–Hood pair, whose stability in three dimensions is established [7]: a valid test must report it stable. A negative control alone cannot detect a test that reports every pair unstable; the negative control, P_1/P_0, gives $\beta_h = 0.333$, 0.229, 0.165, 0.129 at $N = 2$ to 5 with 45, 138, 303, 558 unresisted pressure modes, the alternating-sign spurious pressure mode of that pair (script `prototype/infsup.jl`).

Pair	$N=2$	3	4	5	6	7	8	null	rate
P_2/P_1 continuous (control)	0.1734	0.2096	0.2186	0.2206	0.2214	0.2215	**0.2216**	1	−0.002
P_2/P_0	0.1001	0.0926	0.0755	0.0633	0.0544	0.0476	0.0418	4	0.97
$\oplus\, b_{\text{int}}$	0.1015	0.0927	0.0756	0.0634	0.0544	—	—	4	0.84
$\oplus\, b_{\text{face}}$	0.4671	0.4383	0.4228	0.4129	0.4059	0.4007	**0.3967**	1	0.076
P_2/P_1^{disc}	0.1132	0.0572	0.0424	0.0335	0.0277	—	—	$n_p - n_u$	1.04
$\oplus\, b_{\text{int}}$	0.0782	0.0692	0.0564	0.0473	0.0406	—	—	4	0.83
$\oplus\, b_{\text{face}}$	0.0938	0.0858	0.0701	0.0583	0.0497	—	—	2	0.88
$\oplus\, b_{\text{int}}, b_{\text{face}}$	0.2875	0.2945	0.2962	0.2967	**0.2968**	—	—	1	−0.003

Table 8: β_h over the Freudenthal mesh sequence of the unit cube with N^3 cubes, whole boundary fixed, ordered as the construction of Boffi et al. [7, Example 8.7.2] runs. "null" is $n_p - \operatorname{rank}\boldsymbol{G}$, the pressure modes no displacement resists, which must be one; "rate" is the local rate $\ln(\beta_N/\beta_{N+1})/\ln((N+1)/N)$ over the last two columns computed, with β_N the constant at refinement N, constant for a power law and decreasing toward zero for a sequence with a positive limit. The rates of an earlier version of this table were taken over refinements not shown in it; every N computed is now shown. Script: `prototype/beta.jl`.

The control is reported stable: β_h rises to 0.2216, and its last three values agree to within 2×10^{-4}, which validates the reading of the remaining rows.

Every row of Table 8 is what Boffi et al. [7] predict. What the sweep establishes is that the scripts are sound and that the theory applies to this mesh family; what it adds to the theory is the constants. The rows are ordered as the construction of Boffi et al. [7, Example 8.7.2] adds degrees of freedom.

P_2 does not control a constant pressure in three dimensions. P_2/P_0 decays at a local rate approaching one, although $n_p/n_u = 0.284$, and carries four pressure null modes at every mesh where one is allowed: the hydrostatic mode and three exact spurious modes that no quadratic displacement resists. That is the statement of Boffi et al. [7, §8.7] quoted in §4.1, and the three modes are the pressure modes that face degrees of freedom would resist.

The interior bubble cannot supply it. $P_2 \oplus b_{\text{int}}/P_0$ reproduces P_2/P_0 to three digits with the same four null modes, for the reason in Remark 7. Over P_1^{disc} the interior bubble changes β_h from 0.0424 to 0.0564 at $h = 1/4$ and from 0.0277 to 0.0406 at $h = 1/6$, and the rate falls from 1.04 to 0.83, but the sequence decays and four null modes remain. This pair has $n_p/n_u = 0.613$ (Table 6) and is unstable.

Face bubbles control the constant pressure. $P_2 \oplus b_{\text{face}}/P_0$ drops the null dimension to one at every mesh and its β_h approaches from above with a local rate that falls from 0.16 to 0.076 over the sequence, below the constant rate of a power law and above the rate -0.002 of the control. Its displacement space contains that of the pair of Bernardi and Raugel [6], whose stability implies its own. The seven refinements measured, of which Table 8 shows four, cannot separate a limit near 0.35 from a slow decay on their own, and the row is reported as consistent with stable; its stability rests on Boffi et al. [7, Example 8.7.2], whose proof constructs, from the face degrees of freedom, a displacement that matches the divergence of any given field in the mean on every face. Over P_1^{disc} the face bubbles alone leave one spurious linear pressure mode per mesh and β_h decays: controlling the linear modes is the function of the interior bubble.

Both together are the three-dimensional Crouzeix–Raviart pair, and it is stable. β_h rises monotonically to 0.2968; its last three values agree to within 7×10^{-4}, and its local rate decreases in magnitude from 0.060 to 0.003, as does that of the control, at a higher constant than the control's 0.2216. The pressure null space is one dimensional at every mesh. Each part of the enrichment controls one part of the pressure space, and neither controls the other's.

Remark 7 (The distinct functions of the two enrichments)**.** The interior bubble has zero pairing with a constant pressure: it vanishes on the whole element boundary, so $\int_{\Omega_e} \operatorname{Div} \boldsymbol{b}_{\text{int}} = 0$ and its columns of $\boldsymbol{G}$ against a P_0 mode are identically zero. It pairs only with the linear pressure modes, which is why it changes P_2/P_1^{disc} and changes the P_2/P_0 constant by at most 0.0014 (the $P_2 \oplus b_{\text{int}}/P_0$ row of Table 8, with the same null dimension): upward, because enlarging U_h cannot lower β_h, and by at most 0.0014, because the bubble's columns of $\boldsymbol{G}$ against P_0 are zero. `checks.jl` reads those columns directly and confirms it. The face bubbles are what supply a displacement response to the pressure jump across a face, which is the mode a discontinuous pressure needs resisted.

P_2/P_0 and $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$ have $n_p/n_u = 0.284$ and 0.613 and are unstable: a ratio above one excludes a pair, and a ratio below one does not admit it.

Remark 8 (Mesh dependence)**.** These results are for the Freudenthal subdivision, a structured family. A pair can degenerate on a structured mesh and remain well behaved on a general one. The paragraph below repeats the measurement on two families without that structure. It shows that the three additional null modes and the decay of P_2/P_0 are properties of the Freudenthal family, and that the constant of the Crouzeix–Raviart pair does not depend on the structure. The affirmative result needs no such confirmation, since Boffi et al. [7, Example 8.7.2] establish it on any shape-regular family; what the measurement adds is the constant.

Meshes without the Freudenthal structure. Two families are measured. In the first, every interior vertex of the Freudenthal mesh is displaced by a random vector with independent components uniform in $[-\delta h, \delta h]$, $h = 1/N$; the boundary vertices are kept, and the edge nodes

Table 9: β_h on meshes without the Freudenthal structure, whole boundary fixed. Perturbed: the Freudenthal mesh with interior vertices moved at random by up to $0.25h$ per component, smallest and largest value over three seeds. Cubit: unstructured meshes of size $1/n$. The null dimension is given in parentheses where it differs from one.

mesh	pair	$N, n = 2$	3	4	5	6
perturbed	P_2/P_1 cont.	0.171–0.173	0.206–0.207	0.214–0.216	0.214–0.218	0.217–0.218
	P_2/P_0	0.104–0.106 (4)	0.043–0.056	0.061–0.068	0.071–0.074	0.071–0.076
	$P_2 \oplus b_{\text{face}}/P_0$	0.466–0.467	0.438	0.423	0.413	0.406–0.407
	TET15-P1	0.287	0.290–0.294	0.289–0.295	0.293–0.294	0.291–0.295
Cubit	P_2/P_1 cont.	0.212 (4)	0.327 (7)	0.328 (6)	0.328 (6)	0.328 (5)
	P_2/P_0	0.076 (4)	0.082 (7)	0.068 (6)	0.099 (6)	0.101 (5)
	$P_2 \oplus b_{\text{face}}/P_0$	0.434	0.429	0.416	0.407	0.403
	TET15-P1	0.233	0.319	0.330	0.331	0.332

are placed at the midpoints, so every element is straight-sided. The shape quality of an element is $q = 6\sqrt{2}\,V/\ell_{\text{rms}}^3$, with V its volume and ℓ_{rms} the root mean square of its six edge lengths; $q = 1$ for a regular tetrahedron and 0.66 for the Freudenthal elements. At $\delta = 0.25$ the smallest q over the three seeds of each mesh is 0.49, 0.13, 0.003, 0.17 and 0.096 at $N = 2$ to 6. The second family consists of unstructured meshes of the unit cube generated by Cubit with mesh size $1/n$ and the condition-number smoother of the punch meshes of §12, with straight edges; their smallest q is between 0.51 and 0.58. Table 9 gives β_h. Scripts: `prototype/jitter.jl` and `prototype/cubit_cube.jl`.

On both families the Crouzeix–Raviart pair (TET15-P1 in the table) has one null mode. Its β_h lies between 0.287 and 0.295 on the perturbed meshes, and between 0.319 and 0.332 on the Cubit meshes from $n = 3$ on. $P_2 \oplus b_{\text{face}}/P_0$ also has one null mode on both. On the perturbed meshes P_2/P_0 has a single null mode from $N = 3$ on, and its β_h does not decrease: the three additional null modes and the decay of P_2/P_0 in Table 8 are properties of the Freudenthal family. On the Cubit meshes, Taylor–Hood and P_2/P_0 have, besides the constant mode, one null mode for each element whose ten nodes all lie on the fixed boundary. There are between three and six such elements, at the corners of the cube. The face bubbles of the other two pairs control the pressure of these elements.

9.6 Stability on a deformed configuration

Every measurement above is at $\boldsymbol{F} = \boldsymbol{I}$. At a finite deformation the pairing is $\int \varpi \,\mathrm{div}_{\boldsymbol{x}} \boldsymbol{\psi}\, \mathrm{d}V$ (§4.1), the divergence pairing on the deformed configuration with the pressure scaled by $1/J$, and whether β_h stays bounded below under a large non-affine deformation is what the regime of §1 requires. Auricchio et al. [4] show, on a two-dimensional finite-strain problem with a loading parameter, that mixed elements stable in linear elasticity (the MINI triangle, a mixed-enhanced quadrilateral and the quadrilateral Q_2/P_1^{disc}) fail to reproduce the range of the parameter over which the continuum problem is stable: the discrete problem becomes unstable where the continuum one is stable, or stays stable where it is not; and Auricchio et al. [5] conclude that for such pairs in finite elasticity the causes are identified but a general remedy is still missing. Their analysis concerns the full linearized operator, with the material and geometric stiffness on the discrete kernel. This section measures a part of that question only, the constant of the divergence pairing on deformed configurations. The stability of the full tangent at deformed

Table 10: β_h at $N = 5$ on deformed configurations, whole boundary fixed, with the local rate over $N = 4 \to 5$ in parentheses for the Crouzeix–Raviart pair; the null dimension is one on every row for the three left-hand pairs. The P_2/P_0 column gives its null dimension and β_h: its three exact spurious modes exist only on the affine mesh.

map	J	P_2/P_1 cont.	$P_2 \oplus b_{\text{face}}/P_0$	$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	P_2/P_0 (null, β_h)
reference	1	0.221	0.413	0.297 (−0.003)	4, 0.063
twist 90°	0.99–1.01	0.189	0.385	0.204 (0.18)	1, 0.027
twist 180°	0.98–1.05	0.137	0.345	0.116 (0.42)	1, 0.051
parabolic shear	1	0.176	0.336	0.183 (0.05)	1, 0.005
inflation	1.0–2.3	0.192	0.368	0.276 (0.01)	1, 0.004
compression	0.5	0.192	0.333	0.213 (0.00)	4, 0.038

and stressed states is not measured here and remains open. The operators (50) are assembled on the isoparametric image of the quadratic mesh under a prescribed map, with the derivatives, the divergence and the volume taken on the image and the pressure basis on the reference cell as the element uses it, and with the boundary conditions of the reference cube; the scaling by $1/J$ changes β_h by at most the ratio of extreme Jacobians and is not applied. Five maps of the unit cube, about its center: a twist about the vertical axis by 90° and by 180° over the height, isochoric and non-affine, with shear strains near 2 at the corners of the latter; a parabolic shear $x_1 = X_1 + \frac{1}{2}X_3^2$, isochoric; a radial inflation with J from 1 to 2.3; and an affine compression to $J = \frac{1}{2}$ as a control, since an affine map changes β_h only through the element aspect ratio. Table 10 gives the result at $N = 5$ with the local rate over the last refinement. Script: `prototype/deformed.jl`.

Three findings follow.

β_h is not uniform in the deformation. It decreases under the non-affine maps for every pair, and most for the Crouzeix–Raviart pair: from 0.297 to 0.204 under the 90° twist, 0.116 under the 180° twist and 0.183 under the parabolic shear, against 0.221 → 0.189, 0.137, 0.176 for Taylor–Hood and 0.413 → 0.385, 0.345, 0.336 for the face-bubbled constant-pressure pair. The inflation, which changes J by a factor of 2.3 but shears little, leaves the constant at 0.276. The affine compression, uniaxial with $x_3 = \frac{1}{2}X_3$, gives 0.213, flat to four digits from $N = 3$. A uniform scaling of the mesh leaves β_h unchanged, so the change under the compression is due to the change of shape of the elements. The data do not separate two explanations of the decrease under the non-affine maps: the shear of the deformation, and the loss of shape regularity of the image elements on coarse meshes. The volume change alone does not explain it.

At a fixed deformation the sequence in h is consistent with a positive limit. The local rate of the Crouzeix–Raviart pair falls with N on every non-affine map: 0.28, 0.22, 0.18 over $N = 2$ to 5 for the 90° twist, 0.67, 0.53, 0.42 for the 180° twist, and 0.05 for the parabolic shear (through $N = 5$). A power-law decay would hold its rate. The 180° row has not flattened by $N = 5$, and its shear strain of about 2 at the corners is beyond what a plastic zone of the problems listed in §13 reaches; the finding is that the constant degrades with the strain, not that it vanishes. At $N = 6$ the pair gives 0.1975, 0.1088 and 0.182 on the three non-affine maps, local rates of 0.17, 0.35 and 0.04, each below the rate before it.

The exact spurious modes of P_2/P_0 belong to the affine mesh. Under every non-affine map the pressure null dimension of P_2/P_0 falls from four to one, the three modes are resisted, and β_h of P_2/P_0 is smaller on every non-affine image, 0.004 to 0.05, than on the reference mesh, 0.063.

9.7 Coercivity under a plastic tangent

The claim of §6 is measured as the constant it names. With $\boldsymbol{Z}$ a basis of $\ker \boldsymbol{G}_h$,

$$\lambda_{\min} = \min_{q} \frac{q^{\mathsf{T}}\, \boldsymbol{Z}^{\mathsf{T}} \boldsymbol{K}^{\mathrm{ep}} \boldsymbol{Z}\, q}{q^{\mathsf{T}}\, \boldsymbol{Z}^{\mathsf{T}} \boldsymbol{K}^{H^1} \boldsymbol{Z}\, q}, \qquad \boldsymbol{K}^{\mathrm{ep}}_{ij} = \frac{1}{2\mu} \int \boldsymbol{\epsilon}_i : \mathbb{C}^{\mathrm{ep}} : \boldsymbol{\epsilon}_j \,\mathrm{d}V, \tag{54}$$

is the smallest generalized eigenvalue of the deviatoric form under the tangent (48) against the H^1 seminorm, restricted to the discretely isochoric subspace. It is dimensionless, it is taken on the exact kernel rather than through a penalty, and it is the constant of the second hypothesis; (49) says what to expect of it, at least $(1-\eta)/2$ for every pair when $\eta < 1$ and a decay to zero for every pair when $\eta = 1$. The quantity that carries the claim is the dilatation of the eigenmodes,

$$r_v(\boldsymbol{v}) := \frac{\int (\mathrm{Div}\, \boldsymbol{v})^2 \,\mathrm{d}V}{\int |\mathrm{Grad}\, \boldsymbol{v}|^2 \,\mathrm{d}V}, \tag{55}$$

zero for an isochoric field. The flow direction is uniform, $\boldsymbol{n} = \mathrm{sym}(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2)/\sqrt{2}$ or $\boldsymbol{n} = \mathrm{dev}(\boldsymbol{e}_3 \otimes \boldsymbol{e}_3)/\,\|\cdot\|$, which is the most unfavorable choice: the continuum operator then admits shear bands of any width, a decaying eigenvalue is expected of every pair, and only the dilatation of the mode says whether the band is physical. The tangent is taken at $\eta = 1$ and at $\eta = 0.9$, the boundary is fixed everywhere, and each pair is measured elastically ($\eta = 0$) on the same kernel. Up to $N = 5$ the eigenpairs are computed on an explicit basis of $\ker \boldsymbol{G}_h$. For $N \geq 6$, where that basis does not fit in memory, they are computed by subspace iteration on the inverse of $\boldsymbol{K}^{\mathrm{ep}} + \gamma\, \boldsymbol{G}^{\mathsf{T}} \boldsymbol{D}^{-1} \boldsymbol{G}$, with $\boldsymbol{D}$ the diagonal of $\boldsymbol{M}$ and $\gamma = 10^9$. The eigenpairs of this penalized operator converge to those on $\ker \boldsymbol{G}_h$ as γ grows, with a violation of the constraint of order $1/\gamma$. At $N = 4$ the two computations agree to three digits for all four pairs. At $N = 6$ and 8, for TET15-P1 and P_2/P_0 under the uniaxial direction, $\gamma = 10^8$, 10^9 and 10^{10} give λ_1, λ_5 and r_v of the lowest mode that agree to six digits. Scripts: `prototype/plastic.jl`, `prototype/plastic_ext.jl` and `prototype/gamma_check.jl`.

The scripts are checked by an identity. By the identity behind (49), the elastic form on any kernel satisfies $\lambda = \frac{1}{2} + \frac{1}{6} r_v$, so its lowest mode has $r_v = 0$ and $\lambda_1 = 0.5$ exactly. Every row of the sweep reproduces this; at $N = 2$ for P_2/P_0, where the kernel has no isochoric member, $\lambda_1 = 0.5024$ with $r_v = 0.0146$, as the identity requires. The identity is also the check on the energy metric of the scripts.

The constant does not separate pairs, as (49) predicts. Under the shear direction λ_1 decays for every pair, the control included, at rates between $h^{1.5}$ and h^2: the discrete spaces resolve narrower shear bands, which the continuum operator admits with zero energy. Under the uniaxial direction no shear band has $\boldsymbol{\epsilon} \propto \boldsymbol{n}$, and λ_1 converges for the three element-local pairs, to 0.10 for P_2/P_0, 0.071 for $P_2 \oplus b_{\mathrm{face}}/P_0$ and 0.13 for the Crouzeix–Raviart pair, while it decays for Taylor–Hood. At $\eta = 0.9$ every λ_1 lies above the lower bound 0.05 of (49), the smallest being the control's 0.067.

Table 11: Lowest mode of the deviatoric form on $\ker \boldsymbol{G}_h$ under the tangent (48) at $\eta = 1$, Freudenthal cube with the whole boundary fixed: the eigenvalue λ_1 of (54) and the dilatation r_v of its mode, (55), for a shear and a uniaxial flow direction. The elastic lowest mode has $\lambda_1 = 0.500$ and $r_v = 0$ on every row, as the identity behind (49) requires (computed for $N \le 5$). At $\eta = 0.9$ every λ_1 exceeds the lower bound 0.05 of (49) and the dilatations agree with those below to two digits. †: the subspace iteration did not reach its tolerance of 10^{-7} in 3000 iterations. Scripts: `prototype/plastic.jl` and `prototype/plastic_ext.jl`.

		$\boldsymbol{n} = \mathrm{sym}(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2)/\sqrt{2}$		$\boldsymbol{n} = \mathrm{dev}(\boldsymbol{e}_3 \otimes \boldsymbol{e}_3)/\,\|\cdot\|$	
Pair	N	λ_1	r_v	λ_1	r_v
P_2/P_0	3	0.109	0.065	0.117	0.208
	4	0.067	0.034	0.106	0.217
	5	0.047	0.029	0.101	0.220
	6	0.035	0.020	0.098	0.220
	7	0.026	0.016	0.097	0.220
	8	0.020	0.012	0.096	**0.220**
$P_2 \oplus b_{\mathrm{face}}/P_0$	3	0.044	0.044	0.082	0.274
	4	0.026	0.028	0.074	0.268
	5	0.017	0.019	0.071	0.261
	6	0.012	0.014	0.069	0.257
	7	0.0084	0.010	0.069	**0.254**
$P_2 \oplus b_{\mathrm{int}} \oplus b_{\mathrm{face}}/P_1^{\mathrm{disc}}$	3	0.074	0.036	0.147	0.018
	4	0.049	0.019	0.139	0.0074
	5	0.032	0.0066	0.134	0.0042
	6	0.022	0.0039	0.131	0.0027
	7	0.016	0.0026	0.130	0.0019
	8	0.012	0.0018	0.129	**0.0014**
P_2/P_1 continuous (control)	3	0.056	0.051	0.053	0.861
	4	0.030	0.029	0.026	0.929
	5	0.019	0.020	0.016	0.956
	6	0.013	0.013	0.011	0.970
	7	0.0093	0.014	$0.0071^\dagger$	$0.875^\dagger$
	8	0.0070	0.010	$0.0044^\dagger$	$0.895^\dagger$

The dilatation of the soft modes separates them, in the direction claimed. Under the uniaxial direction the lowest modes of both constant-pressure pairs carry r_v between 0.20 and 0.29 at every N, without decrease, and so do their five lowest modes from $N = 6$ on. These are the modes of §6: the element assigns them no energy for a dilatation the continuum resists with κ, and they are the softest modes the element has. The lowest mode of the Crouzeix–Raviart pair carries 0.040 at $N = 2$, 0.018 at $N = 3$ and 0.0014 at $N = 8$, at local rates of 3.1, 2.6, 2.4, 2.3 and 2.2 from $N = 3$ to 8, and its five lowest modes carry at most 0.0017 at $N = 8$. At every $N \geq 3$ the three lowest modes of the Crouzeix–Raviart pair have r_v below 0.023 and those of the constant-pressure pairs above 0.19, so any threshold between these two values gives the same classification of them. Under the shear direction the lowest modes of every pair are shear bands whose dilatation decreases: from $N = 5$ to 8 by a factor of 2.5 for P_2/P_0 and 3.6 for the Crouzeix–Raviart pair. The separation of the pairs appears under the uniaxial direction only.

A reference for the eigenvalues. On divergence-free fields in the whole space, the Fourier symbol of the form in (54) at $\eta = 1$ for a plane wave of unit amplitude $\boldsymbol{a}$ and unit wave vector $\boldsymbol{k}$, orthogonal to each other, is $\frac{1}{2} - (\boldsymbol{n} : \operatorname{sym}(\boldsymbol{a} \otimes \boldsymbol{k}))^2$. Its minimum is $\lambda_\infty = \frac{1}{2} - \frac{1}{4}(n_{\max} - n_{\min})^2$, with $n_{\max}$ and $n_{\min}$ the extreme eigenvalues of $\boldsymbol{n}$: $\lambda_\infty = \frac{1}{8}$ under the uniaxial direction and 0 under the shear direction. Under the uniaxial direction, λ_1 of the Crouzeix–Raviart pair decreases to 0.129 at $N = 8$, above $\frac{1}{8}$. λ_1 of P_2/P_0 and of $P_2 \oplus b_{\text{face}}/P_0$ converges to about 0.096 and 0.069, below it. The softest modes of the constant-pressure pairs thus have a smaller energy than any divergence-free continuum field: they are spurious by energy as well as by dilatation.

A criterion independent of the number of eigenpairs. An earlier version of this note counted the modes with $r_v > 0.1$ among the twenty lowest. Such a count depends on the number of eigenpairs computed: the spectrum near λ_∞ becomes dense under refinement, and a fixed number of eigenpairs covers a narrowing interval of eigenvalues. A dilatational member can therefore lie above the twentieth eigenvalue and still lie in the low part of the spectrum. The criterion used instead is the smallest energy of a member with a prescribed dilatation,

$$\lambda_*(r_0) = \inf \Big\{ \frac{a(\boldsymbol{v}, \boldsymbol{v})}{\int |\operatorname{Grad} \boldsymbol{v}|^2 \, \mathrm{d}V} : \boldsymbol{v} \in \ker \boldsymbol{G}_h, \ r_v(\boldsymbol{v}) \geq r_0 \Big\}, \tag{56}$$

with a the form in the numerator of (54). For every $s \geq 0$, let $g(s)$ be the smallest eigenvalue of $a(\boldsymbol{v}, \boldsymbol{v}) - s \int (\operatorname{Div} \boldsymbol{v})^2 \, \mathrm{d}V$ on $\ker \boldsymbol{G}_h$ against $\int |\operatorname{Grad} \boldsymbol{v}|^2 \, \mathrm{d}V$. Every admissible $\boldsymbol{v}$ has a quotient of at least $g(s) + s\, r_0$, so $\max_s [g(s) + s\, r_0]$ is a lower bound of $\lambda_*(r_0)$. The constraint is a single quadratic inequality, and by the convexity of the joint range of two quadratic forms on the unit sphere [9] the bound is attained. $g(s)$ is computed with the penalized subspace iteration above, and the maximizing s by bisection on the dilatation of the minimizer, which increases with s, to a relative width of 10^{-4}. The value reported is the bound. Where the minimizer at the maximizing s is unique, a field with $r_v = 0.1$ attains it to three digits. On the Freudenthal meshes with $N \geq 6$ the minimizer changes at that s between a smooth and a dilatational eigenvector, and the field that attains the bound is a combination of the two. Table 12 gives $\lambda_*(0.1)$ under the uniaxial direction on the Freudenthal meshes and on the two families of §9.5 without that structure. Script: `prototype/dil_bound.jl`.

Table 12: Smallest energy $\lambda_*(0.1)$ (56) of a member of $\ker \boldsymbol{G}_h$ with $r_v \geq 0.1$, uniaxial flow direction, $\eta = 1$, whole boundary fixed; $\lambda_\infty = 0.125$. For the Crouzeix–Raviart pair (TET15-P1): the smallest shape quality $q_{\min}$ of the mesh, its lowest eigenvalue λ_1, and the share of $\int(\operatorname{Div} \boldsymbol{v})^2$ of the minimizing field in its three largest elements, where the minimizer is unique. Where the minimizer is a combination of a smooth and a dilatational eigenvector, "share" gives in parentheses the eigenvalue and r_v of the dilatational one. *: the minimizer is the lowest eigenmode. For the two constant-pressure pairs the minimizer is the lowest eigenmode on every mesh. Perturbed meshes: seed 1.

		TET15-P1				P_2/P_0	$P_2 \oplus b_{\text{face}}/P_0$
mesh		$q_{\min}$	λ_1	$\lambda_*(0.1)$	share	$\lambda_*(0.1)$	$\lambda_*(0.1)$
Freudenthal	$N = 4$	0.66	0.139	0.167	0.14	0.106	0.074
	$N = 6$	0.66	0.131	0.163	(0.226, 0.27)	0.098	0.069
	$N = 8$	0.66	0.129	0.162	(0.226, 0.28)	0.096	0.068
$\delta = 0.1$	$N = 6$	0.46	0.134	0.160	0.33	0.099	0.067
	$N = 7$	0.49	—	0.156	0.47	0.097	0.065
$\delta = 0.25$	$N = 6$	0.15	0.138	0.139	0.74	0.080	0.048
	$N = 7$	0.21	0.106	0.106*	0.80	0.064	0.037
Cubit	$n = 3$	0.58	0.160	0.160*	0.98	0.190	0.115
	$n = 4$	0.57	0.152	0.152*	0.76	0.119	0.065
	$n = 5$	0.55	0.143	0.144*	0.97	0.115	0.065
	$n = 6$	0.54	0.132	0.132*	0.96	0.088	0.044
	$n = 7$	0.53	0.144	0.149	0.75	0.092	0.053
	$n = 8$	0.53	0.142	0.144	0.95	0.088	0.044
	$n = 10$	0.52	0.138	0.144	(0.155, 0.25)	0.089	0.051

The constant-pressure pairs on every mesh. For the constant-pressure pairs the lowest eigenmode has r_v between 0.17 and 0.55 on every mesh of Table 12. Their $\lambda_*(0.1)$ lies below λ_∞ on every mesh except the coarsest Cubit mesh for P_2/P_0. For $P_2 \oplus b_{\text{face}}/P_0$ it decreases from 0.074 on the Freudenthal meshes to 0.037 at $\delta = 0.25$. Their dilatational softest modes are spread over the mesh on the Freudenthal meshes, with 1 to 9% of $\int(\operatorname{Div} \boldsymbol{v})^2$ in the three largest elements, and have between 6 and 84% in three elements on the perturbed and unstructured meshes.

The Crouzeix–Raviart pair on unstructured meshes. For the Crouzeix–Raviart pair, $\lambda_*(0.1)$ is above λ_∞ on every mesh except the strongly perturbed mesh at $N = 7$, where the lowest eigenmode itself has $r_v = 0.28$ at $\lambda_1 = 0.106$. On the Freudenthal meshes and the mildly perturbed meshes $\lambda_*(0.1)$ exceeds λ_1 by 19 to 26%. On the Cubit meshes the minimizer is the lowest eigenmode up to $n = 6$. At $n = 7$ and 8 the minimizer is unique, and $\lambda_*(0.1)$ exceeds λ_1 by 4 and 1%. At $n = 10$ the bound 0.144 is attained by a combination of a smooth and a dilatational eigenvector; the dilatational one, with $r_v = 0.25$, lies at 0.155, 12% above λ_1, and has 89% of its dilatation in three elements. On the Cubit meshes the minimizing field has 75 to 98% of its dilatation in three elements, whose smallest shape quality is between 0.54 and 0.68. The kernel of the Crouzeix–Raviart pair on unstructured meshes thus contains element-scale members with $r_v \geq 0.1$ whose energy lies within 12% of the lowest eigenvalue at every n measured. These members do not disappear under refinement up to $n = 10$, and they are spurious dilatational modes in the sense of §6. An earlier conclusion of this study, that such modes are absent from $n = 7$ on, rested on the count among the twenty lowest eigenpairs

Table 13: Spurious modes among the twenty lowest of the plastic form on $\ker \boldsymbol{G}_h$ ($r_v > 0.1$) in a confined plastic zone with a varying flow direction, at $N = 3, 4, 5$.

field	boundary	P_2/P_0	$P_2 \oplus b_{\text{face}}/P_0$	$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	P_2/P_1 cont.
indentation	whole fixed	16, 15, 7	18, 17, 5	4, **0, 0**	20, 18, 17
indentation	base fixed	11, 1, 0	13, 5, 0	**0, 0, 0**	14, 7, 5
twist	whole fixed	14, 18, 15	18, 13, 8	1, 4, **0**	20, 20, 20
twist	base fixed	5, 2, 0	13, 7, 4	**0, 0, 0**	12, 13, 12

and is withdrawn: from $n = 7$ on the members lie above the twentieth eigenvalue and still lie within 12% of the lowest one. Their energy remains above λ_∞, the infimum of the continuum, on every mesh except the strongly perturbed one; the energy of the dilatational modes of the constant-pressure pairs lies below it. The separation of the pairs is therefore a separation by energy on unstructured meshes, and a separation by both energy and dilatation on structured ones.

A confined zone with a varying flow direction. The counts in this paragraph are taken over the twenty lowest eigenpairs and are subject to the limitation stated above; they are kept as a record of the lowest part of the spectrum at $N \le 5$. The flow direction and the plastic zone are taken from a prescribed displacement field: $\boldsymbol{n} = \operatorname{dev} \boldsymbol{\epsilon}(\boldsymbol{u}) / \|\operatorname{dev} \boldsymbol{\epsilon}(\boldsymbol{u})\|$ at each quadrature point and $\eta = 1$ where $|\operatorname{dev} \boldsymbol{\epsilon}(\boldsymbol{u})|$ exceeds 0.35 of its peak, $\eta = 0$ elsewhere. Two fields: an indentation of the top face, $u_3 = -\exp(-d^2/w^2)\, X_3$ with d the distance from the vertical axis and $w = \frac{1}{4}$, whose zone is 51% of the volume with elastic surroundings; and the displacement of the 90° twist of §9.6, whose zone is 97%. Both are run with the whole boundary fixed and with the base alone fixed. Table 13 counts the spurious modes among the twenty lowest. Script: `prototype/plastic_zone.jl`.

With the whole boundary fixed the constant-pressure pairs retain between five and fifteen spurious modes among their twenty softest at $N = 5$, and under the twist the softest mode of P_2/P_0 is spurious at every N up to 4 ($r_v = 0.22$–0.31) and its third at $N = 5$ (0.20). The Crouzeix–Raviart pair has none from $N = 4$ on under the indentation and none at $N = 5$ under the twist, and none at any N with the base alone fixed. Taylor–Hood keeps seventeen to twenty, and under the twist its softest mode's dilatation grows with refinement, 0.29, 0.37, 0.66, 0.74. With the base alone fixed the softest modes of every pair are long-wavelength motions of the block and the counts fall for the constant-pressure pairs too, to zero at $N = 5$ for P_2/P_0 and four for the face-bubbled pair, so the spurious modes are then above the twentieth; they are not absent, as the whole-boundary rows show. The number of modes with $r_v > 0.1$ among the twenty lowest decreases with N for every pair with the whole boundary fixed, and in this configuration the dilatational modes of the constant-pressure pairs lie above physical modes; the separation is weaker than under the uniform uniaxial direction. Whether a nonlinear solve excites them is not settled: the indentation benchmark of §12 finds no persistent element-to-element pressure mode in CT10, but its measure compares element averages and does not resolve a dilatation that varies within an element (§13).

Taylor–Hood is the worst pair on this measure. Its pressure is continuous, so its kernel is divergence-free only against continuous linear test functions, and under the uniaxial direction

its lowest modes are dilatation almost entirely: r_v between 0.86 and 0.97 at $N = 3$ to 6, where its iteration converged, with λ_1 decaying to 0.011. Those modes have no counterpart in the continuum, which resists them with κ. Taylor–Hood is therefore an alternative for elasticity only (§14).

CT10 on the same measure. The LCM CT10, the element §10 compares against, is measured with the same scripts (`prototype/composite_tet.jl`) as Albany builds it: ten nodes, a displacement that is linear on each of twelve subtetrahedra, the gradient of that displacement projected onto the linear functions of the element for the constitutive update, and a five-point rule of degree three. Two volumetric variants exist in Albany: J replaced by its element mean (the option `Weighted Volume Average J`), whose constraint as $\kappa \to \infty$ is a constant pressure on the projected dilatation, and J at each quadrature point, whose constraint at the five points is a linear discontinuous pressure. With the averaged J, under the uniaxial flow direction, the three lowest modes carry $r_v = 0.22$ to 0.25 at $N = 3$, 4, 5 with no decay, all twenty lowest modes are spurious from $N = 4$ on, and in the confined zones of Table 13 with the whole boundary fixed the count at $N = 5$ is 17 under the indentation and 19 under the twist, against 7 and 15 for P_2/P_0. With J at each point the discretely isochoric subspace has dimension 0, 0, 2 and 16 at $N = 2$ to 5, against 72 for P_2/P_1^{disc} at $N = 4$: the element locks, and too few modes exist to count. The average removes the locking and admits the spurious dilatational modes of a constant pressure, in larger numbers than P_2/P_0. The measurement is linearized, like every other in this section, and models no stabilization of the element's own zero-energy modes, because the Albany runs of §10 use none.

9.8 Locking measured against a known solution

The rank count of §9.4 and the constant β_h are indirect measures of locking. This section measures locking directly, by the error against a known solution as κ/μ grows. The problem is isotropic linear elasticity on the unit cube with the whole boundary fixed, on the Freudenthal meshes, with the manufactured displacement

$$\boldsymbol{u} = \operatorname{curl}(B\,\boldsymbol{a}) + \frac{\nabla\phi}{\lambda+\mu}, \qquad B = \big(x(1-x)\,y(1-y)\,z(1-z)\big)^2, \tag{57}$$

with $\phi = 3B$, $\boldsymbol{a} = (1, 2, 3)$, the Lamé constant $\lambda = \kappa - \frac{2}{3}\mu$ and $\mu = 1$. Then $\operatorname{Div}\boldsymbol{u} = \Delta\phi/(\lambda+\mu)$, and the mean stress $\kappa \operatorname{Div}\boldsymbol{u}$ tends to $\Delta\phi$ as $\kappa \to \infty$. The body force $-\mu\Delta\boldsymbol{u} - (\lambda+\mu)\nabla\operatorname{Div}\boldsymbol{u}$ is bounded uniformly in κ, and $\boldsymbol{u}$ vanishes on the boundary because B and ∇B do. The load is integrated with a rule of degree 12. The measured quantities are the relative H^1-seminorm error e_u of the displacement and the relative L^2 error e_p of the mean stress. The mean stress is κ times the projected dilatation for the projected elements and κ times the pointwise dilatation for the others. Five elements are compared: TET10 and TET15 with the volumetric response at each quadrature point ("pointwise"), P_2/P_0, TET15-P0 and TET15-P1. Table 14 gives the result. Scripts: `prototype/kappa_sweep.jl`, `prototype/quasiopt.jl` and `prototype/pressure_bestapprox.jl`.

The pointwise elements lock. The error of the two pointwise elements at $\kappa/\mu = 10^6$ is 1.8 (TET10) and 2.0 (TET15) times their error at $\kappa/\mu = 1$ at $N = 2$, and 4.0 and 5.0 times at $N = 6$. For TET10 the ratio reaches 7.5 at $N = 12$. There its local rate is 1.01 at $\kappa/\mu = 10^6$

Table 14: Linear elasticity with the manufactured solution (57): relative H^1-seminorm error e_u of the displacement for four values of κ/μ, and relative L^2 error e_p of the mean stress at $\kappa/\mu = 10^4$, on the Freudenthal mesh with N^3 cubes. "Rate" is the local rate of e_u between the last two meshes of each element, at $\kappa/\mu = 1$ and at 10^6.

		e_u at $\kappa/\mu =$				e_p	rate of e_u	
element	N	1	10^2	10^4	10^6	10^4	1	10^6
TET10, pointwise	2	0.557	0.921	0.999	1.000	3.15		
	3	0.304	0.616	0.742	0.744	3.88		
	4	0.189	0.424	0.555	0.557	3.80		
	6	0.092	0.230	0.365	0.369	3.74		
	8	0.053	0.141	0.270	0.275	3.67		
	10	0.035	0.093	0.213	0.219	3.58		
	12	0.024	0.065	0.175	0.182	3.49	1.95	1.01
P_2/P_0	2	0.623	0.814	0.892	0.894	3.36		
	3	0.377	0.527	0.600	0.601	2.89		
	4	0.261	0.381	0.446	0.448	2.58		
	6	0.158	0.241	0.294	0.296	2.35	1.24	1.02
TET15, pointwise	2	0.497	0.767	0.995	1.000	6.97		
	3	0.244	0.426	0.729	0.744	10.3		
	4	0.150	0.269	0.531	0.557	10.8		
	6	0.073	0.131	0.331	0.367	9.46	1.78	1.03
TET15-P0	2	0.580	0.670	0.671	0.671	0.924		
	3	0.342	0.448	0.449	0.449	0.634		
	4	0.243	0.340	0.341	0.341	0.479		
	6	0.152	0.229	0.229	0.229	0.321		
	8	0.111	0.172	0.173	0.173	0.241		
	10	0.087	0.138	0.139	0.139	0.193		
	12	0.072	0.116	0.116	0.116	0.160	1.05	0.99
TET15-P1	2	0.504	0.630	0.642	0.642	1.06		
	3	0.247	0.308	0.314	0.314	0.543		
	4	0.151	0.190	0.195	0.195	0.339		
	6	0.073	0.094	0.096	0.096	0.171		
	8	0.0430	0.0560	0.0575	0.0575	0.104		
	10	0.0282	0.0371	0.0381	0.0381	0.0696		
	12	0.0199	0.0264	0.0271	0.0271	0.0500	1.92	1.87

against 1.95 at $\kappa/\mu = 1$, and its mean stress has an error of several hundred percent that does not decrease.

TET15-P1. For TET15-P1 the ratio of the errors at $\kappa/\mu = 10^6$ and 1 is 1.27 at $N = 2$ and 3, and 1.29, 1.32, 1.34, 1.35 and 1.36 at $N = 4$ to 12. The local rates at $\kappa/\mu = 1$ and 10^6 are 1.79 and 1.73 from $N = 4$ to 6, 1.85 and 1.80 from $N = 6$ to 8, 1.89 and 1.84 from $N = 8$ to 10, and 1.92 and 1.87 from $N = 10$ to 12. Both rates increase toward the order 2 of the space, and their difference decreases slowly, from 0.055 to 0.051 to 0.047 over $N = 6$ to 12. The mean stress converges at a local rate of 1.82 from $N = 10$ to 12, and its errors at $\kappa/\mu = 10^4$ and 10^6 agree to three digits on every mesh. The L^2 projection error of the exact mean stress onto the linear discontinuous pressures converges at a local rate between 1.96 and 2.00 from $N = 8$ on, so the approximation of the pressure does not explain the lower rate of the displacement.

Quasi-optimality. The exact displacement (57) depends on κ through its gradient part, so the ratio of the errors compares two different solutions. The quasi-optimality ratio $Q = e_u/e_{\text{best}}$, with e_{best} the relative error of the best approximation of the same solution in the same space in the H^1 seminorm, removes that dependence. For TET15-P1, Q is 1.04 to 1.05 at $\kappa/\mu = 1$ on every mesh, and 1.45, 1.47, 1.49, 1.52, 1.55, 1.58 and 1.59 at $\kappa/\mu = 10^6$ at $N = 2$, 3, 4, 6, 8, 10 and 12. For a pair that satisfies the inf-sup condition with a constant bounded below, this ratio is bounded uniformly in κ and h [7, Chapter 5]. The measurement shows it growing by about $N^{0.06}$ up to $N = 12$, and it does not show where the growth ends.

The definition of locking. A definition by the ratio of the errors at large and small κ/μ is met by TET15-P1 itself, whose ratio grows slowly. The definition used in this note is therefore by rate: an element locks when its rate of convergence at large κ/μ falls below its rate at $\kappa/\mu = 1$ by an amount that does not vanish under refinement. On this definition the measured loss of TET15-P1 is 0.047 at the finest pair of meshes and decreasing, against 0.94 for TET10. A slowly decreasing loss is not a proof that the loss vanishes. TET15-P0 does not lock either, but its errors converge at first order, the order of the constant pressure. P_2/P_0, whose pressure has three uncontrolled modes on this mesh family (Table 8), converges at first order in the displacement and has a mean-stress error above 200% on every mesh.

9.9 Quadrature rule and basis

Two implementation choices set the number of operations per element and its explicit stable step, and neither is decided by the stability results: the quadrature rule, and the basis that carries the enriched space. Both are measured on one element. Scripts: `prototype/quadrature.jl` and `prototype/basis.jl`.

Quadrature. The scripts integrate the enriched element to degree 6 so that the linear measurements are exact. For the element the criterion is weaker: a rule is admissible when the condensed stiffness $\boldsymbol{K}_e = \boldsymbol{K}^{\text{dev}} + \kappa \boldsymbol{G}^{\mathsf{T}} \boldsymbol{M}^{-1} \boldsymbol{G}$ has exactly six zero-energy modes, and its effect is the softening it introduces, measured as the extreme generalized eigenvalues of $\boldsymbol{K}_e$(rule) against $\boldsymbol{K}_e$(exact) on the complement of the rigid modes, which are 1 for an exact rule. Table 15 gives both for the rules considered, with a degree-11 rule of 216 points as the reference. The two rules supplied by `ReferenceFiniteElements` (RFE), the Julia package that provides

Table 15: Quadrature rules on the enriched element: zero-energy modes of the condensed stiffness (six is correct) and of the deviatoric stiffness (ten: six rigid modes and the four volumetric directions of §9.2), and the extreme softening factors against a 216-point rule, on the reference tetrahedron and on a Kuhn tetrahedron of the Freudenthal mesh. $\kappa/\mu = 10^4$. Script: `prototype/quadrature.jl`.

		reference			Kuhn		
rule	points	zero($\boldsymbol{K}_e$)	min	max	zero($\boldsymbol{K}_e$)	min	max
RFE degree 2	4	21	0	1.30	21	0	1.30
RFE degree 3	5	21	< 0	1.13	21	< 0	1.12
conical degree 3	8	6	0.10	1.14	6	0.03	1.21
Walkington degree 5	14	6	1.00	1.35	6	1.00	1.35
conical degree 5	27	6	0.91	1.00	6	0.88	1.00
conical degree 7	64	6	1.00	1.00	6	1.00	1.00

the reference elements and their quadrature, four and five points, leave 21 zero modes and are inadmissible. The eight-point conical rule (a tensor-product Gauss rule mapped onto the tetrahedron) is rank sufficient and softens the element to 0.03–0.10 of its stiffness in the worst direction, which is also inadmissible. Two degree-5 rules are admissible: the 14-point symmetric rule of Walkington [33], which integrates every monomial through degree 5 to 4×10^{-16} and over-stiffens the interior bubble's own block by 35% since its gradient-squared is of degree 6, and the 27-point conical rule, which under-stiffens the same block by 9–12%. Assembled, both give the Crouzeix–Raviart pair's constant within 1% (0.293 and 0.298 against 0.297 at $N = 5$), a one-dimensional pressure null space, no spurious mode among the twenty lowest under the uniaxial plastic tangent, and the same dilatation of the lowest mode (0.0073, 0.0074 and 0.0074). The 14-point rule satisfies the criteria — six zero-energy modes, positive weights, 4.6 times fewer points than the 64-point degree-7 rule — and is used, with a stiffer interior bubble.

Basis. The space $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$ can be carried by the hierarchical basis of the scripts, ten Lagrange functions and five bubbles, or by the nodal basis with unit value at one of fifteen nodes, the `TETRA15` arrangement: vertices, edge midpoints, face centroids, centroid. The space is the same, so every stiffness and stability result is the same; what differs is conditioning, lumping, and the coefficients of a rigid translation. Table 16 gives the measurements at $\kappa/\mu = 2$. The nodal basis conditions the consistent mass eight times better (94 against 766 with exact integration; an earlier value of 778 came from a rule of degree seven) and the stiffness better (559 against 797). Under HRZ lumping the hierarchical basis loses 51% of the momentum of a rigid translation unless the normalization runs over the Lagrange functions alone (§7); the nodal basis conserves it without qualification, and its lumped masses are positive on every element measured. The explicit stable step from the element bound, as the Courant number $c_p \Delta t/h$, is 0.476 for TET4, 0.197 for TET10 under HRZ, 0.145 for the enriched element in the nodal basis under HRZ, and 0.114 in the hierarchical basis under Lagrange-only HRZ, with the same ordering on the distorted and Kuhn elements. The nodal basis is used: every unknown is attached to a node, so no unknown of a new kind is introduced; its mass lumps correctly by the standard construction; and it takes a 20–27% larger stable step.

Table 16: Basis choice on the reference tetrahedron, $\kappa/\mu = 2$, $\mu = \rho = 1$: condition numbers of the condensed stiffness on the complement of the rigid modes and of the consistent mass, the smallest lumped mass over ρV, the momentum of a unit translation under the lumped mass over ρV, and the Courant number $c_p \Delta t/h$ from the element bound. Script: `prototype/basis.jl`.

basis, lumping	cond $\boldsymbol{K}_e$	cond $\boldsymbol{M}$	min $m/\rho V$	momentum	Courant
hierarchical, consistent	797	766	—	—	0.092
hierarchical, HRZ over all 15	—	—	0.014	0.486	0.097
hierarchical, HRZ over Lagrange 10	—	—	0.028	1.000	0.114
nodal, consistent	559	94	—	—	0.092
nodal, HRZ	—	—	0.021	1.000	0.145
nodal, row sum	—	—	0.020	1.000	—
TET10, HRZ	—	—	0.028	1.000	0.197
TET10, row sum	—	—	−0.050	1.000	—
TET4, row sum	—	—	0.250	1.000	0.476

10 Nonlinear results: Cook's membrane

Every measurement of §9 is on the linearized operator. This section measures the element in nonlinear solves, against the LCM CT10 on the same meshes, on Cook's membrane: a tapered panel clamped on one edge and sheared on the opposite one [12], in its nearly incompressible form [31] and in its elastoplastic form, both in three dimensions.

Problem. The membrane is the trapezoid with corners $(0, 0)$, $(48, 44)$, $(48, 60)$ and $(0, 44)$ in the x–y plane, of thickness 10 in z, with both z faces free. The face $x = 0$ is clamped in all three components and the face $x = 48$ carries a uniform shear traction q in y, applied in ten equal load steps. The material is the Simo–Hughes J_2 model (Remark 5) in both codes. Two cases: the elastic one, with Young's modulus $E = 240.565$, Poisson's ratio $\nu = 0.4999$ ($\kappa/\mu = 5000$), a yield stress that is never reached and $q = 6.25$, the plane-strain load $F = 100$ per unit thickness divided by the height 16 of the loaded edge; and the plastic one, with the material of Simo and Armero [31], $E = 206.9$, $\nu = 0.29$, yield stress $\sigma_y = 0.45$, linear isotropic hardening modulus $K = 0.12924$, and $q = 0.14$. Their load $q = 0.1125$ leaves the three-dimensional membrane with free faces almost elastic (tip displacement 0.21 with no difference between elements); the quadratic tetrahedron without projection collapses between $q = 0.16$ and $q = 0.20$, and Albany's Newton iteration on the CT10 stops converging at $q = 0.158$ on the middle mesh and $q = 0.137$ on the finest, so the load is 0.14. The measured quantity is the mean of the displacement u_y over the nodes of the loaded face at full load.

Meshes and elements. Three `TETRA10` meshes with element size $h = 8$, 4 and 2 (327, 1860 and 14 475 elements; 647, 3208 and 22 059 nodes) are generated by Cubit and used by both codes; the `TETRA15` meshes are their conversions (§4.3; 1721, 9127 and 66 813 nodes). Six elements are compared. In Carina: TET10 with the volumetric response evaluated at each quadrature point (pointwise); TET15 pointwise; TET15-P1, the element of this note, with the linear projection; and TET15-P0, the same displacement element with the constant projection. In Albany-LCM: TET10 pointwise, which serves as the cross-code control; and CT10 with the

Table 17: Cook's membrane, mean u_y of the loaded face at full load. Rows: element size h with the element count. Elastic case: $E = 240.565$, $\nu = 0.4999$, $q = 6.25$. Plastic case: $E = 206.9$, $\nu = 0.29$, $\sigma_y = 0.45$, $K = 0.12924$, $q = 0.14$. TET15-P1 and TET15-P0: the projected strain $\log J$ (Carina 29c9606); the pointwise columns were run with the volumetric energy $\kappa/2\,(J-1)^2$, which differs from the present $\kappa/2\,(\log J)^2$ at third order in $J-1$, below the digits shown. Driver: `benchmark/tet15-p1/cook/run.jl`; records in `results.tsv`.

h (elements)	TET10 pointwise	TET15 pointwise	TET15-P1	TET15-P0	TET10 Albany	LCM CT10
elastic						
8 (327)	8.1960	8.2235	8.4099	8.5639	8.1960	8.4596
4 (1860)	8.3241	8.3490	8.4536	8.5124	8.3241	8.4643
2 (14 475)	8.4203	8.4300	8.4709	8.4955	8.4203	8.4772
plastic						
8 (327)	0.2689	0.2700	0.2706	0.2752	0.2689	0.2716
4 (1860)	0.2710	0.2717	0.2721	0.2744	0.2710	0.2725
2 (14 475)	0.2720	0.2723	0.2724	0.2734	0.2720	0.2728

volume-averaged J and pressure. The displacement unknowns are 1941, 9624 and 66 177 on the `TETRA10` meshes and 5163, 27 381 and 200 439 on the `TETRA15` meshes. Both codes converge each load step to a residual norm of 10^{-6} absolute or 10^{-8} relative with a direct linear solver; both halve a load step whose Newton iteration fails and grow it back after successful steps.

Cross-code control. The pointwise TET10 gives the same tip displacement in Carina and in Albany to six digits on every mesh and in both cases (8.19605 elastic at $h = 8$; 8.4202946 and 8.4202943 at $h = 2$; 0.27204 plastic at $h = 2$), although the two codes take the volumetric stress as $\kappa(J-1)$ (Carina before 2026-10-16) and $\kappa(J-1/J)/2$: at these volumetric strains the two laws agree to the digits shown. The comparison between the other elements rests on this check.

Tip displacement. Table 17 gives the results. In the elastic case the pointwise elements converge from below, 8.196, 8.324, 8.420 for TET10 and 8.224, 8.349, 8.430 for TET15, and TET15-P0 from above, 8.564, 8.512, 8.496. TET15-P1, 8.410, 8.454, 8.471, and the CT10, 8.460, 8.464, 8.477, are within 0.6% of each other on every mesh and change least with refinement; the limit lies between 8.477 and 8.496. On the coarsest mesh the pointwise TET10 is 3% below the limit and TET15-P1 0.8% below it. In the plastic case every element converges to about 0.273, with the spread between elements shrinking from 2.3% at $h = 8$ to 0.5% at $h = 2$, and the projected and composite elements closest to the limit on every mesh; at $\nu = 0.29$ and this load the plastic zone does not lock the quadratic elements. On the finest mesh the elastic runs took 1099 s for TET10 and 3214 to 3639 s for the TET15 variants in Carina on 16 threads of one host, and 26 496 and 28 867 s in Albany with a serial direct solver; at $\nu = 0.4999$ the residual grows with κ times the square of the volumetric strain of a Newton step, so every element needs load steps of a fraction of a percent of the load in both codes.

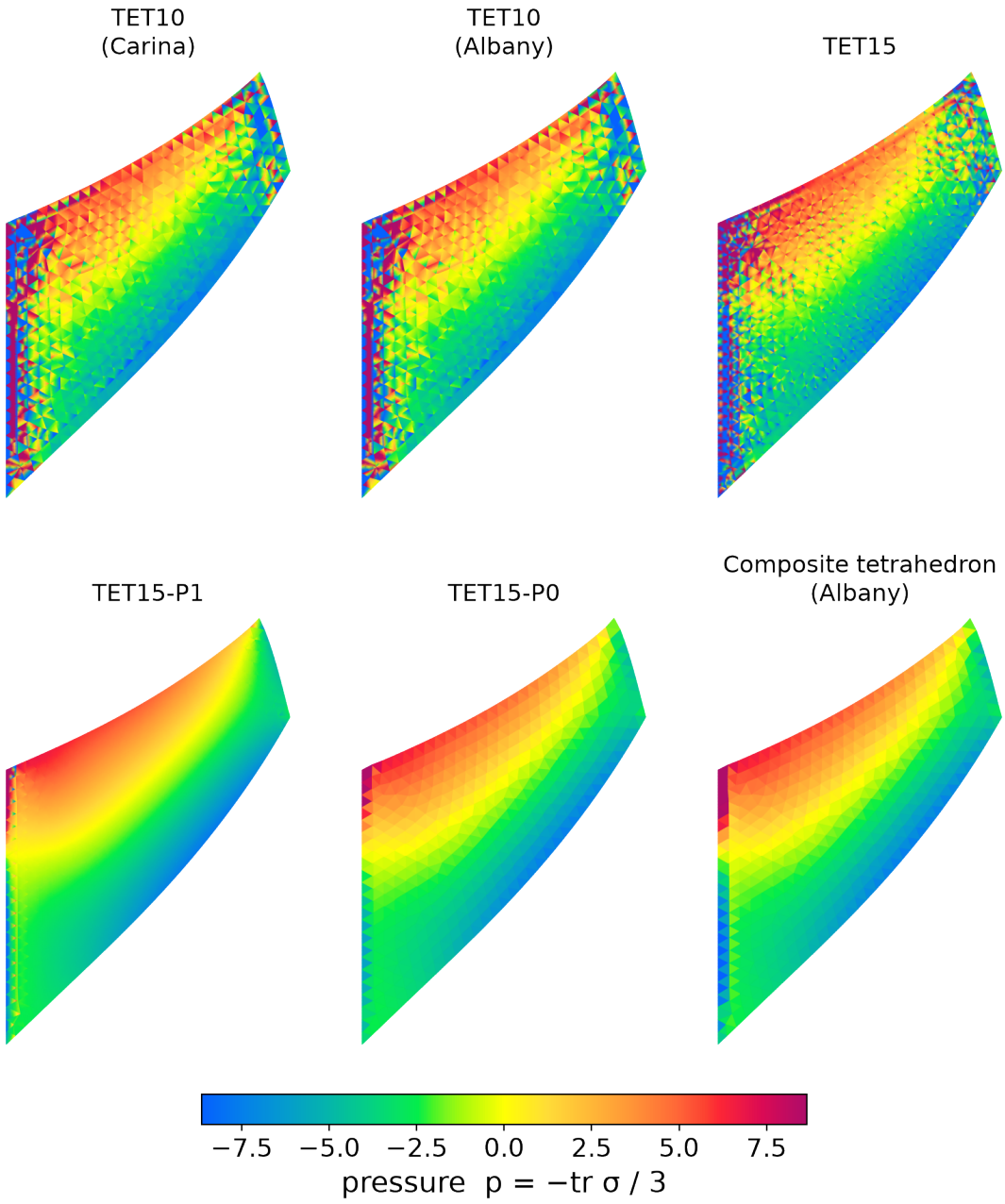


Figure 1: Elastic case, $h = 2$: pressure $p = -\operatorname{tr}\boldsymbol{\sigma}/3$ on the face $z = 10$ of the deformed membrane at full load, for the six elements. One color scale for all panels, symmetric about zero and bounded by the 99th percentile of $|p|$ over TET15-P1, TET15-P0 and CT10; values beyond the scale saturate. Inside each element the pressure drawn is the polynomial fitted to the quadrature-point values (text).

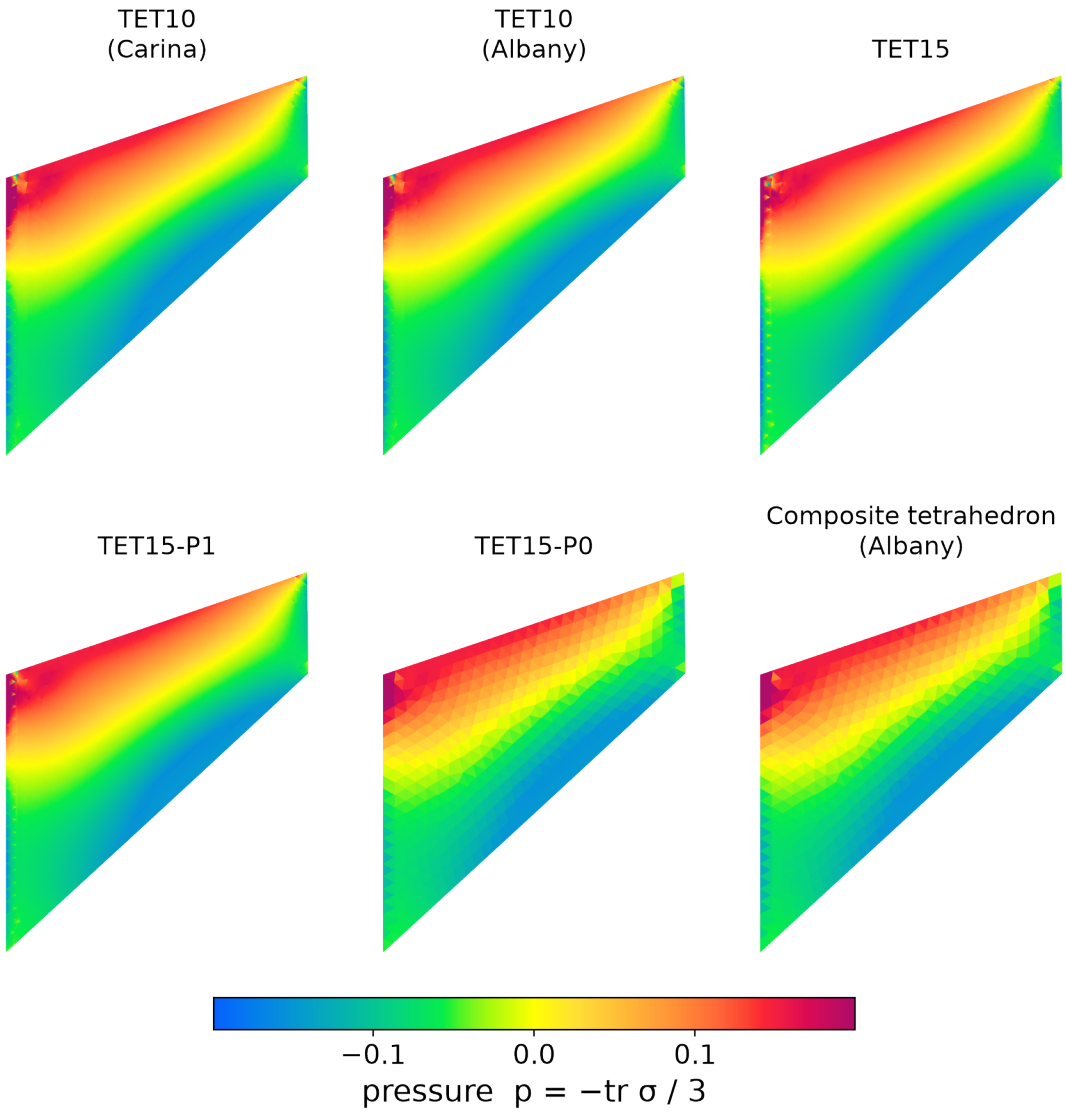


Figure 2: Plastic case, $h = 2$: pressure on the face $z = 10$ at full load, as in Figure 1.

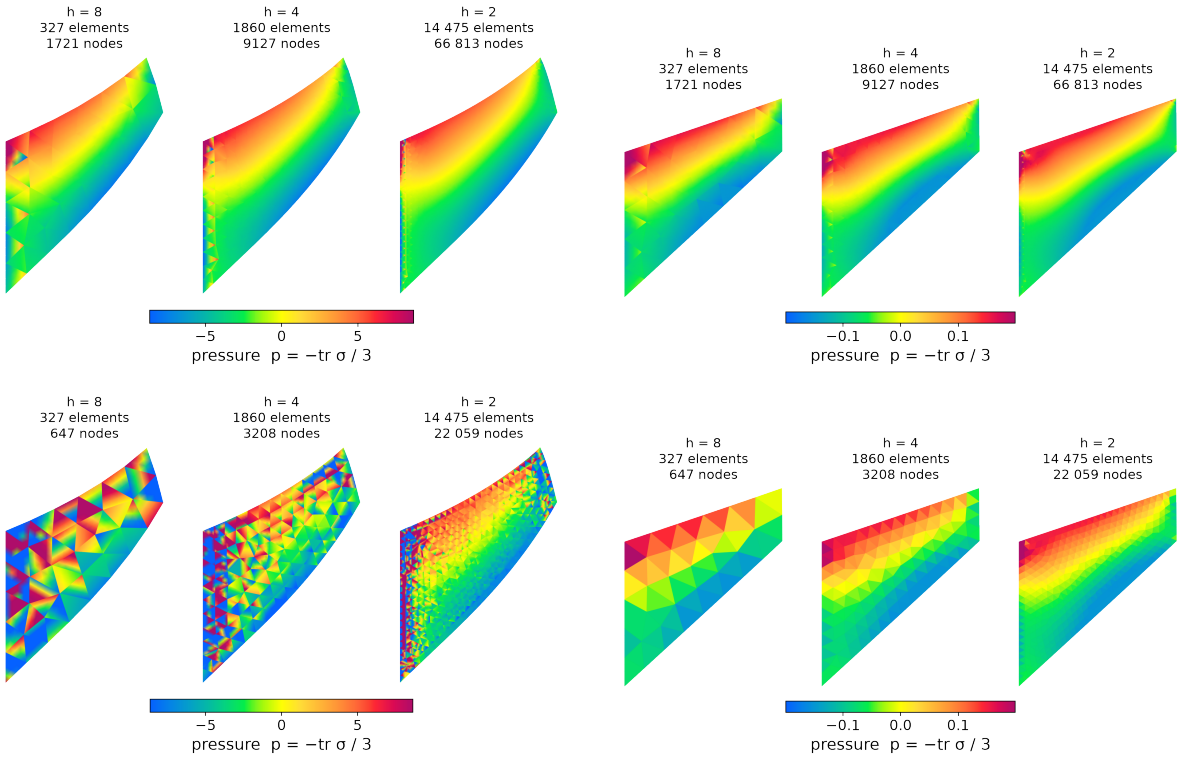


Figure 3: Pressure on the face $z = 10$ under mesh refinement, $h = 8$, 4, 2 from left to right in each panel, with the color scale of the case. Top: TET15-P1, elastic (left) and plastic (right). Bottom left: TET10 pointwise, elastic. Bottom right: the CT10, plastic.

Pressure fields. The pressure $p = -\operatorname{tr}\boldsymbol{\sigma}/3$, with $\boldsymbol{\sigma}$ the Cauchy stress, is read at the quadrature points of every element at full load and drawn on the face $z = 10$ of the deformed membrane (Figures 1–3). Inside each element the field drawn is the polynomial fitted to the quadrature-point values in the least-squares sense with the weights of the rule: quadratic for the fourteen-point rule of TET15, which reproduces the linear and the constant projection to rounding error and the pointwise TET15 pressure, of higher degree, to 0.11 of its largest value; linear for the four-point rule of TET10, which interpolates; and the single value Albany writes per CT10. In the elastic case the pointwise TET10 and TET15 give a pressure that changes sign inside elements over the whole membrane, on every mesh, with a range on the face of -226 to 211 and -512 to 1010 at $h = 2$. TET15-P1 is smooth except in the layer of elements along the clamped edge, with a range of -43 to 61; TET15-P0 and CT10 are smooth and piecewise constant, with ranges of -7.5 to 22 and -9.4 to 35. The clamped edge is a singular line of the continuum solution: where a clamped face meets a free face the stress is unbounded, and it grows faster as $\nu \to 1/2$. The jumps of TET15-P1 between neighboring elements confirm that the discontinuities are the element's error at that line and nothing else: at every h the six largest jumps on the face lie within one element layer of the clamped edge, the largest $|p|$ of TET15-P1 on the face grows from 22 to 35 to 61 over the three meshes as an unbounded field requires, and away from that layer and from the loaded corner the median jump falls from 3.9% to 0.35% of the color limit from $h = 8$ to $h = 2$, faster than h, whereas for the two piecewise constant elements it is 12 to 15% of the color limit at $h = 2$, in proportion to h. In the plastic case no element oscillates, the six fields agree away from the upper corner of the clamped edge, and the largest pressure on the face at $h = 2$ is 0.715 for TET10, 1.15 for TET15 pointwise, 0.519 for TET15-P1, 0.292 for TET15-P0 and 0.407 for CT10. Scripts: `benchmark/tet15-p1/cook/extract_pressure.jl` and `render_pressure.py`.

A second implementation. An implementation of TET15-P1 in Sierra/SM, Sierra/SM TET15-P1, was run on the elastic case on the same meshes, with a neo-Hookean material of the same κ and μ (deviatoric part as in Carina, volumetric pressure $\kappa(J - 1/J)/2$, which differs from Carina's at second order in $J - 1$, of order 10^{-4} here), the traction applied as consistent nodal forces, and a preconditioned conjugate-gradient solver in the Newton iteration. The mean tip displacement is 8.4139, 8.4555 and 8.4722 at $h = 8$, 4 and 2, against 8.4099, 8.4536 and 8.4709 in Carina, a difference of 0.05%, 0.02% and 0.015%, and the maximum agrees to five digits at every level; both codes project $\log J$, and the remaining difference in the mean is of the order of the difference of the two volumetric laws; without the projection the two codes' TETRA15 agree to 0.05%. The Sierra/SM CT10 is within 0.4% of Albany's at $h = 8$ and 4, and its Newton iteration with that solver stalled at $h = 2$ (`benchmark/tet15-p1/cook/README.md`). Sierra/SM TET15-P1 writes one stress per element; its element pressures at $h = 2$ correlate with the element means of Carina's TET15-P1 to 0.999999, with an RMS difference of 0.0054 against a largest $|p|$ of 30.3 (Figure 4). Without the projection the two fields are not comparable element by element: Carina's pressure oscillates in sign inside each element, and the Sierra/SM value is its average.

What the benchmark establishes. The element of this note does not lock at $\nu = 0.4999$ on any of the three meshes, agrees in the tip displacement with CT10 within 0.6% and in the limit of refinement, and gives a pressure that is smooth wherever the continuum pressure is, on a mesh where the quadratic tetrahedron without projection gives a pressure that oscillates

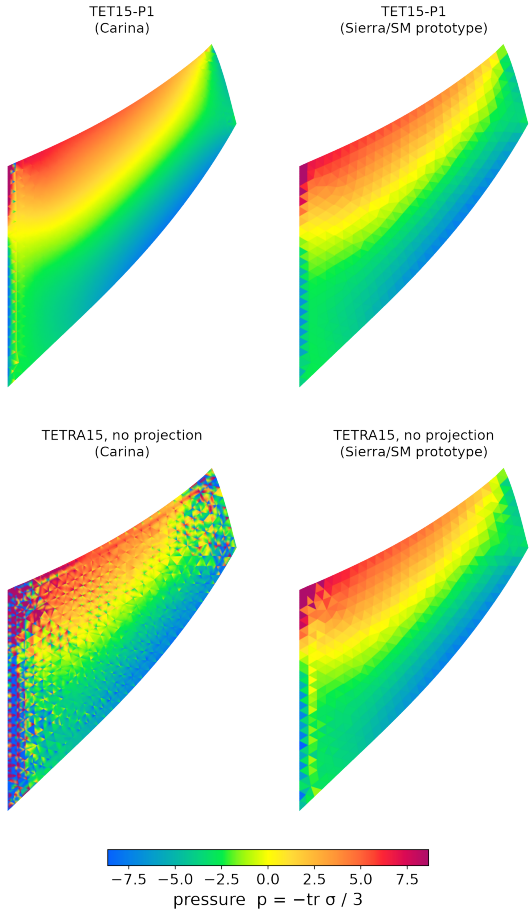


Figure 4: Cook's membrane, elastic case, $h = 2$: pressure on the front face at full load, on the color scale of Figure 1. Top, TET15-P1 in Carina (the quadratic of each element) and in Sierra/SM (one value per element); bottom, TETRA15 without the projection in the two codes.

in sign everywhere. Its unknowns are 3.0 times those of CT10 on the same mesh, the interior bubble not being condensed in the implementation, and the run times above are those of one implementation with a direct solver; a comparison at equal accuracy is not made. The equivalent plastic strain is not measured here: at $q = 0.14$ the plastic zone of the membrane does not lock any of the quadratic elements, so the membrane does not separate them on that quantity.

11 Explicit dynamics: the Taylor bar impact

This section measures the element in explicit dynamics with plastic strains above 3, against the Sierra/SM CT10, the code of Foulk III et al. [16], on the same meshes. The quantity measured is the final radius of the impact face, for which a converged reference value exists; the equivalent plastic strain is not measured.

Problem. A copper bar of length 32.4 mm and radius 3.2 mm, with its axis along z, strikes a rigid frictionless wall at $z = 0$ with velocity 227 m/s [16, 28, 34]. The material has density $\rho = 8930$ kg/m^3, Young's modulus $E = 117$ GPa, Poisson's ratio $\nu = 0.35$ ($\kappa/\mu = 3.0$; the plastic flow is isochoric, so the response in the plastic zone is nearly incompressible), yield stress 400 MPa and a linear isotropic hardening modulus of 100 MPa; the Simo–Hughes J_2 model (Remark 5) in Carina and the finite-deformation J_2 model of the Sierra material library, with the same constants. The wall is the condition $u_z = 0$ on the impact face, which is in contact at $t = 0$ and remains in contact while the bar moves toward the wall. The equations are integrated by central differences with the row-sum lumped mass of §7 to $t = 80\boldsymbol{\mu}$s, without artificial bulk viscosity. The measured quantities are the radius of the impact face, the largest distance of its nodes from the axis, and the length of the bar, the largest z of its free end, at every microsecond. Foulk III et al. [16] give 7.22 mm for the final radius of the eight-node hexahedron with constant pressure (Q1/P0) on their finest mesh, $h = 0.047$ mm, which is the reference here; Simo [28] gives 7.12 mm and Zienkiewicz et al. [36] 7.11 mm on coarser meshes.

Meshes and elements. Four levels, $h = 1.5$, 0.75, 0.38 and 0.19 mm, of the full bar are meshed by Cubit with four-node tetrahedra, the Cubit size being adjusted until the element count is within 3% of that of Foulk III et al. [16, Figure 14]: 3522, 25 029, 190 727 and 1 521 212 elements, with 5601, 36 666, 267 150 and 2 083 320 nodes as `TETRA10` meshes and 16 587, 113 331, 845 573 and 6 672 877 as `TETRA15` meshes. Each mesh is smoothed by the energetic mesh smoothing of Norma, which minimizes a strain energy of the mesh over the positions of its nodes with every boundary node held on its surface; this raises the smallest shape quality $q = 12\,(3V)^{2/3}/\sum_e l_e^2$, with V the volume and l_e the six edge lengths, from 0.58–0.65 to 0.69–0.71 at every level. The edge midpoints are added with those on the lateral surface projected onto the cylinder (`TETRA10`, the mesh of CT10), and the face and interior nodes are added by the conversion of §4.3 (`TETRA15`). Three elements are compared: TET15-P1 and TET15-P0 in Carina, and the Sierra/SM CT10 with its stabilization parameter $\alpha = 0.1$ and the stabilization exponent 0 of Foulk III et al. [16]. The displacement unknowns of TET15 are 49 761, 339 993 and 2 536 719 on the first three levels; those of CT10 are 109 998, 801 450 and 6 249 960 on the second to fourth.

Table 18: Taylor bar, final radius of the impact face and final length of the bar in mm at 80μs, per element and mesh level h in mm with its element count. Reference: 7.22 for the converged Q1/P0 hexahedron [16]. TET15-P1 and TET15-P0 project $\log J$ (Carina 29c9606; $h = 0.19$ on an A100); TET15-P0 at $h = 0.38$ stopped with an inverted element at 1μs. With $J - 1$ projected, TET15-P1 gave 7.2399, 7.2225 and 7.2255 at $h = 1.5$, 0.75 and 0.38. CT10 at $h = 1.5$ ended at 41.6μs. Driver: `benchmark/tet15-p1/taylor/run.jl`; histories in `benchmark/tet15-p1/taylor/data`.

		TET15-P1		TET15-P0		CT10	
h	elements	radius	length	radius	length	radius	length
1.5	3522	7.2405	21.4262	8.0915	21.5434	—	—
0.75	25 029	7.2217	21.4199	8.0249	21.5075	7.2820	21.4760
0.38	190 727	7.2211	21.4186	—	—	7.2453	21.4395
0.19	1 521 212	7.2242	21.4193	—	—	7.2326	21.4304

Stable time step. The step of central differences is stable when $\Delta t \leq 2/\sqrt{\lambda_{\max}}$, with $\lambda_{\max}$ the largest eigenvalue of $M_L^{-1}K$, M_L the lumped mass and K the tangent stiffness. The runs of Table 18 set the step to a Courant number of 0.25 times an element estimate, the smallest distance between two nodes of an element divided by the dilatational wave speed c_p, recomputed every ten steps on the current configuration. That estimate is not a bound: $\lambda_{\max}$ computed by power iteration on the TET15-P1 run at $h = 0.75$ mm gives a ratio of the critical step to the element estimate of 1.14 at 1μs, 1.02 at 10μs, 0.69 at 20μs, 0.42 at 28μs, 0.31 at 40μs and 0.29 at 60 and 80μs. At a Courant number of 0.5 the run stopped with a non-finite residual at 28μs; at 0.25 it reached 80μs with a step 0.82 times the critical one at 40μs; TET15-P0 at $h = 0.38$ mm stopped at 78.7μs, and its radius at 78μs is reported, the radius being constant within 0.06% after 50μs for every element (Figure 5). Carina also offers the step $0.8 \cdot 2/\sqrt{\lambda_{\max}}$, with $\lambda_{\max}$ computed every 200 steps and the element estimate scaled by the ratio of the two in between; on the same run it takes 43 863 steps against about 54 000 at the Courant number 0.25, and the final radius differs by 0.0007 mm.

Accuracy. Table 18 and Figure 5. CT10 converges from above: its differences between successive levels are 0.0367 and 0.0127 mm, a ratio of 2.89, which corresponds to an order of 1.53 in h; the extrapolation of its three levels by that ratio gives 7.2259 mm. TET15-P1 is at 7.2217, 7.2211 and 7.2242 mm at $h = 0.75$, 0.38 and 0.19, within 0.07% of that value, and at 7.2405 on the coarsest mesh, where CT10 did not reach 80μs. CT10 is 0.78%, 0.27% and 0.09% from the extrapolated value at $h = 0.75$, 0.38 and 0.19. Both elements approach the reference within 0.1% in the limit. The extrapolation and the Q1/P0 value of Foulk III et al. [16], 7.22 mm, given to three significant digits, differ by 0.08%, which is taken as the uncertainty of the limit. The sequence of TET15-P1 is not monotone, and its differences from the two reference values at $h \leq 0.75$ mm, and the difference of CT10 at $h = 0.19$ mm from the extrapolation, are of the order of that uncertainty. TET15-P0 differs from TET15-P1 only in the constant pressure, and it is sensitive to the projected measure where TET15-P1 is not: it is 11% above the extrapolated value at $h = 0.75$ and inverts an element at 1μs at $h = 0.38$; projecting $J - 1$ instead of $\log J$ changes the TET15-P1 radius by 0.01% at $h = 1.5$ and 0.75 and by 0.06% at $h = 0.38$. A constant pressure leaves the pointwise J free to vary across the

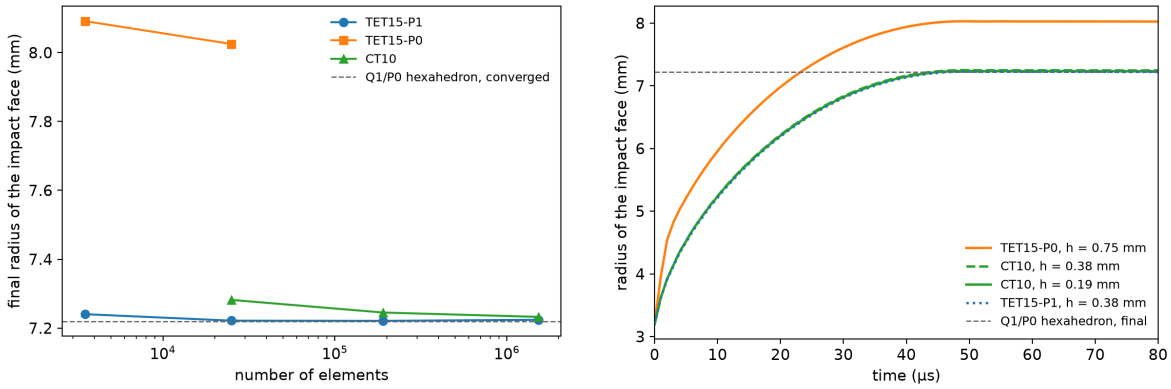


Figure 5: Taylor bar. Left: final radius of the impact face against the number of elements. Right: radius of the impact face against time for TET15-P1 at $h = 0.38$ mm, TET15-P0 at $h = 0.75$ mm and CT10 at $h = 0.38$ and 0.19 mm. Dashed gray: the converged Q1/P0 hexahedron of Foulk III et al. [16]. Carina projects $\log J$. Script: `benchmark/tet15-p1/taylor/plot.py`.

element, and the element mean of $\log J$, unbounded as $J \to 0$ at a single quadrature point, is then far from that of $J - 1$; the linear pressure of TET15-P1 keeps the pointwise J close to the projected one.

Operations. The two codes were run on the same machine, a two-socket AMD EPYC 9634 with 168 cores, one run at a time with no other job, at $h = 0.75$ mm (Table 19). Per element and time step on one core, TET15-P1 takes 1.40 times the time of CT10; per element degree of freedom, 45 against 30, it takes 0.93 times as long. On 16 cores the ratio per element and step is 3.0, because Carina's time per step decreases by a factor of 6.5 from 1 to 16 threads, a parallel efficiency of 0.40, against 0.88 for Sierra/SM on 16 processes; the difference is in the implementation of the time step in Carina, not in the element. The time step of TET15-P1 at $t = 0$ is 3.86×10^{-9} s under the element estimate at the Courant number 0.25, and its critical step is 1.77×10^{-8} s; that of the CT10 is 2.44×10^{-8} s.

The comparison above is between two codes. Within one code, Sierra/SM at $h = 0.75$ mm, Sierra/SM TET15-P1 takes 20.4μs and CT10 7.8μs per element and time step on one core, a factor of 2.6. The time of TET15-P1 is an upper bound, because that run shared the machine with other work. Per element degree of freedom, 45 against 30, the factor is 1.7. In Sierra/SM at $h = 0.75$ mm the time step at 1μs is 3.86×10^{-9} s for TET15-P1 and 2.76×10^{-8} s for CT10. The step of CT10 decreases with the deformation, by a factor of 24 at $h = 0.75$ mm and 15 at $h = 0.38$ mm by 80μs, against 2.9 and 5.5 for TET15-P1. At $h = 0.75$ mm the two runs take 50 832 and 44 260 steps. From $h = 0.75$ to 0.38 mm the step count of CT10 grows by a factor of 1.42, not by the factor of 2 of a step proportional to h. This is consistent with a step set by the deformation of the most compressed elements rather than by the initial element size, and it makes a ratio of costs between the two elements depend on the deformation.

Comparison at equal accuracy. TET15-P1 at $h = 0.75$ mm is closer to the extrapolated radius (0.06%) than CT10 at $h = 0.19$ mm (0.09%). On the machine above, each run alone, the first takes 1303 s on 16 cores in Carina, 2.1×10^4 core-seconds, and the second 10 983 s on

Table 19: Taylor bar at $h = 0.75$ mm (25 029 elements), both codes on the same two-socket AMD EPYC 9634, each run alone. One-core figures from runs to 5μs. Core-μs: wall time times the number of cores. Memory of Sierra/SM: the largest of its processes.

	TET15-P1 (Carina)	CT10 (Sierra/SM)
per element and step, 1 core (μs)	6.27	4.47
per element and step, 16 cores (core-μs)	15.5	5.09
steps to 80μs	52 100	44 260
time stepping to 80μs on 16 cores (s)	1265	352
largest resident memory (GB)	1.63	0.27
final radius (mm)	7.2225	7.2820

Table 20: Taylor bar: cost and accuracy of every run. "Unknowns" counts displacement unknowns. "Deviation" is the final radius relative to the extrapolation of CT10, 7.2259 mm. All runs on the same two-socket AMD EPYC 9634 machine; the runs marked * were made alone on the machine; the others were not, and their times are upper bounds. n.r.: not recorded.

element	code	h	unknowns	steps	processes	wall (s)	core-s	deviation
TET15-P1	Sierra/SM	1.5	49 761	24 480	16	99	1.6×10^3	+0.12%
TET15-P1	Carina	0.75	339 993	52 100	16	1303*	2.1×10^4	−0.06%
TET15-P1	Sierra/SM	0.75	339 993	50 832	96	271	2.6×10^4	−0.10%
TET15-P1	Sierra/SM	0.38	2 536 719	187 209	160	6473	1.0×10^6	−0.11%
CT10	Sierra/SM	0.75	109 998	44 260	96	90	8.6×10^3	+0.78%
CT10	Sierra/SM	0.38	801 450	62 821	160	626	1.0×10^5	+0.27%
CT10	Sierra/SM	0.19	6 249 960	n.r.	128	10 983*	1.4×10^6	+0.09%

128 processes in Sierra/SM, 1.41×10^6 core-seconds: a factor of 67 in core-seconds and of 8.4 in wall time, on 25 029 elements against 1 521 212. That factor compares two codes. Table 20 gives the cost of every run with the deviation of its radius from the extrapolation. The first level at which the radius is within 0.11% of the extrapolation is $h = 0.75$ mm for TET15-P1 in both codes and $h = 0.19$ mm for CT10. Within Sierra/SM, the first costs 2.6×10^4 core-seconds and the second 1.4×10^6, a factor of 54 on 61 times fewer elements and 18 times fewer displacement unknowns. The factor depends on the tolerance, and the tolerance of 0.11% is close to the uncertainty of the limit, 0.08%. With a tolerance of 0.3%, TET15-P1 qualifies at $h = 1.5$ mm in both codes (+0.12% in Sierra/SM, +0.19% in Carina), at 1.6×10^3 core-seconds in Sierra/SM, and CT10 at $h = 0.38$ mm, at 1.0×10^5 core-seconds, a factor of 63. The core-second counts at different numbers of processes include differences of parallel efficiency, and the comparison is indicative. At equal mesh CT10 takes 3.6 times less time to 80μs on 16 cores, with an error 13 times larger.

The same run on GPUs. The explicit integration of TET15-P1 runs on a GPU through the same kernels, with the stable step recomputed on the device. At $h = 0.75$ mm with the global stable step, the radius and the length of every output frame of the run are the same numbers to every written digit on the CPU, on an AMD RX 7600 and on NVIDIA L4, V100, A100 and H100 cards (the last four on one mesh, the first two on another smoothing of the same Cubit mesh, which changes the radius by 2.8×10^{-7} relative), and so are the 221 stable-step estimates.

Table 21 gives the time per step from the kernel timings and the wall time of the run. Three defects were found and corrected in making these runs: the stable-step recomputation read the block sizes and the material properties through accessors defined for host arrays only, so it had never run on a device; the temporaries of the explicit step (about 0.65 MB per step at 340 000 unknowns) are released only when the garbage collector runs, which host pressure did not trigger, so the run exhausted the 8 GB of the RX 7600 after about 5 000 steps, and the collector is now run every 100 steps on a device; and the J_2 tangent of the material library built its pull-back in mutable buffers that CUDA places on the device heap, which is never freed, so the matrix-free solves of the implicit tests failed on the L4 after some thousands of tangent calls (the buffers are now immutable, and the tangent is 5% faster on every device). The eigenvalue estimate of the stable step runs 20 to 40 power iterations, each two residual evaluations, every 200 steps. Its share of the run, from the saving of the adaptive interval below, is 14.8% on the RX 7600, 14.4% on the L4, 11.4% on the V100 and about 3% on the H100. An earlier estimate of 25 to 40% on the GPUs, from a single timed estimate, was too high: timed once right after the warm-up of the timing script, the estimate took 1.5 to 4.4 times the median of five. A fixed longer interval is not safe: the ratio of the eigenvalue step to the element estimate falls by up to 11% between two estimates 200 steps apart while the bar spreads, between 8 and 30μs, and by up to 23% within 1000 steps, more than the 20% margin of CFL 0.8; after 30μs it varies by at most 1.3% per 200 steps, and after 60μs by 0.04%. Carina therefore sets the interval from the observed change (input key `stable time step eigenvalue change` c): after each estimate the next interval is the one over which the relative decrease of the ratio observed over the last interval would amount to c, at least the given interval, at most twice the last one and at most 16 times the given one. With $c = 0.02$ the run at $h = 0.75$ mm on the RX 7600 takes 52 estimates instead of 220, at intervals from 200 to 3200 steps, 43 557 steps instead of 43 865, and 443 instead of 500 s, with the radius and length histories equal to within 7×10^{-7} relative; the same runs take 399 instead of 449 s on the L4, 367 instead of 402 s on the V100 and 155 instead of 159 s on the H100 (a difference within the variation between runs), with the same histories, frame for frame, on the four NVIDIA cards. Applied to the ratios recorded with the fixed interval, the rule keeps the largest decrease of the ratio within an interval at the one over 200 steps.

A second implementation. An implementation of TET15-P1 in Sierra/SM, Sierra/SM TET15-P1, was run on the same TETRA15 meshes with the finite-deformation J_2 model of the Sierra material library and linear hardening, as in the CT10 runs, and a stable step of 0.25 times the smallest node distance over c_p (the Sierra/SM element estimate, 0.5 times that distance over c_p, with an input factor of 0.5). The final radius is 7.2344, 7.2186, 7.2181 and 7.2217 mm at $h = 1.5$, 0.75, 0.38 and 0.19 mm: 0.084%, 0.043%, 0.042% and 0.035% below Carina's at the four levels, both projecting $\log J$ (with $J - 1$ projected in Carina the differences were 0.08%, 0.05% and 0.10% at the first three), and within 0.12% of CT10's extrapolated limit at every level (0.06% below it at $h = 0.19$ mm). Both codes give a sequence that is not monotone at the finest level: the radius at $h = 0.19$ mm is 3.6μm above the $h = 0.38$ mm value in Sierra/SM and 3.1μm above it in Carina. Over the run the radii of the two codes differ by at most 0.16%, in the first 10μs, and after 40μs by a nearly constant 0.035 to 0.084% that decreases with h (Figure 6); the two codes' J_2 implementations and time steps differ. Without the projection, the plain `TETRA15` element in Sierra/SM ends at 7.1268 mm at $h = 1.5$ mm, 1.57% below TET15-P1 in Carina at the same h and 1.37% below the extrapolation, and at 7.1900 mm

Table 21: TET15-P1, split form, explicit Taylor bar: time per step of a run from the kernel timings (median of 20, deformed state with plastic strains to 1.0; includes the stable-step estimates at their intervals), in microseconds per element-step, at $h = 0.75$ and 0.38 mm, the ratio of the general form's time to the split form's at $h = 0.75$ mm, and the wall time of the run to 80μs at $h = 0.75$ mm. GPU rows and the Ryzen row: Carina 36b4a9f and 178e811, the eigenvalue estimate as the median of five (2026-10-06). EPYC rows (*): Carina 9e4bfaf, the eigenvalue estimate timed once (2026-10-04), which on a CPU agrees with the median (on the Ryzen 459.6 against 457.1 ms). Wall times: Carina 9e4bfaf, the H100 36b4a9f. One process alone on each GPU. Scripts: `benchmark/tet15-p1/taylor/kernel_timing.jl`, `timing_table.py`; records in `data/`.

device	$h = 0.75$	$h = 0.38$	general/split	run (s)
AMD Ryzen 9 9900X, 12 threads	0.704	—	1.39	839
2×AMD EPYC 9634, 32 threads*	0.700	0.772	1.36†	1283†
2×AMD EPYC 9634, 64 threads*	0.836	0.652	—	—
AMD RX 7600	0.430	—	1.42	513
NVIDIA L4	0.321	0.279	1.71	406
NVIDIA V100	0.271	0.209	2.27	404
NVIDIA A100	0.119	0.097	2.18	227
NVIDIA H100	0.069	0.052	2.43	159

†at 16 threads.

at $h = 0.75$ mm, 0.40% below Sierra/SM TET15-P1 and 0.50% below the extrapolation. Its deficit decreases by a factor of 2.7 over this halving of h. Two levels do not separate locking from discretization error. The step used makes the Sierra/SM TET15-P1's step 7 times smaller than CT10's at 1μs. The critical step $2/\sqrt{\lambda}$, with λ from a converged global estimate, is 2.2 to 3.1 times the element estimate alone before 20μs and 0.60 to 0.62 times from 40μs on, at $h = 1.5$ and 0.75 mm: the element estimate alone is not a bound on the deformed meshes, as Carina's node-distance estimate was not (§11), and the runs were stable through the input factor, with a margin of about 1.2. The global estimate of Sierra/SM, a power iteration for the largest eigenvalue of $M^{-1}K$, diverged on these meshes. It forms $K\,x$ from the internal force under a trial displacement of fixed size, 10^{-5} in model length units (10μm here); once the iterate concentrates on a few nodes, that displacement inverts quadrature points of the 15-node elements, whose nodes are 0.1 to 0.4 mm apart, and the estimate rises from 2.66×10^{15} to 6.6×10^{17} s^{-2}. The same size makes CT10's first estimate 22% low at $h = 0.75$ mm. With the trial displacement set to 10^{-5} of the local element length, and the iteration stopped only when the remaining change extrapolated from successive changes is below 5×10^{-4}, the global estimate runs to 80μs with 22% and 15% fewer steps than the element estimate at $h = 1.5$ and 0.75 mm and a final radius within 0.007%; the power iterations add 3 to 5% to the number of steps. Its pressure at 12μs, one value per element, has the distribution of Carina's on every mesh, with a largest value of 1.56, 1.62, 1.63 and 1.63 GPa against Carina's 1.67, 1.58, 1.56 and 1.54 GPa at $h = 1.5$, 0.75, 0.38 and 0.19 mm (Figure 7). TET15-P1 took 271 and 6 473 s at $h = 0.75$ and 0.38 mm on 96 and 160 MPI ranks, against 90 and 626 s for CT10, and 598 681 steps at $h = 0.19$ mm on 168 ranks, 71 h on a machine shared with other work (`benchmark/tet15-p1/taylor/README.md`).

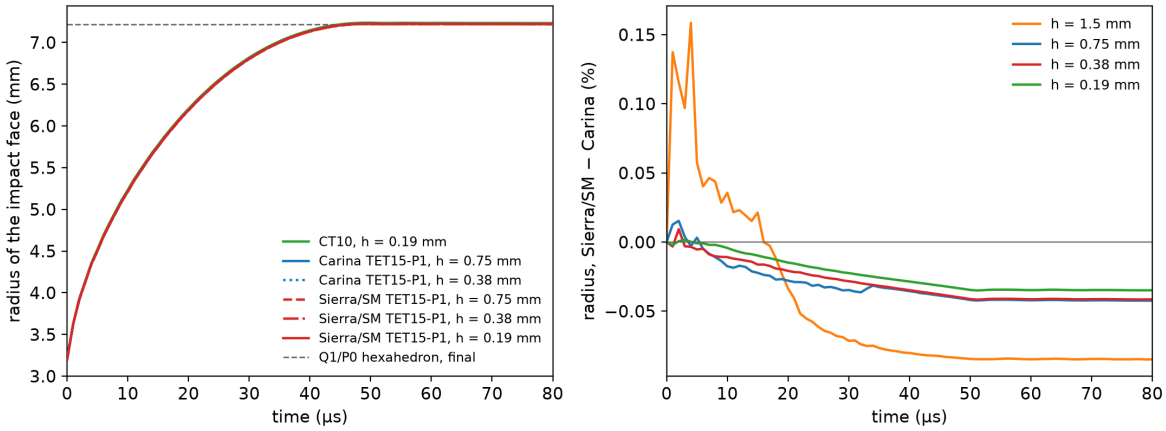


Figure 6: Taylor bar: TET15-P1 in Carina and in Sierra/SM on the same meshes. Left, the radius of the impact face against time at $h = 0.75$ and 0.38 mm in both codes, Sierra/SM TET15-P1 at $h = 0.19$ mm, and CT10 at $h = 0.19$ mm; right, the relative difference of the two codes' radii at the four levels, both projecting $\log J$.

Pressure field. Figure 8 shows the pressure $p = -\frac{1}{3}\operatorname{tr}\boldsymbol{\sigma}$, positive in compression, on the impact face at $t = 12\boldsymbol{\mu}$s, the time of Figure 12 of Foulk III et al. [16], for both elements on the same meshes. CT10 has one value per element, the element mean of the stress that Sierra/SM writes; for TET15-P1 the pressure in each element is the quadratic fitted to its fourteen quadrature-point values, as in §10. The largest pressure on the impact end is 0.33, 0.48, 1.37 and 1.59 GPa for the CT10 at $h = 1.5$, 0.75, 0.38 and 0.19 mm, and 1.67, 1.58, 1.56 and 1.54 GPa for TET15-P1, between 3% below and 5.2% above CT10's value on its finest mesh at every level; on that mesh the two differ by 3%. On the two coarser meshes the pressure of CT10 varies from element to element without a maximum on the axis; TET15-P1 gives the maximum on the axis on every mesh. The radius of the impact face at $12\boldsymbol{\mu}$s is 5.4944, 5.4957, 5.4750 and 5.4661 mm for the CT10 and 5.4517, 5.4507, 5.4533 and 5.4570 mm for TET15-P1. Foulk III et al. [16] plot the L^2 projection of CT10's pressure onto a continuous field, and the range of their color scale, −1.2 to 0.32 GPa, indicates the mean stress $\frac{1}{3}\operatorname{tr}\boldsymbol{\sigma} = -p$. Scripts: `benchmark/tet15-p1/taylor/pressure.jl`, `render_pressure.py` and `compose_pressure.py`.

What the benchmark establishes. In explicit dynamics with plastic strains above 3, TET15-P1 gives the final radius of the Taylor bar within 0.06% of the extrapolated value on a mesh of 25 029 elements, where the CT10 needs 1.5 million elements to reach 0.09%; both differences are of the order of the uncertainty of the limit, 0.08%. The constant-pressure variant on the same displacement space is 11% from it at $h = 0.75$ mm. On 3522 elements its largest pressure on the impact face at $12\boldsymbol{\mu}$s is 5.2% above that of CT10 on 1.5 million, and on 1.5 million elements the two differ by 3%. Within Sierra/SM its time per element and step is at most 2.6 times that of CT10 on one core, and the cost at a tolerance of 0.11% is 1/54 of that of CT10. The element estimate of the stable step used by Carina is not a bound on distorted elements; the eigenvalue estimate replaces it. The equivalent plastic strain was not written in these runs and is not compared.

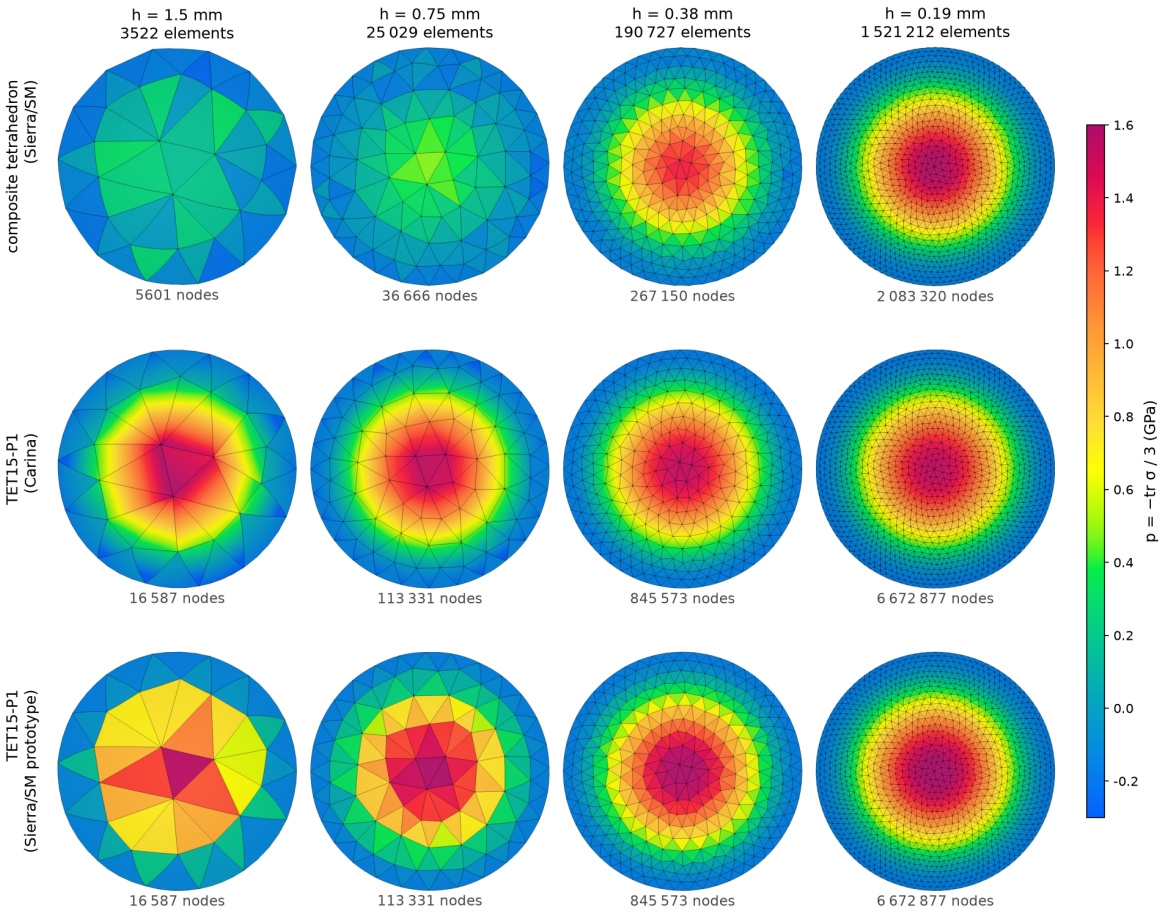


Figure 7: Taylor bar: pressure on the impact face at 12μs on the scale of Figure 8, for the Sierra/SM CT10 (), TET15-P1 in Carina and TET15-P1 in Sierra/SM (one value per element).

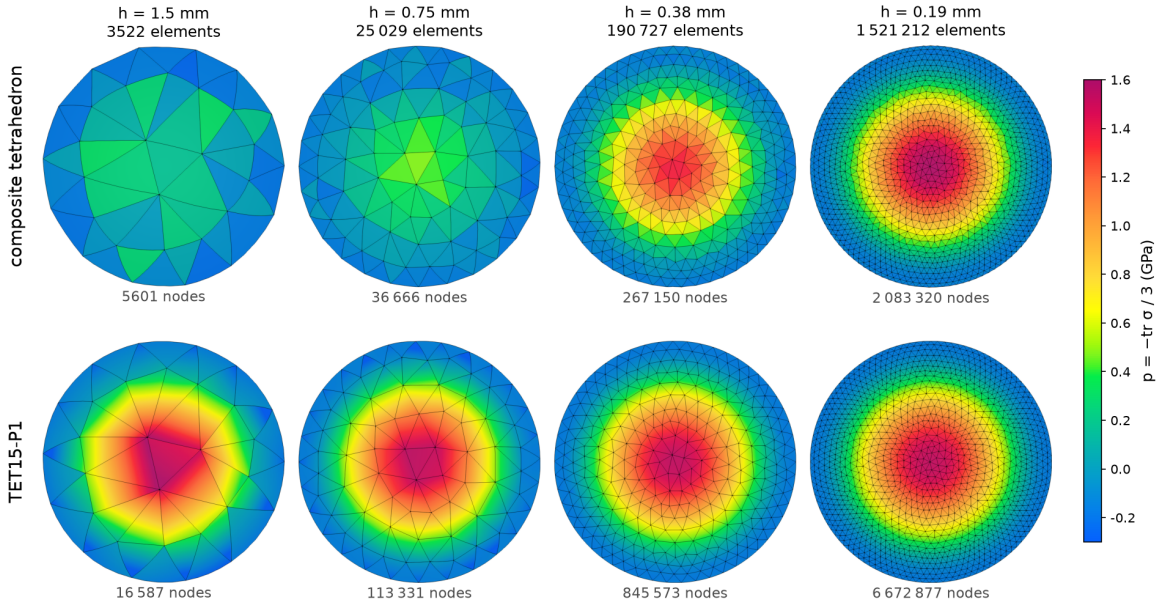


Figure 8: Taylor bar, pressure $p = -\frac{1}{3}\operatorname{tr}\boldsymbol{\sigma}$ on the impact face at $t = 12\boldsymbol{\mu}$s, seen along the axis, on the same meshes for both elements, with the element count of each level and the node count of each mesh (`TETRA10` for CT10, `TETRA15` for TET15-P1); one color scale, values beyond it saturate. Top: CT10 (Sierra/SM, $\alpha = 0.1$, exponent 0), one value per element. Bottom: TET15-P1, the quadratic fitted in each element.

12 Quasi-static plasticity: indentation by a flat punch

This section measures the element in quasi-static finite-deformation plasticity with a plastic zone confined by elastic material at $\kappa/\mu = 10^4$, the regime in which Foulk III et al. [16] report a soft mode of constant-pressure elements and the linearized analysis of §9.7 finds the dilatation of their softest modes not to decay, against CT10, Sierra/SM TET15-P1 and the plain fifteen-node tetrahedron, on the same meshes. The problem has no published three-dimensional tetrahedral instance; its definition is the reference. The measured quantities are the punch force, the element-to-element jumps of the pressure and of the equivalent plastic strain near the punch, and the departure of the surface displacement from axisymmetry, on four levels of refinement.

Problem. A cylinder of radius $R = 4a$ and height $H = 4a$, $a = 1$ mm, rests on a frictionless rigid plane ($u_z = 0$ on the base, u_x and u_y free) and is indented on its top face by a frictionless rigid flat punch of radius a centered on the axis: $u_z = -d(t)$ on every node of the footprint faces, including the nodes on the circle $r = a$, with u_x and u_y free, and $d(t) = 0.2a\,t$ for $t \in [0, 1]$, with output at the twenty times $t = 0.05, 0.10, \ldots, 1$. The problem is a displacement-controlled problem with a fixed footprint, not a contact problem. It coincides with frictionless contact by a rigid punch if every footprint node stays in compression and the material outside the footprint does not meet the punch. The sign of the nodal reactions was not checked, so this coincidence is not established. The computational domain is the quarter $x \geq 0$, $y \geq 0$, with $u_x = 0$ on the plane $x = 0$ and $u_y = 0$ on the plane $y = 0$; these conditions remove the rigid-body translations in x and y and the rotation about z, so that both codes solve the same discrete problem with no further constraint, and the full punch force is four times the reaction of the quarter. The material is the J_2 model with linear isotropic hardening: $\mu = 10^9$ Pa, $\kappa = 10^{13}$ Pa ($E = 2.9999 \times 10^9$ Pa, $\nu = 0.49995$), yield stress $\sigma_y = 10^6$ Pa, hardening modulus $H = 10^7$ Pa (case A) or $H = 0$ (case B); the Simo–Hughes model (Remark 5) in Carina and both the hypoelastic rate form and the multiplicative hyperelastic form of the Sierra material library, which agree with each other to five or six digits in force and to three in the field statistics in every completed pair of runs but one (see Results). The mean pressure under a smooth circular punch on a rigid-perfectly-plastic half-space at incipient flow is about $5.7\,\sigma_y/\sqrt{3}$ [27], a full force of about 10.3 N in case B; at small depth the force is the elastic flat-punch value $8\mu ad$ of an incompressible half-space. The computed forces of case B at $t = 1$, about 14.0 N, exceed 10.3 N by 36%. The estimate does not include the elastic deformation, the finite depth of $0.2a$ with its pile-up, or the finite body of radius $4a$.

Meshes and elements. The quarter is meshed by Cubit with four-node tetrahedra of size h within $r \leq 2a$, $z \geq H - 2a$ and $4h$ elsewhere, the quarter circle of the footprint imprinted on the top face so that its edge lies on element edges, and smoothed by Cubit's condition-number smoother only, so that the footprint circle and the symmetry planes stay exact. Four levels, $h = a/2$, $a/4$, $a/8$ and $a/16$, have 944, 6187, 38 818 and 277 530 tetrahedra, 1566, 9146, 54 582 and 380 564 nodes as `TETRA10` meshes and 4537, 28 129, 172 391 and 1 218 149 as `TETRA15` meshes. The `TETRA10` meshes place the edge nodes on the lateral surface on the cylinder and those on the footprint circle on that circle. The smallest shape quality $6\sqrt{2}\,V/\ell_{\mathrm{rms}}^3$ of the elements (§9.5) is between 0.45 and 0.49 on the three finer levels. Four elements are compared on each mesh: the Sierra/SM CT10 with its stabilization parameter $\alpha = 0.1$ and with $\alpha = 0$ (`TETRA10` meshes), Sierra/SM TET15-P1 and TET15-P1 in Carina (`TETRA15` meshes), and the

plain fifteen-node tetrahedron without the projection in both codes, the locking reference.

Time stepping. The first iterate of a step from the converged solution with only the footprint displacement advanced carries the whole increment Δd in the first element layer: a volumetric strain $\Delta d/h$ and a pressure $\kappa\,\Delta d/h$, 400μ at the nominal step of $0.01a$ at $h = a/4$. Newton converges from that state only while $(\Delta d/h)(\kappa/\mu)$ is below about 1, that is Δt below about 1.2×10^{-4} at $h = a/4$ (measured: steps converge at 9.8×10^{-5} and fail at 1.95×10^{-4}), which would take 10^4 steps to $t = 1$. Carina therefore starts each step from $U_n + \Delta U$ with $K_n \Delta U = -K_n[0; \Delta g]$, the response of the tangent at the converged state to the increment Δg of the prescribed values, computed by the matrix-free action with Δg in the Dirichlet slots (`predictor: tangent`); the right-hand side is not the residual at the advanced boundary values, whose second-order part at $\kappa/\mu \gg 1$ has a component outside the range of the volumetric stiffness that only μ resists and gives an increment of order $(\Delta d/h)(\kappa/\mu)\,\Delta d$, $400\,\Delta d$ at $h = a/4$, in place of the lift of order Δd. From the predicted state the error is of second order, and the admissible step is set by $(\Delta d/h)^2\kappa/\mu \lesssim 1$: both codes start at $\Delta t = 0.05/4096$, double after each converged step up to a maximum of 0.00625 at $h = a/4$ halved per level, halve a failed step and regrow by 1.5, with the twenty output times as block boundaries; Sierra/SM uses its load-step predictor, the previous increment scaled by the ratio of the steps. At 0.0125 on $h = a/4$ Newton needs about fifty iterations with cutbacks; at 0.00625, between two and twenty. The cost of the nonlinear iteration is the change of the active set near the punch edge: quadrature points switch between elastic and plastic states under the iteration, the merit function rises by two orders of magnitude at the full Newton step, and the line search accepts $\alpha = 1/8$ or less with a decrease of 2 to 10% per iteration until the active set stops changing. The algorithmic tangent of the J_2 model, the symmetric part of the derivative of the return map, differs from that derivative by 4 to 6×10^{-5} of its norm at a homogeneous plastic state in both elements and is not the cause: the homogeneous plastic cube converges in two to three iterations per step with it. The exact linear solves of a direct solver give the same counts as the iterative ones. Applying the Armijo condition to the first Newton iteration after a kept prediction lowers the count in case A from 124 to 77 iterations over the first fifteen steps.

Linear solver. Carina solves the Newton systems by conjugate gradients with the smoothed-aggregation multigrid V-cycle of §7 applied on the device; Sierra/SM by conjugate gradients with a FETI preconditioner on 8 to 16 ranks. At $\kappa/\mu = 10^4$ the multigrid hierarchy built on the tangent $K(\kappa)$ coarsens on the volumetric couplings and its damped Jacobi smoother, scaled to the diagonal of $K(\kappa)$, leaves the deviatoric error to the Krylov iteration: on the tangent of the $h = a/4$ full cylinder (325 414 unknowns) the count to a relative residual of 10^{-8} grows as $\kappa^{0.4}$, 274, 747, 1283 and 1982 for $\kappa/\mu = 10^2$ to 10^5, against 65 to 75 for the FETI preconditioner on the same meshes. The same V-cycle built on $K(\kappa')$, the tangent of the material with its bulk modulus set to $\kappa' = \mu$, and applied to $K(\kappa)$ needs 330, 483, 523 and 678; the saddle-point form in displacement and projected pressure with the augmented-Lagrangian block preconditioner $\mathrm{diag}\big(K(\kappa'),\, (1/\kappa + 1/\kappa')M\big)$ needs 326 to 340 independent of κ but new device kernels. The first is one property substitution in the host assembly of the hierarchy and in the fine smoothing action (`operator: compressible`); on the quarter $h = a/4$ mesh to $t = 0.025$ it takes 34 637 conjugate-gradient iterations in 55 solves against 115 416 in 57, and 700 to 1000 iterations per solve at the plastic steps against 2500 to 3700.

The damped Jacobi smoother of the V-cycle uses the weight $\omega = 4/(3\lambda_{\max})$, where $\lambda_{\max}$

is an estimate of the largest eigenvalue of $D^{-1}K'$ and D is the diagonal of K'. Under plastic flow this eigenvalue rises from 2.64 in the elastic state to 4.2 to 5.2. With the estimate of the elastic state the V-cycle is not positive definite, and 46 to 60% of the solves in case B had $r^{\mathsf{T}}M^{-1}r < 0$, where M^{-1} is the action of the V-cycle. Carina now estimates $\lambda_{\max}$ again before every solve. A solve that still meets $r^{\mathsf{T}}M^{-1}r \leq 0$ is logged and continued with the diagonal preconditioner. No production run with the new estimate has met that condition.

In case B the tangent loses positive definiteness near $t = 0.08$, and the conjugate-gradient iteration assumes a positive definite operator. A direct solver measures the cost of this property. The case B run at $h = a/4$ (82 521 free unknowns) was repeated with the sparse direct solver of Carina on 36 threads of one CPU node. The solver factors the tangent by Cholesky decomposition (CHOLMOD) and changes to LU decomposition when the Cholesky factorization fails. The Cholesky factorization failed for the first time at the step from $t = 0.0813$ to 0.0875, and every later solve used LU decomposition. A factorization and solve took 1.9 s on average with Cholesky decomposition and 9.6 s with LU decomposition. The peak resident memory was 28.7 GiB. The run reached $t = 1$ in 168 steps without a cutback, with 1612 Newton iterations, in 4.7 h. To $t = 0.40$ the summed time of the linear solves was 7079 s with the direct solver and 16 422 s with the conjugate-gradient iteration on one NVIDIA A100 GPU. Over the same interval the two runs took 812 and 839 Newton iterations and 0 and 5 cutbacks. Of the 953 conjugate-gradient solves, 850 stopped at the limit of 1000 iterations. The difference in cost is therefore in the linear solve, not in the nonlinear iteration. To $t = 1$ the conjugate-gradient run took 173 steps, 5 cutbacks, 1656 Newton iterations and 1866 linear solves, of which 1666 stopped at the limit, with a summed solve time of 32 747 s and a wall time of 9.3 h. Its force at $t = 1$ is 14.1124 N, 0.002% above that of the direct solver, and its jump statistics agree with those of the direct solver to four digits. The preconditioner, not the Krylov method, limits the iterative solver on this problem.

The Carina runs of case B at $h = a/8$ use the conjugate-gradient iteration with the re-estimated smoother bound, on one NVIDIA A100 GPU each. TET15-P1 uses the general form of the projected volumetric term (§3.5) with the V-cycle built on the tangent with $\kappa' = \mu$, and plain `TETRA15` the same V-cycle. On 2026-10-20 at 05:10 UTC they had reached $t = 0.59$ and $t = 0.63$. The Carina runs at $h = a/16$ were stopped on 2026-10-17, and Carina results are reported up to $h = a/8$. Per conjugate-gradient iteration, the operator of the general form with $\theta = \log J$ costs 1.9 times that of the default form.

The jump measure. Each code writes one value of the pressure $p = -\operatorname{tr}\sigma/3$ and of the equivalent plastic strain per element: in Carina the mean of the fourteen quadrature-point values, in Sierra/SM the element value of the code. Over the interior faces whose two elements have their centroids within $r \leq 1.5a$ and $z \geq H - 1.5a$ in the reference configuration, the region around the plastic zone (2548 faces at $h = a/4$, 20 491 at $a/8$), the jump of a quantity q across a face is $|q_L - q_R|$ of the two element values divided by the largest $|q|$ over the elements of the region; the statistics are the mean, the 99th percentile and the maximum over the faces, at $t = 0.5$ and $t = 1$. For a field that the mesh resolves, two neighboring elements differ by about h times the gradient, so the normalized jump is of order $h\,|\nabla q|/\max|q|$ and halves when h halves; for an oscillation that alternates from element to element, a pressure checkerboard, the two neighbors differ by the amplitude of the oscillation whatever h is, and the jump does not decrease. The measure therefore separates a resolved gradient from a spurious mode by its behavior under refinement, not by its value on one mesh: the quantity of interest is the ratio of

Table 22: Punch indentation, quarter model: full punch force in N at $t = 1$. Carina TET15-P1 projects $\log J$; its case B run at $h = a/8$ is in progress ($t = 0.59$ on 2026-10-20 at 05:10 UTC), and the Carina runs at $h = a/16$ were stopped before $t = 1$ on 2026-10-17. The Sierra/SM TET15-P1 runs of case B at $h = a/16$ are in progress. Plain `TETRA15`: Carina (the Sierra/SM runs at $a/8$ ended at a time limit at $t \approx 0.08$); the Carina case B run at $a/8$ is in progress ($t = 0.63$ on 2026-10-20 at 05:10 UTC); the two codes agree to 0.02% at $a/2$ and $a/4$. Entries marked — are in progress or were not computed.

	Case A, $H = 10^7$ Pa				Case B, $H = 0$			
element	$a/2$	$a/4$	$a/8$	$a/16$	$a/2$	$a/4$	$a/8$	$a/16$
CT10, $\alpha = 0.1$	25.775	24.954	24.310	24.058	17.373	14.645	14.076	14.576
CT10, $\alpha = 0$	24.450	24.202	24.034	24.010	15.380	14.025	14.094	14.051
Sierra/SM TET15-P1	24.442	24.160	24.056	24.007	14.666	14.116	14.044	—
Carina TET15-P1	24.434	24.155	24.052	—	14.657	14.112	—	—
plain `TETRA15`	28.206	27.221	25.557	—	19.328	18.863	—	—

the statistics between two levels. The mean is dominated by the bulk of the region, where the field is smooth, and measures its roughness relative to the resolution; the 99th percentile is the roughest one percent of the faces, 25 at $a/4$ and 205 at $a/8$, where a localized mode appears before it reaches the mean; the maximum is a single face, under a flat punch the face at the punch edge where the pressure changes sign through the edge singularity, of order one for every element and every mesh, and carries no information about modes. The measure has three limitations. The normalization divides by the peak of the region, which a mode contaminates: the peak pressure of the plain element at $h = a/4$ is 146 MPa in Sierra/SM and 594 MPa in Carina against 7.5 MPa for TET15-P1, so its normalized jumps understate its mode, and the peak of CT10 is 1.7 to 2.7 times that of TET15-P1 at both levels, so the normalized ratio between the two understates the ratio in pressure units. The measure sees element averages only: the linear variation of the TET15-P1 pressure within an element and the variation over the sub-tetrahedra of CT10 are invisible to it, so a mode confined within elements would not register. Two levels give one ratio, which distinguishes a decrease from no decrease but not first order from second.

Results. Tables 22 and 23 give the full punch force at $t = 1$ and the jump statistics of the pressure at $t = 1$ on the four levels; Carina projects $\log J$, as Sierra/SM TET15-P1 does (Remark 5). The Sierra/SM rows are the hypoelastic model; the multiplicative model agrees with it to five digits in force and three in the statistics in every completed pair but one, and replaces it in CT10 $\alpha = 0$, case B, $h = a/8$, where the hypoelastic run stalled. In Sierra/SM TET15-P1, case B, $h = a/8$, the multiplicative model gives 14.0457 N and the hypoelastic model 14.0440 N. This difference of 0.012% is the exception to the agreement to five digits. The multiplicative run used an absolute residual target of 10^{-6} N to $t = 0.85$, 10^{-5} N from $t = 0.85$ to 0.95, and 10^{-4} N from $t = 0.95$ to 1. At the target of 10^{-4} N the accepted residual is about 10^{-3} N, 7×10^{-5} of the force. Two steps recomputed with the target of 10^{-5} N differ in force by 3.1×10^{-8} and 4.2×10^{-8} relative to the steps at 10^{-4} N.

Table 23: Punch indentation: the element-to-element jump of the pressure at $t = 1$, normalized by the largest $|p|$ of the region; mean and 99th percentile over the faces of the region, with the 99th percentile in MPa (normalized value times the largest $|p|$) and the largest $|p|$ in MPa. Carina TET15-P1 agrees with Sierra/SM TET15-P1 to 0.001 in the mean and 0.01 in the 99th percentile at $h = a/2$ and $a/4$, and in case A at $a/8$. Plain `TETRA15`: Carina, whose element value is the mean of the fourteen quadrature-point values; Sierra/SM writes one value per element and gives a largest $|p|$ of 64 and 146 MPa at $a/2$ and $a/4$.

element		$a/2$	$a/4$	$a/8$	$a/16$
Case A					
CT10, $\alpha = 0.1$	mean / p99	0.158 / 1.073	0.048 / 0.690	0.021 / 0.388	0.008 / 0.142
	p99, max $\|p\|$ (MPa)	10.6, 9.9	14.0, 20.3	8.7, 22.4	3.7, 26.4
CT10, $\alpha = 0$	mean / p99	0.115 / 0.776	0.044 / 0.648	0.019 / 0.370	0.006 / 0.105
	p99, max $\|p\|$ (MPa)	4.6, 6.0	8.1, 12.5	5.0, 13.6	2.0, 19.2
TET15-P1	mean / p99	0.065 / 0.505	0.025 / 0.253	0.010 / 0.117	0.003 / 0.039
	p99, max $\|p\|$ (MPa)	2.5, 4.9	1.9, 7.5	1.2, 10.4	0.6, 16.3
plain `TETRA15`	mean / p99	0.237 / 1.189	0.152 / 0.857	0.058 / 0.525	—
	p99, max $\|p\|$ (MPa)	603, 507	508, 594	635, 1210	—
Case B					
CT10, $\alpha = 0.1$	mean / p99	0.197 / 1.229	0.078 / 0.763	0.030 / 0.409	0.013 / 0.240
CT10, $\alpha = 0$	mean / p99	0.146 / 0.961	0.055 / 0.616	0.024 / 0.343	0.010 / 0.180
TET15-P1	mean / p99	0.074 / 0.504	0.033 / 0.210	0.016 / 0.129	—
plain `TETRA15`	mean / p99	0.345 / 1.504	0.198 / 1.034	—	—

The two implementations. With the same projected measure and different J_2 models, hyperelastic in Carina and hypoelastic in Sierra/SM, the two implementations of TET15-P1 agree in force to 0.035% (case A) and 0.06% (case B) at $h = a/2$ and to 0.017% at $a/4$, at every output time to 0.09%, and in the jump statistics and the axisymmetry measure to two or three digits. At $h = a/8$ in case A the Carina force is 0.015% below that of Sierra/SM. The agreement is a consistency check between two implementations with different material models, not a verification to the tolerance of the nonlinear solve. The source of the remaining difference, 0.02 to 0.06%, has not been identified. The Sierra/SM multiplicative model gives the same force as its hypoelastic model to five digits, which leaves the volumetric law inside the Sierra/SM model and the implementation as candidates. In case B at $h = a/4$ the Carina runs with the direct solver and with the conjugate-gradient solver give 14.1121 and 14.1124 N, against 14.1153 N (multiplicative model) and 14.1158 N (hypoelastic model) in Sierra/SM, differences of -0.023% and -0.026%. With $J - 1$ projected in Carina the difference was 0.25% and 0.35% at $a/2$, 0.05% at $a/4$ and 0.06% at $a/8$: the measure changes the force at second order in the variation of J within an element, largest on the coarsest mesh at the punch edge. The plain element agrees between the codes to 0.02% at $a/2$ and $a/4$.

Locking. The plain fifteen-node tetrahedron is stiffer than TET15-P1 by 15.4%, 12.7% and 6.2% at $h = a/2$, $a/4$ and $a/8$ in case A, and by 32% and 34% at $a/2$ and $a/4$ in case B; the excess falls under refinement. Its pressure carries a mode that does not decay: the largest $|p|$ of the region rises from 507 to 1210 MPa between $a/2$ and $a/8$ in Carina, against about 10 MPa for the projected elements, and the 99th percentile of its jump, in MPa, grows from 508 to 635

Table 24: Punch indentation: ratio of the jump statistics of Table 23 between successive levels, a factor of 2 per halving of h being first order, computed from the unrounded statistics. For CT10 with $\alpha = 0$ in case B the $a/8$ level is the multiplicative hyperelastic model, whose statistics at $a/16$ agree with those of the hypoelastic model to three digits. Some ratios quoted in an earlier version of this note were computed from rounded statistics; the values below replace them.

		normalized 99th percentile			mean		
case	element	$\frac{a/2}{a/4}$	$\frac{a/4}{a/8}$	$\frac{a/8}{a/16}$	$\frac{a/2}{a/4}$	$\frac{a/4}{a/8}$	$\frac{a/8}{a/16}$
A	CT10, $\alpha = 0.1$	1.55	1.78	2.74	3.28	2.33	2.45
A	CT10, $\alpha = 0$	1.20	1.75	3.52	2.60	2.29	3.07
A	TET15-P1	1.99	2.17	3.03	2.67	2.43	2.99
B	CT10, $\alpha = 0.1$	1.61	1.87	1.71	2.52	2.59	2.37
B	CT10, $\alpha = 0$	1.56	1.79	1.90	2.65	2.26	2.44
B	TET15-P1	2.40	1.63	—	2.26	2.00	—

between $a/4$ and $a/8$. Its normalized jumps decrease only because the peak by which they are divided grows faster, which is the limitation of the normalization stated above.

Pressure jumps of TET15-P1 and CT10. Table 24 gives the ratios of the statistics between levels. TET15-P1 has the smallest jumps of the four elements at every level and in both cases: at $h = a/8$ its mean jump is 0.52 (case A) and 0.67 (case B) of that of CT10 without stabilization, and its 99th percentile in MPa is 0.24 and 0.27 of it. At $h = a/16$ in case A its mean jump is 0.53 and its 99th percentile in MPa is 0.31 of those of CT10 without stabilization. In case A its normalized 99th percentile falls by 1.99, 2.17 and 3.03 per halving of h, about first order or faster from $h = a/2$ on; in MPa it falls by 1.3, 1.6 and 1.9, because the largest $|p|$ rises as the punch edge is resolved (4.9, 7.5, 10.4, 16.3 MPa). Its mean falls by 2.99 from $a/8$ to $a/16$. CT10 without stabilization falls by 1.20, 1.75 and 3.52 per halving in the normalized 99th percentile, and in MPa rises from $a/2$ to $a/4$ and falls by 1.6 and 2.5 from $a/4$ to $a/8$ and from $a/8$ to $a/16$; its mean falls by 2.60, 2.29 and 3.07. With the stabilization the normalized 99th percentile falls by 1.55, 1.78 and 2.73, and in MPa by 0.76, 1.61 and 2.32. Up to $h = a/8$ the decay of CT10 is pre-asymptotic, and from $a/8$ to $a/16$ it is faster than first order in both measures, with and without stabilization. In case B the mean of TET15-P1 falls at first order (2.26, 2.00) and its normalized 99th percentile by 2.40 and then 1.63, while that of CT10 without stabilization falls by 1.56 and 1.79. From $a/8$ to $a/16$ in case B the mean of CT10 falls by 2.44 without stabilization and by 2.37 with it, and its normalized 99th percentile by 1.90 and 1.71. In case B the means of the three elements decrease by factors between 2.00 and 2.65 per halving at every pair of levels computed. The equivalent plastic strain shows no mode in any element: its jump statistics are of the same size for all four elements and decay by 2.1 to 3.9 per halving.

The stabilization of CT10. The parameter $\alpha = 0.1$ raises the force of CT10 by 5.4%, 3.1% and 1.1% at $h = a/2$, $a/4$ and $a/8$ and 0.20% at $a/16$ in case A and by 13%, 4.4%, −0.1% and 3.7% in case B, and its normalized 99th percentile by 5 to 38%, with the largest $|p|$ 1.4 to 2.5 times that without it. On this problem there is no oscillation in CT10 for the stabilization to remove; it adds stiffness and pressure extremes.

Mesh dependence of the force in case B. In case B the CT10 force at $h = a/16$ differs from that at $a/8$ by more than in case A. With $\alpha = 0.1$ the difference ranges from -4.8% at $t = 0.25$ to $+4.1\%$ at $t = 0.90$, and it is $+3.6\%$ at $t = 1$. Without stabilization it ranges from -2.5% at $t = 0.25$ to $+2.3\%$ at $t = 0.65$, and it is -0.3% at $t = 1$. In both rows the $a/16$ force is below the $a/8$ force up to about $t = 0.45$ and above it after. Two repetitions of the $a/8$ run with $\alpha = 0.1$ test whether the difference comes from the time step. The maximum step at $a/8$ is 3.125×10^{-3}; the repetitions use 1.5625×10^{-3}, the maximum step at $a/16$, and 7.8125×10^{-4}. Their forces differ from that of the original $a/8$ run by at most 0.019% and 0.035% over the twenty output times. The difference between $a/8$ and $a/16$ is therefore an effect of the mesh, not of the time step. In the CT10 runs the equilibrium residual is at most 3×10^{-5} of the force, so the difference is not an effect of the nonlinear tolerance either. The force of TET15-P1 in case B changes by -0.51% from $a/4$ to $a/8$; its value at $a/16$ is not yet available.

Loss of positive definiteness of the tangent in case B. In the case B run of Carina at $h = a/4$ with the direct solver (§12, linear solver), the Cholesky factorization first failed in the step from $t = 0.0813$ to 0.0875. A second computation of the same case wrote the tangent at every factorization and was stopped after the first failure. In it the tangent loses positive definiteness in the step from $t = 0.0625$ to 0.0688. The tangent at the converged state at $t = 0.0625$ and the tangents of the predictor and of the first two Newton iterations of the step have a Cholesky factor, and the tangent of the third Newton iteration has none. The loss therefore occurs at an unconverged Newton iterate. The last two assembled tangents were analyzed by shift-and-invert subspace iteration for the smallest eigenvalues, with the inertia, the numbers of positive, zero and negative eigenvalues, from an LDL^{T} factorization. The first is positive definite. Its six smallest eigenvalues, 7.1×10^{-9} to 3.3×10^{-8} times the largest diagonal entry, belong to modes with r_v between 0.04 and 0.13, 80 to 87% of whose dilatation lies in three elements, with the largest displacement at $r = 1.06a$ to $1.09a$ on the top surface. The second has exactly one negative eigenvalue, -8.7×10^{-8} times the largest diagonal entry. Its mode has $r_v = 0.075$ and 84% of both its gradient energy and its dilatation in three elements. These elements are adjacent to the punch edge, at $r = 1.07a$ to $1.12a$ on the top surface, and their shape quality is 0.72, 0.73 and 0.94, against 0.49 for the worst element of the mesh. The loss of definiteness is therefore accompanied by an element-scale mode localized at the punch edge, of the kind found in §9.7 on unstructured meshes, with a dilatation below the threshold $r_v = 0.1$, and not by a mode on the scale of the plastic zone. The same tangents give $\left\|\boldsymbol{K} - \boldsymbol{K}^{\mathsf{T}}\right\|_F / \|\boldsymbol{K}\|_F = 1.7 \times 10^{-15}$ in a plastic state in which 65% of the elements have yielded. Script: `prototype/tangent_modes.jl`.

The punch edge is a singular point, at which perfect plasticity can also localize, and the only comparator available locks. The plain fifteen-node tetrahedron, the same displacement element without the projection, keeps a tangent with a Cholesky factor at every iterate to $t = 0.125$ on the same mesh. The tangent of CT10 cannot be extracted from Sierra/SM, and a constant-pressure comparator in Carina was not computed.

A third computation of the same case, whose step sequence differs from that of the second, loses definiteness at the second Newton iterate of the step from $t = 0.0125$ to 0.0187. Its tangent has one negative eigenvalue, -8.9×10^{-8} times the largest diagonal entry, and the largest contribution to its mode comes from the same element. For this tangent the energy $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{v}$ of the unit eigenvector $\boldsymbol{v}$ is split, element by element, into the three terms of the tangent of the split form (§4.4): the isochoric material term, the pressure term $\bar{p}_q \operatorname{Grad} \boldsymbol{v}_q : (\partial \boldsymbol{g}/\partial \boldsymbol{F})_q : \operatorname{Grad} \boldsymbol{v}_q$,

Table 25: Punch, case B, $h = a/4$, Carina TET15-P1: the energy $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{v}$ of the unit eigenvector of the negative eigenvalue of the tangent at the second Newton iterate of the step from $t = 0.0125$ to 0.0187, split into the three terms of the tangent, in N/m, over the mesh and in the element with the largest contribution. Last column: the largest magnitude of the projected pressure at a quadrature point of that element, in units of σ_y.

	total	isochoric	pressure	projected volumetric	$\max \lvert\bar{p}_q\rvert/\sigma_y$
mesh	-1.835×10^3	$+7.19 \times 10^3$	-9.03×10^3	0.14	316
largest element	-6.61×10^3	$+2.29 \times 10^3$	-8.90×10^3	0.021	316

and the projected volumetric term $\kappa\,(\boldsymbol{D}^{\mathsf{T}}\boldsymbol{v}) \cdot \boldsymbol{M}^{-1}\boldsymbol{D}^{\mathsf{T}}\boldsymbol{v}$. Here $\bar{p}_q$ is the projected pressure at quadrature point q, and $\partial\boldsymbol{g}/\partial\boldsymbol{F}$ is the derivative of the volumetric direction $\boldsymbol{g} = \partial\theta/\partial\boldsymbol{F}$. Table 25 gives the split. The sum of the element contributions equals the eigenvalue, -1.835×10^3 N/m, to four digits. Script: `prototype/pd_split.jl`.

In the element with the largest contribution the largest magnitude of the projected pressure at a quadrature point is $316\,\sigma_y$ at the iterate that fails and $278\,\sigma_y$ at the preceding iterate. At the converged states of a computation of the same case, at $t = 0.05$ to 0.3, the largest magnitude of the pressure at a quadrature point of that element is 1.8 to $2.3\,\sigma_y$, and the largest over the mesh is 2.8 to $4.0\,\sigma_y$. The minimum over unit vectors $\boldsymbol{a}$ and $\boldsymbol{k}$ of the acoustic quantity $(\boldsymbol{a} \otimes \boldsymbol{k}) : \partial\boldsymbol{P}^{\text{iso}}/\partial\boldsymbol{F} : (\boldsymbol{a} \otimes \boldsymbol{k})$ of the isochoric tangent is negative at 128 of the 6187 elements, with a least value of $-6.1 \times 10^{-4}\,\mu$, of the order of $\sigma_y/\mu = 10^{-3}$. With the pressure term added it is $-6.8 \times 10^{-2}\,\mu$ in the element that carries the mode. The negative eigenvalue therefore comes from the pressure term of the tangent, evaluated at a projected pressure about 150 times its value at the converged states, on a displacement that the projection leaves free of volumetric energy and on which the isochoric tangent of the perfectly plastic material is close to singular. In the three computations the tangents at all converged states up to the step in which definiteness is lost have a Cholesky factor. The plain fifteen-node tetrahedron, whose pressure at converged states reaches 770 to $2140\,\sigma_y$ at $t = 0.05$ to 0.3, keeps a positive definite tangent because its pointwise volumetric term gives the same displacement an energy proportional to κ. The stalled nonlinear iterations of the Sierra/SM TET15-P1 runs at $h = a/16$ in case B were localized at the face-bubble nodes of the first ring of elements outside the punch edge, the same location. After the loss of definiteness the conjugate-gradient iteration of Carina reaches its limit of 1000 iterations in 1666 of its 1866 solves at $h = a/4$, and the direct solver changes from Cholesky to LU decomposition. These are properties of the nonlinear and linear solvers at unconverged iterates, and the converged solutions of the two solvers agree in force to 0.002%.

What the benchmark establishes, and what it does not. The two implementations of TET15-P1, with different J_2 models, agree in force within 0.06%; TET15-P1 does not lock, carries the smallest pressure jumps of the elements compared, and its jumps decay under refinement from the coarsest level on. The plain element locks and carries a pressure mode that grows under refinement. CT10, with and without stabilization, does not show an element-to-element pressure mode that persists under refinement: in case A on the finest level its jumps decay faster than first order, and in case B they decay at about first order from $a/8$ to $a/16$. The soft modes of constant-pressure elements found in the linearized analysis of §9.7 carry a dilatation that varies within an element and averages to zero over it; the jump measure

compares element averages and does not see such a mode (§12, the jump measure), so the benchmark neither confirms nor refutes them. A measure that resolves the pressure within an element (the quadrature-point values of TET15-P1 and the sub-tetrahedron values of CT10) is the measurement that would. In case B the tangent of TET15-P1 loses positive definiteness at unconverged iterates through the pressure term at large projected pressures; the tangents at the converged states are positive definite. Sierra/SM TET15-P1 in case B at $h = a/16$ and Carina TET15-P1 and plain `TETRA15` in case B at $h = a/8$ are in progress. The definition of the problem, the drivers, the meshes and the extraction are in `benchmark/tet15-p1/punch`.

Case B at $h = a/16$ in Sierra/SM. Sierra/SM solves each step by a nonlinear conjugate-gradient iteration preconditioned by the full tangent, which is applied by a linear solver. With the initial settings every case B run at $h = a/16$ stalled in the nonlinear iteration between $t = 0.04$ and 0.14. Four changes of the solver settings, applied in restarts, remove the stalls. The first raises the limit on the nonlinear iterations of a step from 400 to 1600, so that slowly converging steps are not cut back. The second sets the target iteration count of the adaptive step control to the same 1600, so that a step accepted at the limit keeps its step size instead of halving it. The third raises the absolute residual target from 10^{-6} N to 10^{-5} N, because the residual of TET15-P1 stagnated at a level between the target and the acceptable residual, ten times the target. The fourth, for TET15-P1 only, replaces the FETI-DP preconditioner by a two-level overlapping Schwarz preconditioner of the generalized Dryja–Smith–Widlund (GDSW) type, under which the linear solves no longer stagnate. With GDSW the nonlinear iteration limit is lowered to 200. An attempt that fails under GDSW keeps a relative residual near 1 (0.93 in the first one observed), and the step converges after the cutback. The lower limit shortens such a failed attempt from about 40 h to about 4.5 h. In single runs the maximum step is halved to 7.8125×10^{-4} after an attempt makes no progress over more than 100 iterations. The three CT10 runs of case B at $a/16$ are complete with these changes: 14.576 N with $\alpha = 0.1$ and the hypoelastic model, 14.051 N with $\alpha = 0$ and the multiplicative model, and 14.0511 N with $\alpha = 0$ and the hypoelastic model, at $t = 1$. The TET15-P1 runs reached $t = 0.42$ (hypoelastic) and $t = 0.29$ (multiplicative) on 2026-10-18.

Two further changes of the solver settings were made for the TET15-P1 runs after 2026-10-18. The overlap of the GDSW subdomains was set to one, which halves the setup time of the preconditioner, 33 s against 65 s, with an unchanged convergence of the nonlinear iteration. Without overlap the Krylov iteration (GMRES) needs about 550 iterations per linear solve. The limit on the nonlinear iterations of a step was lowered from 200 to 30, so that an attempt that does not converge is cut back sooner. Two other changes were tested and gave no improvement: the reuse of the preconditioner over several linear solves, and a variant of the line search. The hypoelastic run was restarted from $t = 0.73125$ with these settings. The multiplicative run with the limit of 200 was stopped at $t = 0.444$; a run restarted from $t = 0.3564453125$ with an overlap of one layer and the limit of 30 replaces it and had reached $t = 0.477$ on 2026-10-20 at 05:10 UTC. Both runs are in progress.

Domain decomposition in Sierra/SM. At 16 ranks and more the default partition of the mesh, recursive inertial bisection, leaves single elements or small groups joined to the rest of their subdomain along an edge or at a node. The subdomain matrix of FETI-DP then has a zero-energy hinge rotation that its corner nodes may leave unconstrained. Sierra/SM computes the element tangents of TET15-P1, CT10 and `TETRA15` by finite differences, and the

hinge pivot of that matrix is the difference error: for TET15-P1 between 10^{-13} and 10^{-12} of its largest entry, above the tolerance at which FETI-DP declares a mode null (10^{-14}) and too small to resolve, so the local solve returns a displacement of 10^{-4} m at an interface node and the first Newton step fails; for CT10 and `TETRA15` between 10^{-10} and 10^{-8}, resolved as stiff and harmless. A partition into face-connected subdomains (multilevel Kernighan–Lin graph partitioning with face adjacency and the connected-domain option) removes the hinges for every element; with it the 16-rank run of case A at $h = a/4$ reproduces the 8-rank run to 9.2×10^{-6} in force. An element tangent in closed form does not remove the defect: in an unstressed hinged element it gives the hinge a pivot at round-off (10^{-17} in a stand-alone script on the hinged element of the $a/4$ partition), which FETI-DP detects, but under stress the hinge carries its physical initial-stress stiffness, small and negative in compression (-6.4×10^{-8} of the largest entry at a strain of -2.4×10^{-6}), the same in every tangent. The connected partition is the remedy; the closed form would reduce the cost of the tangent, which the probe takes 30%, 17% and 4 to 8% of the step time at $h = a/4$, $a/8$ and $a/16$, and costs no Newton iteration in the steady regime, where both codes take three per step.

13 Summary of the evidence and open measurements

The formulation rests on six claims. For each, the section that establishes it and the number that carries it are given; the measurements that remain are listed at the end.

The pair is inf-sup stable in three dimensions. §9.5: the Crouzeix–Raviart pair has $\beta_h = 0.2968$ at a local rate of -0.003, above the Taylor–Hood control's 0.2216, with a one-dimensional pressure null space at every mesh; the constant-pressure pair with face bubbles is stable with a constant decreasing toward a limit near 0.35, and the five other element-local pairs decay. On randomly perturbed meshes the constant of the Crouzeix–Raviart pair lies between 0.287 and 0.295, and on unstructured Cubit meshes between 0.319 and 0.332 from $n = 3$ on, with one null mode on both (Table 9); the decay of P_2/P_0 is a property of the Freudenthal family. The face bubbles are the necessary part of the enrichment; the interior bubble alone is not sufficient (Remark 7). Uniformity in κ/μ follows from stability of the incompressible limit [7, §5.5.2, §8.12.1, Remark 8.12.1]. On deformed configurations (§9.6) the constant decreases under the non-affine maps, to 0.204 at a 90° twist and 0.116 at 180°, and by 7% under a volume change of 2.3, and at each fixed deformation the sequence in h is consistent with a positive limit; the constant is not uniform in the deformation.

No spurious soft modes. In elasticity the second hypothesis of Brezzi's theorem holds for every conforming pair (§6). The three exact zero-energy modes per element of a constant pressure (§9.2) do not persist after assembly (§9.4). Under a plastic tangent (§9.7) the coercivity constant is bounded below by (49) for every pair, and the dilatation of the soft modes separates them on the Freudenthal meshes: 0.20–0.29 without decay for the constant-pressure pairs, CT10 with its averaged J among them at 0.22–0.25, and a decay at local rates of 3.1 to 2.2 for the Crouzeix–Raviart pair, to 0.0014 at $N = 8$. The smallest energy $\lambda_*(0.1)$ of a discretely isochoric member with $r_v \geq 0.1$ (Table 12) lies above the continuum infimum $\lambda_\infty = \frac{1}{8}$ for the Crouzeix–Raviart pair on every mesh except one strongly perturbed mesh, and below it for the constant-pressure pairs. The heading of this paragraph holds in that sense only.

On unstructured meshes the Crouzeix–Raviart pair has element-scale members with $r_v \geq 0.1$ within 12% of its lowest eigenvalue up to $n = 10$; they are spurious dilatational modes by the definition of §6, with an energy above λ_∞. In the nonlinear indentation benchmark (§12) the element-to-element pressure jumps of TET15-P1 are the smallest of the elements compared and decay under refinement, and those of CT10 decay as well, in case A faster than first order on the finest level; that measurement compares element averages and does not resolve the modes above. In case B the tangent of TET15-P1 loses positive definiteness at an unconverged Newton iterate, through its pressure term at a projected pressure of $316\,\sigma_y$, with a negative mode of the element-scale kind localized at the punch edge (Table 25).

No volumetric locking. §9.4: the discretely isochoric fraction $\dim\ker \boldsymbol{G}/n_u$ is 0.61–0.63 for the Crouzeix–Raviart pair, at the bound $1 - n_p/n_u$, against 0.03–0.07 for P_2/P_1^{disc}, which locks. §10: on Cook's membrane at $\nu = 0.4999$ the element agrees with CT10 in the tip displacement within 0.6% on meshes of 327 to 14 475 elements, and its pressure is smooth away from the singular clamped edge, where the quadratic tetrahedron without projection oscillates in sign inside every element. §9.8: against a manufactured solution in linear elasticity, the rate of convergence of TET15-P1 at $\kappa/\mu = 10^6$ is 0.047 below its rate at $\kappa/\mu = 1$ from $N = 10$ to 12, a difference that decreases under refinement, against 0.94 for TET10 with the volumetric response at each quadrature point. Its quasi-optimality ratio at $\kappa/\mu = 10^6$ grows from 1.45 to 1.59 over $N = 2$ to 12, about as $N^{0.06}$.

Accuracy in explicit dynamics. §11: on the Taylor bar the final radius is within 0.06% of the extrapolated value on 25 029 elements, against 0.09% for CT10 on 1 521 212, both of the order of the uncertainty of the limit, 0.08%. Within Sierra/SM the first costs 1/54 of the core-seconds of the second, and 1/63 at a tolerance of 0.3%. The constant-pressure variant is 11% from it at $h = 0.75$ mm.

Element-local elimination. §4.2: the pressure is discontinuous, so its elimination is element by element and the tangent has the sparsity of the displacement mesh; the face bubbles add global displacement unknowns, 2.6 times those of P_2/P_0 at $N = 12$ (§9.3).

Quadrature and basis. §9.9: the 14-point degree-5 rule keeps six zero-energy modes and the assembled constant within 1%; the nodal basis has a positive row-sum lumped mass that conserves the momentum of a rigid translation, and a Courant number of 0.145 under HRZ lumping against 0.476 for TET4. The consistent mass is singular under the 14-point rule and is integrated in both implementations with the 125-point conical rule of degree nine, with condition number 93.6 (Table 4).

Open measurements. Six remain. A measurement showing whether the quasi-optimality ratio of TET15-P1 at large κ/μ levels off; it grows through $N = 12$ (§9.8). The finest level of the indentation benchmark of §12: TET15-P1 in case B at $h = a/16$, which Sierra/SM solves only with changed solver settings, and Carina TET15-P1 and plain `TETRA15` in case B at $h = a/8$; these runs are in progress. A comparator for the loss of positive definiteness of the case-B tangent: the tangent of CT10 cannot be extracted from Sierra/SM, and a constant-pressure element in Carina has not been computed. The source of the force difference of 0.02 to 0.06%

between the two implementations of TET15-P1. A measurement of the pressure within each element on that benchmark, which alone can show the soft modes of constant-pressure elements found in the linearized analysis: their dilatation varies within an element and averages to zero, and the element-to-element jumps measured so far, which show no persistent mode in CT10 up to $h = a/16$ in case A, do not resolve it. The large-rotation twist of a bar and a weak shock in a material of high κ/μ, measured with the implemented Simo–Hughes instance (Remark 5) against CT10 on the same meshes. The comparison at equal accuracy against CT10 has been made on one problem, the Taylor bar (§11); a problem that the quadratic element does not resolve on a coarser mesh would give a different result.

14 Limitations

This section states what the formulation requires, what it does not establish, and what its parts are.

The number of unknowns. The face bubbles of the Crouzeix–Raviart pair are shared between neighbors and do not condense: at $N = 12$ the enriched space carries 96,117 global displacement degrees of freedom against 36,501 for P_2/P_0 and for CT10 on the same mesh, a factor of 2.6, and 2.5 times Taylor–Hood's 38,698 (§9.3); the implementation does not condense the interior bubble, and on the Cook meshes of §10 the factor is 3.0. On the Taylor bar (§11) the accuracy on a coarser mesh more than compensates for that factor: the final radius within 0.06% on 25 029 elements, against 1.5 million elements for CT10 at 0.09%. On other problems it is not established; for elasticity that is the comparison with Taylor–Hood, whose pressure is global and whose displacement space is 2.5 times smaller, and for plasticity Taylor–Hood's softest modes are dilatation almost entirely (§9.7), so the comparison there is with CT10.

Operations per element. With the 14-point rule of §9.9 the enriched element evaluates the constitutive law at 14 points on 45 displacement degrees of freedom, against 4 or 5 points on 30 for TET10 and one on 12 for TET4. On the Taylor bar the time per element and explicit step on one core of Carina TET15-P1 is 1.40 times that of CT10 in Sierra/SM, and within Sierra/SM the time of Sierra/SM TET15-P1 is at most 2.6 times that of CT10 (§11).

The explicit stable step. Row-sum lumping in the nodal basis gives positive masses and conserves momentum, and the stable step is 3.3 times smaller than TET4's on the same element, of which a factor of 2.4 is the quadratic element and 1.4 the enrichment (§9.9). On the Taylor bar at equal accuracy the smaller element count outweighs the smaller step (§11). On a deformed mesh an element estimate from node distances does not bound the critical step; a step of 0.8 times $2/\sqrt{\lambda_{\max}}$, with $\lambda_{\max}$ the largest eigenvalue of $M_L^{-1}K$, stays below it, at a cost of 3 to 15% of the run on the GPUs with the estimate every 200 steps (§11); a fixed longer interval is unsafe on the Taylor bar, and the interval set from the observed change of the ratio reduces the time of the run by 11% on the RX 7600 and the L4, 9% on the V100 and 3% on the H100. If weak shocks are run with the implicit integrator the factor does not apply.

The material. The element places no requirement on the material: the split form of §4.4 calls it through the split of §3.6, which of the seven material models of Table 1 five have, and

the general form of §3.5 calls any material at the modified deformation gradient (21), at 1.36 to 1.60 times the split form's time per residual. What the general form requires is a positive volumetric tangent of the material over the states reached, and what it changes for a material whose isochoric response depends on J is that this response is evaluated at the projected volume. The element matrix and the diagonal of the general form take 0.9 and 1.2 times the split form's time on a CPU with 12 threads, and the diagonal 2.5 times on the RX 7600 (§3.5). For the split form, the action of the tangent on a vector takes 2.4 times the residual's time on the RX 7600 (20.8 against 8.6 ms on the Taylor mesh at $h = 0.75$ mm).

The parts. Mixed volumetric treatment [30], logarithmic-strain plasticity [20] and displacement–pressure pairs satisfying (38) uniformly [7] are established, and the pair itself has been used in large-deformation hyperelasticity [10]. The contribution is their combination for finite-deformation plasticity: an element-local pair satisfying (38) uniformly, a projected volumetric strain, and a deviatoric law evaluated pointwise, which removes the stabilization that the CT10 and variational-multiscale methods carry; together with the measurements that support it, of the dilatation of the soft modes under a plastic tangent (§9.7), of the inf-sup constant on deformed configurations (§9.6), and of the nonlinear response (§10, §11, §12). The evidence is that of §9, §10, §11 and §12; the open measurements are those of §13. Hybrid high-order methods, with unknowns on element faces and element interiors coupled only through the faces, are locking-free and eliminate their interior unknowns element by element, and exist for small-deformation plasticity [1] and for finite plasticity in logarithmic strain [2]; they are the alternative, at the price of face unknowns that are not the nodal displacements of a standard element.

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